Bambu Lab P1 Series

HMS Error Code Manual

Health Management System (HMS) Troubleshooting Guide

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Table of Contents

Error Code: 0300-0100-0001-000A

HMS_0300-0100-0001-000A: The heatbed temperature control is abnormal, the AC board may be broken.

Observation

The AC board provides alternating current for the heatbed to heat up. When the AC board module is abnormal, the heating bed's temperature rise is abnormal. Possible causes of this abnormality:

- The AC board malfunction
- The MC board malfunction
- The heatbed malfunction

Troubleshooting

- H1.5 and H2.0 Hex Wrench
- Phillips Screwdriver
- Tweezers

Fault Isolation Procedure

This chart illustrates the general troubleshooting procedure, including critical judgments, checks and actions.

Actions

Operators may use discretion in determining the actual procedures used, and the order in which these steps are applied.

- HeatBed Resistance Measurement
  A1
  A1 mini
  X1 Series
  P1 Series
  P2S
- Replace AC Board
  A1
  A1 mini Mainboard
  X1 Series
  P1 Series
  H2D
  P2S
- Replace MC Board
  A1 Mainboard
  A1 mini Mainboard
  X1 Series
  P1 Series
  H2D
  P2S
- Replace Heatbed
  A1
  A1 mini
  X1 Series V3 Heatbed/V2 Heatbed
  P1 Series

H2D
P2S
If the issue persists after troubleshooting, please submit a customer support ticket.

Error Code: 0300-0100-0001-0003

The heatbed temperature is abnormal; the heater is over temperature.

Issue Description

This alert indicates that the actual temperature of the heatbed is higher than the set temperature.

Example of heatbed temperature control anomaly, heater over temperature

The temperature control system of the heatbed for the X/P1/A1/H/P2 series printers is shown below, mainly including: MC board, AC board, power supply, and heatbed (including heatbed temperature sensor). The MC board issues heatbed temperature control commands to control the AC board to supply power and heat the heatbed. Meanwhile, the temperature sensor on the heatbed feeds back the actual temperature of the heatbed to the MC board, achieving a closed-loop temperature control of the heatbed. The A1mini printer mainly includes: motherboard, power supply, and heatbed (including heatbed temperature sensor), where the motherboard integrates the functions of the MC board and AC board.

Schematic diagram of the heatbed temperature control system

Common reasons for the actual temperature of the heatbed exceeding the set temperature are:

• Abnormal heating power supply chain of the heatbed (AC board/motherboard failure), causing continuous power supply and heating to the heatbed
• Heatbed temperature sensor failure, resulting in incorrect identification of the heatbed temperature**

Troubleshooting

Fault Isolation Procedure

This chart illustrates the general troubleshooting procedure, including critical judgments, checks, and actions.

Heatbed over-temperature fault isolation process

1. Abnormal heating power supply chain of the heatbed

1. Abnormal temperature measurement chain of the heatbed

Error Codes: 0300-1000-0002-0001, 0300-1100-0002-0001

The 1st order mechanical resonance mode of X axis is low.

What the error is

As time goes by, the pretension force of the timing belt will drop gradually, and the stiffness will get low. High print acceleration will influence the print quality if the stiffness is too low. So when the printer starts the resonance calibration, it will calculate the pretension status of the belt; if it is loose, this error will be present.

This wiki article equally applies to the HMS_0300_1100_0002_0001: The resonance frequency of the Y axis is low. The timing belt may be loose.

- IPA-Isopropyl alcohol (Or ethanol)
- Clean cloth

Safety warning

Due to the need to manually move the toolhead and spray the cleaning agent, it is necessary to disconnect the power supply in advance.

Troubleshooting guide for X1 and P1 series printers:

Follow the instructions in this video to re-tension the belt and clean the carbon rods:

These wiki guides can also help: Belt tensioning procedure | Bambu Lab Wiki; How to clean the carbon rods for Bambu Lab Printers | Bambu Lab Wiki

Troubleshooting guide for A1 series printer:

The possible problems are ranked by probability: 1. Software bug. 2. The timing belt may be loose.

1. Solution for software bug:

We expect to release a new version of Bambu Studio in mid-November 2023. Until then, you can ignore this error message.

Alternatively, you can replace the "start gcode" in the current latest version of Bambu Studio (click here to download).

Click here to download the start gcode

If you encounter this error message with the built-in model for printing, you can click here to download the latest version of the built-in slice file. Then, copy it to the micro SD card to replace the factory-installed built-in model.

2. Tighten the belt as the guide below:

Use an H2 Hex key to loosen the screw on the back of the toolhead (as shown below), and tighten it again. Then perform the printer calibration again, and the error should disappear.

Do not operate the screw on the side of the toolhead, as shown below:

Get more information on A1 series timing belt tensions:

A1 Belt Tensioning

A1 mini Belt Tensioning

Error message

HMS_0300-1000-0002-0001: The 1st order mechanical resonance mode of X axis is low.

"The 1st order mechanical resonance mode of X axis is low."

The error code below shows the same problem of different axes:

0300-1000-0002-0001

0300-1100-0002-0001

Error Codes: 0300-1000-0002-0002, 0300-1100-0002-0002

The resonance frequency of the X axis differs greatly from last calibration. Please clean the carbon rod and rerun the machine calibration afterward

What it is

In the default start gcode, the printer will perform a fast mechanical resonance mode checking, if the result shows that the resonance mode has drifted away from the last auto-calibration, this error message will present.

Troubleshooting guide

step 1

Follow the instructions in this video to re-tension the belt and clean the carbon rods:

These wiki guides can also help: Belt tensioning procedure | Bambu Lab Wiki; How to clean the carbon rods for Bambu Lab Printers | Bambu Lab Wiki

Error message

HMS_0300-1000-0002-0002: The resonance frequency of the X axis differs greatly from last calibration. Please clean the carbon rod and rerun the machine calibration afterward

“The resonance frequency of the X axis differs greatly from last calibration. Please clean the carbon rod and rerun the machine calibration afterward”

These error codes below show the same problem of different axes:

0300-1100-0002-0002

0300-1000-0002-0002

Error Code: 0300-0F00-0001-0001

HMS_0300-0F00-0001-0001: Abnormal accelerometer data detected. Please try to restart the printer.

What it is

This shows that the machine control board cannot get the accelerometer data from the tool head.

Possible reason:

1. The instantaneous high load on the MC board CPU causes the accelerometer data to be lost.
2. The connection between the tool head and the MC board is broken. For example :For X1C, the connection includes TH -> AP ->MC For P1P, the connection includes TH -> MC
3. For X1C, the connection includes TH -> AP ->MC
4. For P1P, the connection includes TH -> MC
5. The toolhead board(TH board) firmware hasn't been successfully upgraded.

- For X1C, the connection includes TH -> AP ->MC
- For P1P, the connection includes TH -> MC

IMPORTANT! It's crucial to power off the printer before performing any maintenance work on the printer and its electronics, including tool head wires, because leaving the printer on while conducting such tasks can cause a short circuit, which can lead to additional electronic damage and safety hazards. When you perform maintenane or troubleshooting on the printer, you may be required to disassemble some parts, including the hotend. This process can expose wires and electrical components that could potentially short circuit if they come into contact with each other or with other metal or electronic components while the printer is still on. This can damage the electronics of the printer and cause further damage. Therefore, it's essential to switch off the printer and disconnect it from the power source before doing any maintenance work. This will prevent any short circuits or damage to the printer's electronics. By doing so, you can avoid potential damage to the printer's electronic components and ensure that the maintenance work is performed safely and effectively. If you have any concerns or questions about following this guide, open a new ticket in our Support Page and we will do our best to respond promptly and provide you with the assistance you need.

IMPORTANT!

It's crucial to power off the printer before performing any maintenance work on the printer and its electronics, including tool head wires, because leaving the printer on while conducting such tasks can cause a short circuit, which can lead to additional electronic damage and safety hazards.

When you perform maintenane or troubleshooting on the printer, you may be required to disassemble some parts, including the hotend. This process can expose wires and electrical components that could potentially short circuit if they come into contact with each other or with other metal or electronic components while the printer is still on. This can damage the electronics of the printer and cause further damage.

Therefore, it's essential to switch off the printer and disconnect it from the power source before doing any maintenance work. This will prevent any short circuits or damage to the printer's electronics. By doing so, you can avoid potential damage to the printer's electronic components and ensure that the maintenance work is

performed safely and effectively.

If you have any concerns or questions about following this guide, open a new ticket in our Support Page and we will do our best to respond promptly and provide you with the assistance you need.

Troubleshooting Guide

X1/P1 Series

1. If it is a momentary alert, you can choose to ignore it and continue printing. If the alert recurs frequently, you can try restarting the printer to solve the the instantaneous high load on the MC board CPU issue.

2. If this error is caused by poor contact, reinstall the USB-C cable or toolhead cable (X1: Type-C cable, P1: Toolhead cable)

X1: Bambu USB-C Cable installation tutorial; USB-C cable connection issues

P1:Toolhead cable installation tutorial

For P1, please also check whether the seat of the tool head cable is soldered. If the seat is floating and falsely welded, please contact our after-sale service.

The MC-TH connector is directional, so please ensure the socket is inserted correctly. If it is inserted in reverse, it will cause a power short circuit, resulting in the printer reporting an HMS error or not being able to power on properly.

For X1, in addition to checking the USB-C cable, you can also check whether the AP-MC cable at both ends is well connected.

A1 Series

1. For the accelerometer data loss caused by some momentary high CPU load, you can choose to ignore it and continue printing. If the alarm reappears frequently, you can restart the printer to eliminate this fault.

2. Please check whether the USB-C cable connection at both ends is normal. You can reconnect it several times to confirm that the connection is stable.

• A1 mini: Please refer to this link to remove the base plate and check the USB-C cable connection on the mainboard. Note: There is no need to cut any zip ties on the cable.

A1 mini installation tutorial

3. If it is caused by a firmware upgrade failure, please restart the printer and update the firmware according to the prompt.

• A1: Make sure the USB-C cable on the bottom of the printer is fully inserted into the connector. After the USB-C cable is inserted into the connector, push the cable box all the way in.

Error message

HMS_0300-0F00-0001-0001: Abnormal accelerometer data detected. Please try to restart the printer.

0300-0F00-0001-0001

Error Code: 0300-0D00-0001-000B

HMS_0300-0D00-0001-000B: The Z axis motor seems to be stuck when moving. Please check if there is any foreign matter on the Z sliders or Z timing belt wheels.

What it is

This error shows that when the Z axis moves up, the Z motor detects an abnormally large force that stops the motor from spinning.

1. There is something on the table that has stuck / partially stuck the Z-axis motor or the Z-axis timing belt wheels, which are at the bottom of the machine
2. There is stuff on the Z-axis slider when the hotbed is moving up, or the bearing on the linear rods is raised, this stuff is stuck between the Z-axis slider and the Z-axis, causing the nozzle unable to touch the heat bed
3. When the heat bed was near the bottom of the chamber, the Z-axis slider was stuck by something on the bottom.
4. The heat bed was interfered with by the excess chute.
5. The heat bed cable is dragging the heat bed from moving up
6. Wrong screw screw has been installed when installing the AMS-HUB.
7. Z-axis slide assembly tolerance is abnormal.
8. The force sensor is abnormal, making the machine unable to detect the touching event when the nozzle touches the heat bed until the Z-axis motor detects a high load and stops.

Please note that these are only some of the possible causes listed above, in reality, there may be other factors causing the error.

Troubleshooting guide

Step 1: Check whether Z slider is stuck by a foreign object

Lower the heat bed, and check if there is anything on the Z-axis slider upper surface, especially the rear one that is partly hidden behind the inner liner :

In addition to checking the surface of the slider for foreign objects, in this step, you will also need to check whether the Z-axis bearing is raised.

If the bearing is raised, please refer to the operation video below to reset the bearing. It is important to note that before following the video, please remove the hotend assembly of the tool head and printer's screen (click here for a guide to disassembling the hotend assembly) and then perform the operation in the video.

- High Resolution Screen Installation Guide
- P1 Screen Installation Guide

If you encounter any difficulties during this process, please contact after-sales service for assistance.

A1 Normal

If you need to disassemble and inspect the A1, please refer to the following two links:

A1 Printer Frame Installation Tutorial

A1 Installation Tutorial

The purpose of this operation is to press the raised bearing back into place using the retaining block:

Raised bearings are usually caused by the presence of foreign material, such as scrap, in the three holes at the bottom of the linear rods, which causes the bearing to be squeezed and raised out of place. If there is any foreign material in these holes, make sure they are thoroughly cleaned out or the failure may recur.

Step 2: Check whether the excess chute is interfering

Check if there is visible interference between the heat bed and the excess chute.

If you find that the excess chute is interfering, click to view this wiki page for a solution.

Step 3: Re-adjust Z sliders

the steel rod may have moved under violent shipping, so the z-axis slider needs to be re-adjusted, see the picture below:

For X1 and P1, refer to steps 3.1 to 3.5.

For A1, refer to steps 3.6 to 3.9.

3.1: Use the control panel, and lift the bed carefully until it almost touches the bed

3.2: Move the toolhead aside and the build plate aside

3.3: Loose both of these 2 screws and re-tighten them (at the back)

3.4: Loose both of these 2 screws and re-tighten them (on the left)

3.5 Loose both of these 2 screws and re-tighten them (on the right)

In some batches of machines, as shown in the image on the right, one of the screws may be concealed beneath a sticker. Please feel free to pierce through the sticker.

The steel rods may shift due to rough handling during transportation, so it is necessary to re-tension the Z-axis synchronization belt. For A1 printers, please refer to Steps 3.6-3.9 for instructions.

3.6 The tensioning position of the Z Belt is located near the right column of the Z-axis, which can be seen by moving the heatbed forward.

3.7 Move the heatbed forward to expose the tensioning position of the Z Belt, and loosen the two tension screws with an H2.0 Allen key by one turn (do not fully loosen them)

3.8 Enter the Control menu and perform Homing; Then hold down the Z-axis up (down) button to move the X-axis up and down once along the Z-axis direction

(Be careful not to make the nozzle hit the heatbed), and then retighten the 2 tensioning screws.

3.9 Select Vibration Compensation in the calibration menu and complete the calibration process.

Step 4: Check for objects at the bottom of the machine

Lift the printer front a bit, and check if there is anything that will stick to the Z-axis timing belt.

Notice that the rubber foot is stuck on to the bottom of the printer, if the printer is pushed on the place surface, the rubber foot may get pushed off and gets stuck into the Z-axis timing belt

(where as shown in the red box below):

Step 5: Check whether Z motor and belt are stuck

Lift the back of the printer a bit, and check if there is anything that will stick the Z-motor and timing belt wheel (as shown in the red box below):

Step 6: Check whether the cable is too tight

raise the hotbed slowly, until it almost reaches the nozzle, then look into the inside of the liner from the chamber, to see if the hotbed cable next to the z-axis lead screw is dragged tight. If the shape of the cable is similar to the picture below, then it may be dragged tight:

Then you have to insert your finger into the gaps and drag the cable out from the inner of the electric case, as shown below,

CAUTION: the liner edge may be a bit sharp, be careful or wear gloves when doing this step

If the hotbed cable is stretched, click on this wiki page for the solution.

Step 7: Check AMS hub/ Filament buffer's screws

If you have installed the AMS-HUB or AMS-BUFFER if the screw is over-tightened, that may also stick the z-axis when the bed is near the nozzle, so try loose these two screws for 1~2 rounds :

Step 8: After completing all the steps above, click “home” again.

After the above steps, retry, tap the "home" button, and watch the homing process to check if the nozzle can touch the hotbed surface normally. If the homing is successful, then the homing z-axis failed error will disappear automatically.

If the issue persists, please contact Bambu Lab Support team.

Error message

HMS_0300-0D00-0001-000B: The Z axis motor seems to be stuck when moving. Please check if there is any foreign matter on the Z sliders or Z timing belt wheels.

“The Z axis motor seems to be stuck when moving. Please check if there is any foreign matter on the Z sliders or Z timing belt wheels.”

0300-0D00-0001-000B

Error Codes: 0300-0D00-0001-0003, 0300-0D00-0001-0004, 0300-0D00-0001-0005, 0300-0D00-0001-0006, 0300-0D00-0001-0007, 0300-0D00-0001-0008, 0300-0D00-0001-0009, 0300-0D00-0001-000A, 0300-0D00-0002-0003, 0300-0D00-0002-0004, 0300-0D00-0002-0005, 0300-0D00-0002-0006, 0300-0D00-0002-0007, 0300-0D00-0002-0008, 0300-0D00-0002-0009, 0300-0D00-0002-000A

HMS_0300-0D00-0001-0003: The build plate may not be properly placed. If this message appears repeatedly, please check the Wiki for more reasons.

X1/P1 series printers

What it is

When doing the Z-axis homing, the printer will calculate the touch position when the nozzle ---- or anything else --- touches the heat bed by the 3 force sensors.

Normally, if it is the very nozzle that touches the heat bed, then the calculated position will be the same as the toolhead position. Otherwise, it is possible that it is not the nozzle that touches the heat bed, which may be:

1. The build plate hasn't been placed correctly and has touched the excess chute
2. The build plate hasn't been placed correctly and has touched the z-axis slider stop on the left
3. The build plate hasn't been placed correctly and has touched the inner liner of the printer
4. Something has been dropped on the top of those 3 z-axis sliders and had made a false signal
5. There is stuff placed at the bottom of the printer chamber and has stuck the heat bed when doing z-axis homing
6. The problem might be that the X and Y axis homing process is disrupted by some foreign matters, which makes the position of the tool head not at the center of the build plate.

Troubleshooting guide

step 1

Lift the heat bed until its height is near the same as the excess chute, and check if the build plate has touched the surface of the excess chute:

step 2

Lift the heat bed until it is near the Z slicer stopper on the left, and check if the edge of the build plate will touch the stopper:

step 3

Check the rear edge of the build plate, to see if it is too close to the inner liner of the printer:

step 4

Lower the heat bed, and check if there is anything on the top surface of the 3 Z slider:

In addition to checking the surface of the slider for foreign objects, you will also need to check whether the Z-axis bearing is raised in this step.

If you find the bearing is raised, please refer to the operation video below to reset the bearing. It is important to note that before following the video, please remove the hotend assembly of the tool head (click here for a guide to disassembling the hotend assembly) and then perform the operation in the video.

If you encounter any difficulties during this process, please contact after-sales service for assistance.

The purpose of this operation is to press the raised bearing back into place using the retaining block:

Raised bearings are usually caused by the presence of foreign material, such as scrap, in the three holes at the bottom of the linear rods, which causes the bearing to be squeezed and raised out of place. If there is any foreign material in these holes, make sure they are thoroughly cleaned out or the failure may recur.

step 5

Clean the bottom of the printer. If the heat bed is near the bottom of the printer and has contact with stuff in the bottom initially, the homing process may get disrupted:

step 6

As the inner space of the XY gantry is very compact, ensure the full movement region of the X and Y sliders is clean, with no 3rd party parts or anything else being installed or placed near the slider movement region.

A1 series printers

A1 series printers are equipped with a printing board detection function, you can refer to this wiki for details: Build plate detection | Bambu Lab Wiki

So, before printing, it is necessary to properly place the printing board on the heat bed. If this error occurs, there may be the following reasons:

1. The build plate was not placed or was placed crooked, resulting in the nozzle not touching the build plate correctly during detection;

2. Using a third-party build plate with different structures results in the nozzle not being able to touch the plate at the detection position;

The build plate detection process is shown in the following GIFs:

Error message:

HMS_0300-0D00-0001-0003: The build plate is not placed properly.

"The build plate is not placed properly."

The error code below shares the same issue as this one:

0300-0D00-0001-0004

0300-0D00-0001-0005

0300-0D00-0001-0006

0300-0D00-0001-0007

0300-0D00-0001-0008

0300-0D00-0001-0009

0300-0D00-0001-000A

0300-0D00-0002-0003

0300-0D00-0002-0004

0300-0D00-0002-0005

0300-0D00-0002-0006

0300-0D00-0002-0007

0300-0D00-0002-0008

0300-0D00-0002-0009

0300-0D00-0002-000A

0300-0D00-0001-0003

Error Codes: 0300-0A00-0001-0005, 0300-0B00-0001-0005, 0300-0C00-0001-0005

HMS_0300-0A00-0001-0005: Force sensor 1 detected unexpected continuous force. The heatbed may be stuck, or the analog front end may be broken.

What it is

X/P Series Printer

When the heatbed is stationary and no force is applied, the voltage signal of the force sensor should be 0V. If the voltage is not 0, it indicates that the force sensor is detecting an unexpected continuous force. During testing, we encountered the following possible causes of failure:

1. There is something on the table that has stuck / partially stuck the Z-axis motor or the Z-axis timing belt wheels, which are at the bottom of the machine
2. There is stuff on the Z-axis slider when the hotbed is moving up, or the bearing on the linear rods is raised, this stuff is stuck between the Z-axis slider and the Z-axis, causing the nozzle unable to touch the heat bed
3. When the heat bed was near the bottom of the chamber, the Z-axis slider was stuck by something on the bottom.
4. The heat bed was interfered with by the excess chute.
5. The heat bed cable is dragging the heat bed from moving up
6. Wrong screw screw has been installed when installing the AMS-HUB.
7. Z-axis slide assembly tolerance is abnormal.
8. The force sensor is abnormal, making the machine unable to detect the touching event when the nozzle touches the heat bed until the Z-axis motor detects a high load and stops.
9. The printer is placed on an unstable platform, such as a cushion or a shaky table.

Please note that these are only some of the possible causes listed above, in reality, there may be other factors causing the error.

A1 Printer

When the heat bed does not move, the voltage signal of the heat bed acceleration sensor should be 0V. If the voltage is not 0, there is a problem.

Possible reasons:

1. The heat bed acceleration sensor is touched by foreign bodies;

2. The sensor electrode is short-circuited.

X/P Series Printer Troubleshooting guide

Step 1: Check whether Z slider is stuck by a foreign object

Lower the heat bed, and check if there is anything on the Z-axis slider upper surface, especially the rear one that is partly hidden behind the inner liner :

In addition to checking the surface of the slider for foreign objects, in this step, you will also need to check whether the Z-axis bearing is raised.

If the bearing is raised, please refer to the operation video below to reset the bearing. It is important to note that before following the video, please remove the hotend assembly of the tool head (click here for a guide to disassembling the hotend assembly) and then perform the operation in the video.

If you encounter any difficulties during this process, please contact after-sales service for assistance.

The purpose of this operation is to press the raised bearing back into place using the retaining block:

Raised bearings are usually caused by the presence of foreign material, such as scrap, in the three holes at the bottom of the linear rods, which causes the bearing to be squeezed and raised out of place. If there is any foreign material in these holes, make sure they are thoroughly cleaned out or the failure may recur.

Step 2: Check whether the excess chute is interfering

Check if there is visible interference between the heatbed and the excess chute.

If you find that the excess chute is interfering, click to view this wiki page for a solution.

Step 3: Re-adjust Z sliders

the steel rod may have moved under violent shipping, so the z-axis slider needs to be re-adjusted, see the picture below:

3.1: Use the control panel, and lift the bed carefully until it almost touches the bed

3.2: Move the toolhead aside and the build plate aside

3.3: Loose both of these 2 screws and re-tighten them (at the back)

3.4: Loose both of these 2 screws and re-tighten them (on the left)

3.5 Loose both of these 2 screws and re-tighten them (on the right)

Step 4: Check for objects at the bottom of the machine

Lift the printer front a bit, and check if there is anything that will stick to the Z-axis timing belt. Notice that the rubber foot is stuck on to the bottom of the printer, if the printer is pushed on the place surface, the rubber foot may get pushed off and gets stuck into the Z-axis timing belt (where as shown in the red box below):

Step 5: Check whether Z motor and belt are stuck

Lift the back of the printer a bit, and check if there is anything that will stick the Z-motor and timing belt wheel (as shown in the red box below):

Step 6: Check whether the cable is too tight

raise the hotbed slowly, until it almost reaches the nozzle, then look into the inside of the liner from the chamber, to see if the hotbed cable next to the z-axis lead screw is dragged tight. If the shape of the cable is similar to the picture below, then it may be dragged tight:

Then you have to insert your finger into the gaps and drag the cable out from the inner of the electric case, as shown below,

CAUTION: the liner edge may be a bit sharp, be careful or wear gloves when doing this step

If the hotbed cable is stretched, click on this wiki page for the solution.

Step 7: Check AMS hub/ Filament buffer's screws

If you have installed the AMS-HUB or AMS-BUFFER if the screw is over-tightened, that may also stick the z-axis when the bed is near the nozzle, so try loose these two screws for 1~2 rounds :

Step 8: Check the printer platform

Check if there is a cushion under the printer or if the platform is shaky. You can try to put the printer on a solid platform like flat ground without any cushion under the printer.

Step 9: After completing all the steps above, click “home” again.

After the above steps, retry, tap the "home" button, and watch the homing process to check if the nozzle can touch the hotbed surface normally. If the homing is successful, then the homing z-axis failed error will disappear automatically.

If the issue persists, please contact Bambu Lab Support team.

A1 Printer Troubleshooting guide

1. Please ensure that the heat bed is not stuck or rubbed during the movement;

2. Remove the build plate, remove the silicone protective cover of the screw holes, expose the screw holes, and remove the screws;

3. Turn the heat bed over and expose the bottom of the heat bed to see the heat bed acceleration sensor. Check whether the sensor is touched by foreign objects. If so, please clean it up.

4. If there is no interference from foreign objects, it is likely that the sensor has been damaged. Please refer to this wiki to replace the sensor: A1 Heatbed Unit Installation Tutorial | Bambu Lab Wiki

Error message

HMS_0300-0A00-0001-0005: Force sensor 1 detected unexpected continuous force. The heatbed may be stuck, or the analog front end may be broken.

“Force sensor 1 detected unexpected continuous force. The heatbed may be stuck, or the analog front end may be broken.`”

The error code below shows the same issue with different force sensors:

0300-0A00-0001-0005

0300-0B00-0001-0005

0300-0C00-0001-0005

Error Codes: 0300-0A00-0001-0004, 0300-0B00-0001-0004, 0300-0C00-0001-0004

HMS_0300-0A00-0001-0004: An external disturbance was detected on force sensor 1. The heatbed plate may have touched something outside the heatbed.

What it is

X/P Series Printer

Before sensing the force that is applied on the heat bed, the printer will test the sensitivity of each force sensor by vibrating the hotbed up and down.

If anything touches the heat bed, the printer will get the wrong sensitivity parameters and make subsequent procedure works abnormally.

Below are the possible reasons that the printer thinks there were external disturbances occurred:

1. The build plate hasn't been placed correctly.
2. The printer is placed on an unstable platform, such as a cushion or a shaky table.
3. There is an object obstructing the Z-axis motor or the Z-axis synchronous belt wheel on the workbench.
4. When the print bed approaches the bottom of the chamber, the Z-axis slider gets stuck due to an object at the bottom.
5. There are foreign objects on the Z-axis slider, or the Z-axis linear bearings are protruding.
6. The print bed is interfered with by the waste chute or the print board is interfered with by the printer liner.
7. The print bed cable is tightened too much, causing interference with the movement of the print bed.
8. Incorrect specification screws were used during the installation of the AMS-HUB.
9. The Z-axis slider assembly has abnormal tolerances.
10. The force sensor or its signal chain is malfunctioning, resulting in the processor not receiving data from the force sensor.

Please note that the above list only includes some possible causes, and there may be other factors contributing to this error in actual situations.

X/P Series Printer Troubleshooting guide

Step 1: Check whether build plate is placed correctly

Please refer to: HMS_0300-0D00-0001-0003: The build plate may not be properly placed. If this message appears repeatedly, please check the Wiki for more reasons.

Step 2: Check for objects at the bottom of the machine

Lift the printer front a bit, and check if there is anything that will stick to the Z-axis timing belt. Notice that the rubber foot is stuck on to the bottom of the printer, if the printer is pushed on the place surface, the rubber foot may get pushed off and gets stuck into the Z-axis timing belt (where as shown in the red box below):

Step 3: Check whether Z motor and belt is stuck

Lift the back of the printer a bit, and check if there is anything that will stick the Z-motor and timing belt wheel (as shown in the red box below):

Step 4: Check whether Z slider is stuck by for foreign object

Lower the heat bed, and check if there is anything on the Z-axis slider upper surface, especially the rear one that is partly hidden behind the inner liner :

In addition to checking the surface of the slider for foreign objects, you will also need to check whether the Z-axis bearing is raised in this step.

If you find the bearing is raised, please refer to the operation video below to reset the bearing. It is important to note that before following the video, please remove the hotend assembly of the toolhead (click here for a guide to disassembling the hotend assembly) and then perform the operation in the video.

If you encounter any difficulties during this process, please contact after-sales service for assistance.

The purpose of this operation is to press the raised bearing back into place using the retaining block:

Raised bearings are usually caused by the presence of foreign material, such as scrap, in the three holes at the bottom of the linear rods, which causes the bearing to be squeezed and raised out of place. If there is any foreign material in these holes, make sure they are thoroughly cleaned out or the failure may recur.

Step 5: Check whether the excess chute is interfering

Check if there is visible interference between the heatbed and the excess chute.

If you find that the excess chute is interfering, click to view this wiki page for a solution.

Step 6: Check whether the cable is too tight

raise the hotbed slowly, until it almost reaches the nozzle, then look into the inside of the liner from the chamber, to see if the hotbed cable next to the z-axis lead screw is dragged tight. If the shape of the cable is similar the the picture below, then it may be dragged tight:

Then you have to insert your finger into the gaps and drag the cable out from the inner of the electric case, as shown below, CAUTION: the liner edge may be a bit sharp, be careful or wear gloves when doing this step

If the hotbed cable is stretched, click on this wiki page for the solution.

Step 7: Check AMS hub/ Filament buffer's screws

If you have installed the AMS-HUB or AMS-BUFFER, if the screw is over tightened, that may also stick the z-axis when the bed is near the nozzle, so try loose these two screws for 1~2 rounds :

Step 8: Re-adjust Z sliders

the steel rod may have moved under violent shipping, so the z-axis slider needs to be re-adjusted, see the picture below:

8.1: Use the control panel, and lift the bed carefully until it almost touches the bed

8.2: Move the toolhead aside and the build plate aside

8.3: Loosen both of these 2 screws and re-tighten them (at the back)

8.4: Loosen both of these 2 screws and re-tighten them (on the left)

8.5 Loosen both of these 2 screws and re-tighten them (on the right)

Step 9: After completing all the steps above, click “home” again.

After the above steps, retry, tap the "home" button, and watch the homing process to check if the nozzle can touch the hotbed surface normally. If the homing is successful, then the homing z-axis failed error will disappear automatically.

If it still shows this error, please contact the customer support team.

1. Please ensure that the heat bed is not stuck or rubbed during vibration;
2. Ensure that the printer is placed on a stable surface without abnormal shaking;
3. Remove the build plate and remove the silicone protective cover of the screw holes, expose the screw holes, and remove the screws;

4. Turn the heat bed over and expose the bottom structure of the heat bed to see the heat bed acceleration sensor, and check whether the device is significantly loose or damaged.

5. If the sensor is not loose, it may be damaged, please refer to this wiki to replace the sensor: A1 Heatbed Unit Installation Tutorial | Bambu Lab Wiki

Error message

HMS_0300-0A00-0001-0004: An external disturbance was detected on force sensor 1. The heatbed plate may have touched something outside the heatbed.

“An external disturbance was detected on force sensor 1. The heatbed plate may have touched something outside the heatbed.”

The error code below shows the same issue with different force sensors:

0300-0A00-0001-0004

0300-0B00-0001-0004

0300-0C00-0001-0004

Error Codes: 0300-0A00-0001-0003, 0300-0B00-0001-0003, 0300-0C00-0001-0003

HMS_0300-0A00-0001-0003: The sensitivity of heatbed force sensor 1/2/3 is too low.

What it is

Before sensing the force that is applied on the heat bed, the printer will measure the sensitivity of each force sensor, after the measurement, the printer will then check if the sensitivity of the force sensors is normal, if not, the error will presented.

Below are the possible reasons that may make the force sensor sensitivity too low

1. The hardware connection between the MC board and the heat bed is off:The heat bed signal cable connector on the MC board side was loosened The heat bed signal cable was damaged;
2. The heat bed signal cable connector on the MC board side was loosened
3. The heat bed signal cable was damaged;
4. the Z-axis motor is malfunctioning:The connector of the z-axis motor is loose from the MC board; The Z-axis motor is burned down;
5. The connector of the z-axis motor is loose from the MC board;
6. The Z-axis motor is burned down;
7. The force sensor is malfunctioning.

- The heat bed signal cable connector on the MC board side was loosened
- The heat bed signal cable was damaged;

- The connector of the z-axis motor is loose from the MC board;
- The Z-axis motor is burned down;

Safety warning and Machine state before starting operation

Make sure the heat bed has been cooled down and the ground wire of the power plug is truly connected to the ground

troubleshooting guide

step 0 - turn the power OFF

Turn off the printer, and disconnect the power cord.

step 1 - Remove screws

Remove the 10 screws and 4 screws from the rear cover shown in the picture. There are 2 types of screws, so keep them separate and remember which ones go where.

step 2 - Remove the rear cover

Remove the rear cover, and unlock the left side belt tension ports first and then the right side one, to avoid getting stuck.

step 3 - check heat bed signal cable connector

As shown below, the heat bed signal cable connector is "J2613", pointed by the red arrow:

Check if each connector of each wire of the heat bed cable is loose. If it is loosened, re-insert it ;

If none of the wire connectors is loose, clean the glues around the whole connector, then re-plug the whole connector.

Then power on the printer, and try home again, to see if this error disappeared

step4

If step3 does not work, then the force sensor might be broken, and need to be replaced. See "Replacing the heat bed force sensor".

Error message

HMS_0300-0A00-0001-0003: The sensitivity of heatbed force sensor 1/2/3 is too low.

" The sensitivity of heatbed force sensor 1/2/3 is too low."

The error codes below show the same problem with different force sensors:

0300-0A00-0001-0003

0300-0B00-0001-0003

0300-0C00-0001-0003

Error Codes: 0300-0A00-0001-0002, 0300-0B00-0001-0002, 0300-0C00-0001-0002

HMS_0300-0A00-0001-0002: The sensitivity of heatbed force sensor 1/2/3 is low.

What it is

X/P Series Printer

Before sensing the force that is applied on the heat bed, the printer with measure the sensitivity of each force sensor, after the measurement, the printer will then check if the sensitivity is normal, if not, this error message will present.

Below are the possible reasons that may make the force sensor sensitivity low

1. The screw that connects the heat bed and the heat bed frame has been over-tightened;
2. There is something stuck between the heat bed and the heat bed frame
3. The force sensor was detached from the strain arm.

A1 Printer

Before compensating the resonant frequency of the Y-axis, the printer measures the sensitivity of the heat bed acceleration sensor, and after the measurement, the printer checks if the sensitivity is normal, if it is not normal, this error message will appear.

Possible reasons:

1. The heat bed acceleration sensor itself is damaged and fails;

2. The sensor has fallen off or become loose from the heat bed bracket;

3. The heat bed itself is stuck with foreign objects and cannot move or the moving range is very small;

4. The shielding line of the sensor signal cable has been damaged, resulting in serious external signal interference.

Safety warning and Machine state before starting operation

Make sure the heat bed has been cooled down and the ground wire of the power plug is truly connected to the ground

Troubleshooting guide

X/P Series Printer

step1

Check the heat bed pre-tensioning screw and nut, and see if they are over-tightened. Usually, the depth of the screw in the nut is shown in the middle of the picture below:

step2

Check the gap between the heat bed and heat bed frame near the Z-axis slider, and see if there is anything that is stuck inside:

step3

After step 1 and step 2, retry the "Home" function and see if this error message will disappear after finishing homing.

If not, then the force sensor may be detached from the strain arm. Then the force sensor needs to be replaced, please refer to Replace the Heat bed force sensor

A1 Printer

1. First power off and unplug the power cable;

2. First check whether there is a foreign object in the guide rail of the heat bed, causing the heat bed to be stuck and unable to move normally;

3. If the heat bed is not stuck, remove the build plate and remove the silicone protective cover of the screw holes, expose the screw holes and remove the screws;

4. Flip the heat bed and expose the bottom structure of the heat bed to see the heat bed acceleration sensor.

5. First check if the acceleration sensor is loose. If so, reconnect the sensor first and secure it;

6. If it is not loose, the sensor is likely faulty, please refer to this wiki to replace the sensor: A1 Heatbed Unit Installation Tutorial | Bambu Lab Wiki

Error message

HMS_0300-0A00-0001-0002: The sensitivity of heatbed force sensor 1/2/3 is low.

"The sensitivity of heatbed force sensor 1/2/3 is low."

The error code below shows the same problem with different force sensors:

0300-0A00-0001-0002

0300-0B00-0001-0002

0300-0C00-0001-0002

Error Codes: 0300-0A00-0001-0001, 0300-0B00-0001-0001, 0300-0C00-0001-0001

HMS_0300-0A00-0001-0001: The sensitivity of heatbed force sensor 1/2/3 is too high.

Error message

HMS_0300-0A00-0001-0001: The sensitivity of heatbed force sensor 1/2/3 is too high.

"The sensitivity of heatbed force sensor 1/2/3 is too high."

The error code below shows the same problem with different force sensors:

0300-0A00-0001-0001

0300-0B00-0001-0001

0300-0C00-0001-0001

Troubleshooting guide

Please refer to the following Wiki for troubleshooting guide steps:
HMS_0300_0D00_0001_000B The Z axis motor seems got stuck when moving up. Please check if there is any foreign matter on the Z sliders or Z timing belt wheels

Error Code: 0300-0300-0002-0002

The speed of hotend fan is slow.

What is this

There is a speed feedback signal from the hotend fan. The hotend fan (or called heatbreak fan) is used to keep the filament outside the nozzle cool and solidify. If the hotend fan speed remains low, the temperature of the hotend heatsink will rise, causing the filament to soften and potentially jam inside the heatbreak.

Below are the possible reasons:

- The hotend fan blades are stuck and cannot rotate freely.
- Poor connection between the hotend fan wiring and the TH board.
- Hotend fan malfunction
- TH board malfunction

& þ IMPORTANT: When cleaning the fan, it is recommended to use a pen or screwdriver to perform the operation to avoid accidental startup of the fan, which could cause injury.

When cleaning the fan, it is recommended to use a pen or screwdriver to perform the operation to avoid accidental startup of the fan, which could cause injury.

Troubleshooting

X1 Series/P1 Series Printers

Step 1: Set the hotend temperature to 0 degC

When the hotend temperature is below 50 degC, the hotend fan will automatically turn off. When the temperature reaches or exceeds 50 degC, the fan will automatically turn on.

Step 2: Check for hotend fan obstruction

After removing the front cover, gently rotate the fan blades with a pen or screwdriver to check for any obstructions. If there are any, please remove them.

Step 3: Reconnect the hotend fan wiring

As shown in the image below, reconnect the 4-pin wiring on the hotend fan.

X1:

P1:

Step 4: Check extruder connection board and TH board connector

As shown in the image below, check the connection between the Extruder Connection Board and the TH board. If the connector is loose, please reinstall it.

For P1 Series and X1 Series TH V9:

For X1 Series TH V8:

As shown in the image below, check for any looseness or damage to the 10-pin connector. If loose, reinsert it. If damaged, replace it.

Step 5: Heat up the hotend and check the fan status

Set the hotend temperature to 51 degC. If the hotend fan starts rotating and the HMS message disappears, the troubleshooting is successful.

Step 6: Replace the hotend fan

If you have a spare hotend fan or a complete hotend assembly (with a fan), you can replace it to see if it resolves the issue.

Make sure to install the hotend fan in the correct orientation. Installing it incorrectly may cause an error.

- Correct installation:
- Incorrect installation:

Step 7: Replace the TH board and Extruder connection board

If the above steps do not resolve the issue, please replace the TH board and extruder connection board.
P Series and X1 Series TH V9
X1 Series TH V8

A1/A1 Mini Printers

Step 1: Set the hotend temperature to 0 degC

When the hotend temperature is below 50 degC, the hotend fan will automatically turn off. When the temperature reaches or exceeds 50 degC, the fan will automatically turn on.

Step 2: Check for hotend fan obstruction

Use a pen or screwdriver to gently rotate the fan blades to check for any obstructions. If any, please remove them.

Step 3: Reconnect the hotend fan wiring

As shown in the image below, reconnect the wiring (marked 3) on the hotend fan.

Step 4: Heat up the hotend and check the fan status

Set the hotend temperature to 51 degC. If the hotend fan starts rotating and the HMS message disappears, the troubleshooting is successful.

Step 5: Replace the hotend fan

If you have a spare hotend fan, you can replace the hotend fan to see if the issue is resolved.

Step 6: Replace the TH board

If the above steps do not resolve the issue, please replace the TH board.

Error message

HMS_0300-0300-0002-0002: The speed of hotend fan is slow.

HMS_0500-0400-0002-0020: The speed of hotend fan is slow.

Error Code: 0300-0300-0001-0001

The hotend cooling fan speed is too slow or stopped

What it is

There is a speed feedback signal from the hotend fan. the hotend fan (or called the heat break fan) is used to keep the filament outside the nozzle cool and stiff; if the hotend fan speed is kept low, then the heat beak temperature will rise and cause the filament to be too soft and jam inside the heatbreak.

Below are the possible reasons that this HMS error occurs:

1. The fan blade is stuck by something (like filament strings) and cannot spin freely;
2. The hotend fan connector is not installed correctly.
3. The hotend fan is broken.
4. The TH board is broken.

Error message

HMS_0300-0300-0001-0001: The speed of the hotend fan is too slow or stopped

"The speed of the hotend fan is too slow or stopped."

0300-0300-0001-0001

IMPORTANT! It's crucial to power off the printer before performing any maintenance work on the printer and its electronics, including tool head wires, because leaving the printer on while conducting such tasks can cause a short circuit, which can lead to additional electrical damage and safety hazards. When you perform maintenance or troubleshooting on the printer, you may be required to disassemble some parts, including the hotend. This process can expose wires and electrical components that could potentially short circuit if they come into contact with each other or with other metal or electrical components while the printer is still on. This can damage the electronics of the printer and cause further damage. Therefore, it's essential to power off the printer and disconnect it from the power source before doing any maintenance work. This will prevent any short circuits or damage to the printer's electronics. By doing so, you can avoid potential damage to the printer's electronic components and ensure that the maintenance work is performed safely and effectively.

If you have any concerns or questions about following this guide, open a new ticket in our Support Page and we will do our best to respond promptly and provide you with the assistance you need.

IMPORTANT!

It's crucial to power off the printer before performing any maintenance work on the printer and its electronics, including tool head wires, because leaving the printer on while conducting such tasks can cause a short circuit, which can lead to additional electrical damage and safety hazards.

When you perform maintenance or troubleshooting on the printer, you may be required to disassemble some parts, including the hotend. This process can expose wires and electrical components that could potentially short circuit if they come into contact with each other or with other metal or electrical components while the printer is still on. This can damage the electronics of the printer and cause further damage.

Therefore, it's essential to power off the printer and disconnect it from the power source before doing any maintenance work. This will prevent any short circuits or damage to the printer's electronics. By doing so, you can avoid potential damage to the printer's electronic components and ensure that the maintenance work is performed safely and effectively.

If you have any concerns or questions about following this guide, open a new ticket in our Support Page and we will do our best to respond promptly and provide you with the assistance you need.

& þ WARNING: pen or screwdriver (avoid touching it with your hands)

pen or screwdriver (avoid touching it with your hands)

Troubleshooting Steps for the X1 Series

The following steps guide you through diagnosing and fixing issues with the Hotend Fan on X1 series printers.

Follow each step in order, starting from the simplest checks.

Preparation

- When the temperature is below 50 degC, the fan automatically turns off.
- When the temperature is above 50 degC, the fan automatically turns on.

2. Remove the Toolhead Front Cover

Carefully remove the front housing and place it on the carbon rod as shown.

Handle gently to avoid damaging the connected cables.

Verify the correct installation of the hotend fan.

Correct installation:

Incorrect installation:

Check the 4-Pin Hotend Fan Cable

Step 1 - Disconnect the Fan

Gently unplug the 4-pin connector from the hotend fan.

Avoid pulling on the wires directly to prevent damage to the connector.

Step 2 - Remove Any Obstructions

Use a pen or screwdriver to manually rotate the fan blades.

Check for debris or foreign objects and remove them if present.

Step 3 - Reconnect and Test

Reconnect the 4-pin connector and the power.

Set the hotend temperature to 51 degC and observe whether the fan starts spinning.

Check the 10-Pin Connector

If the problem persists, disconnect and reconnect the 10-pin connector on the toolhead. Inspect the connector for visible damage.

> **!9þ NOTE: Note: In the V9 version, the X1 toolhead board and the extruder interface board are connected via an FPC cable, without a 10-pin connector. Some printer versions may not have this connector, so if you don’t see it on your machine, you can skip this step.**

Note: In the V9 version, the X1 toolhead board and the extruder interface board are connected via an FPC cable, without a 10-pin connector. Some printer versions may not have this connector, so if you don’t see it on your machine, you can skip this step.

If repeated reconnections do not fix the issue, take clear photos of the 10-pin connector and the hotend fan, submit a support ticket, and upload your log files for the support team.

Check the FPC Cable

If the error still occurs after the above checks, reseat the FPC cable.

This requires removing the toolhead rear cover and mid-frame - see Replacing the Toolhead Rear Cover and Mid-Frame.

Why this matters:

- The FPC cable connects the Extruder Connection Board to the TH Board.
- Poor contact at either end can cause the fan to malfunction.
-
- Reseating the cable can rule out loose connections, misaligned plugs, incomplete insertion, or cable damage.
-
- If reseating does not restore function, the fan or TH Board may need replacement.

> **!9þ NOTE: For X1 printers with V9 TH Boards, reseat both ends of the FPC cable.**
>
> **Due to the small BTB socket size, FPC installation can be tricky - reseat one end at a time and avoid excessive force.**

For X1 printers with V9 TH Boards, reseat both ends of the FPC cable.

Due to the small BTB socket size, FPC installation can be tricky - reseat one end at a time and avoid excessive force.

Related Maintenance Links for X1 V9 TH Boards:

- Replace the Toolhead Extruder Board Assembly
- Replace the Toolhead PCB (V9)

Fan Not Blocked but Still Not Spinning

If the fan spins freely but will not run when powered, the hotend fan is likely defective.

Follow Replace the Hotend and Related Components | Bambu Lab Wiki for replacement instructions.

None of the Above Fixes the Issue

If none of the above steps work, the TH Board is likely faulty.

Contact Bambu Lab support for a replacement.

Troubleshooting Steps for the P1 Series

The process for P1 series printers is similar but with differences in cover removal and cable handling.

Preparation

1. Set Hotend Temperature to 0 degC and disconnect the power

Set the hotend temperature to 0 degC.

- Fan turns off below 50 degC.
- Fan turns on above 50 degC.

2. Remove the Toolhead Front Cover

From the lower sides of the toolhead, gently pull the front housing outward.

The cover is still attached via the cooling fan cable.

Unplug the black cooling fan cable to fully detach the front housing.

Verify the correct installation of the hotend fan.

Correct installation:

Incorrect installation:

Check the 4-Pin Hotend Fan Cable

Step 1 - Disconnect the Fan

Gently unplug the 4-pin connector from the hotend fan.

Step 2 - Remove Any Obstructions

Use a pen or screwdriver to rotate the fan blades and clear any debris.

Step 3 - Reconnect and Test

Reconnect the 4-pin connector.

Set hotend temperature to 51 degC and check if the fan starts.

> **Note: Ensure the connector is fully seated on the correct pin row to avoid misalignment.**

Note: Ensure the connector is fully seated on the correct pin row to avoid misalignment.

Check the FPC Cable

If repeated reseating of the fan connector does not fix the issue, reseat the FPC cable.

Removing the toolhead rear cover and mid-frame is required - see Replacing the Toolhead Rear Cover and Mid-Frame.

Why this matters:

• The FPC cable connects the Extruder Connection Board to the TH Board.
• Poor contact at either end can cause the fan to malfunction.
•
• Reseating the cable can rule out loose connections, misaligned plugs, incomplete insertion, or cable damage.
•
• If reseating does not restore function, the fan or TH Board may need replacement.

For P1P printers, reseat both ends of the FPC cable.

Handle with care - reseat one end at a time to avoid damaging the small BTB socket.

For P1P printers, reseat both ends of the FPC cable.

Handle with care - reseat one end at a time to avoid damaging the small BTB socket.

Related Maintenance Links for P1P:

- Replace the Toolhead PCB

Fan Not Blocked but Still Not Spinning

If the fan spins freely but will not operate, it is likely defective.

See Replace the Hotend and Related Components | Bambu Lab Wiki for replacement steps.

None of the Above Fixes the Issue

If the fan still fails after all checks, the TH Board may be faulty.

Contact Bambu Lab support for assistance.

Troubleshooting Steps for the A1 Series

The following steps guide you through diagnosing and fixing issues with the Hotend Fan on A1 series printers.

How it should behave:

- When the temperature is below 50 degC, the fan automatically turns off.
- When the temperature is above 50 degC, the fan automatically turns on.

Follow each step in order, starting from the simplest checks.

Step 1 - Set Hotend Temperature to 0 degC and disconnect the power

Click on the "Control" button, as shown below:

Then click on the "Nozzle" and set the temperature to 0ºC

Step 2 - Remove Any Obstructions

Use a pen or screwdriver to manually rotate the fan blades.

Check for debris or foreign objects and remove them if present.

Step 3 - Check the fan connector

Power the machine off before handling any electronic components.

Begin by removing the rear toolhead cover, as shown in the demonstrated:

Locate the Hotend Fan connector on the toolboard.

Carefully unplug the connector from the board, making sure not to pull on the wires or damage the connector pins.

Step 4 - Reconnect and Test

Power the machine off and disconnect the hotend fan connector.

Reconnect the hotend fan connector.

Set hotend temperature to 51 degC and check if the fan starts.

Step 5 - None of the Above Fixed the Issue

If the fan still fails after all checks, the hotend fan may be faulty.

Contact Bambu Lab support for assistance. Alternativly follow the Hotend Fan Replacement Guide for A1 Series

Error Code: 0300-0400-0002-0001

The Part Cooling Fan speed is too slow or stopped. It may be stuck, or the connector may not be plugged in properly.

What it is

This error occurs when your printer's part cooling fan is not spinning at the expected speed. The printer monitors the fan's feedback signal and triggers this error when the actual speed is slower than designed specifications.

Error applies to

X1 Series, P Series (P1 & P2), H2 Series (H2D/H2D Pro & H2S)

Common causes:

1. Fan Blade Obstruction

The fan blades can get stuck due to small pieces of filament, dust, or other debris that accumulate over time inside the printer's toolhead or fan duct.

2. Loose Connector

The wire connecting the part cooling fan to the printer's control board can become loose or even slightly unplugged, causing improper or intermittent power delivery.

3. Failed Fan Motor

If neither physical blockage nor a bad connection is present, the fan's motor itself may be defective or worn out.

& þ Safety Warning
Always ensure the part cooling fan is turned OFF before any maintenance work. A spinning fan can cause injury if it starts unexpectedly during cleaning. Key Safety Reminders Always power off printer before maintenance Wait for hotend cooling if printer was recently used Handle connectors gently to avoid damage Remove the connector perpendicular to the PCB board, without pulling or twisting at an angle

& þ Safety Warning
Always ensure the part cooling fan is turned OFF before any maintenance work. A spinning fan can cause injury if it starts unexpectedly during cleaning.

Key Safety Reminders

- Always power off printer before maintenance
- Wait for hotend cooling if printer was recently used
- Handle connectors gently to avoid damage Remove the connector perpendicular to the PCB board, without pulling or twisting at an angle
- Remove the connector perpendicular to the PCB board, without pulling or twisting at an angle

Always power off printer before maintenance

Wait for hotend cooling if printer was recently used

Handle connectors gently to avoid damage

- Remove the connector perpendicular to the PCB board, without pulling or twisting at an angle

Step-by-Step Troubleshooting

1. Turn off the part cooling fan via printer controls
2. Manually rotate the fan blade to check for resistance
3. Look for debris: filament strings, dust, or foreign objects
4. Clean if needed: Gently remove any obstructions
5. Test: Turn the fan back on to see if error clears

If the issue persists after Step 1, proceed to check the electrical connections as follows:

1. Open the front housing assembly: X1 / P1 / P2 Series: Front cover is attached magnetically for easy removal. H2 Series: The front cover is secured differently and does require additional steps. Please refer to the official H2D/H2D Pro, H2S disassembly instructions to avoid damage and follow model-specific guidance.
2. X1 / P1 / P2 Series: Front cover is attached magnetically for easy removal.
3. H2 Series: The front cover is secured differently and does require additional steps. Please refer to the official H2D/H2D Pro, H2S disassembly instructions to avoid damage and follow model-specific guidance.
4. Locate the part cooling fan connector
5. Disconnect and reconnect the connector firmly
6. Ensure proper seating of all pins

Open the front housing assembly:

- X1 / P1 / P2 Series: Front cover is attached magnetically for easy removal.
- H2 Series: The front cover is secured differently and does require additional steps. Please refer to the official H2D/H2D Pro, H2S disassembly instructions to avoid damage and follow model-specific guidance.

Locate the part cooling fan connector

Disconnect and reconnect the connector firmly

Ensure proper seating of all pins

Step 3: Fan Replacement (If Required)

If Steps 1-2 don't resolve the issue, the fan motor is likely damaged and requires replacement.

- Part needed: Front Housing Assembly (includes integrated fan)
- Replacement guide: Replace Front Housing Assembly - X1 Series

- Part needed: Front Housing Assembly (includes integrated fan)
- Replacement guide: Replace Front Housing Assembly - P1 Series

- Part needed: Individual part cooling fan
- Official guide: H2D/H2D Pro Replace Part Cooling Fan, H2S Replace Part Cooling Fan

Verification Steps

1. Power on the printer
2. Navigate to: Control > Fans > Part Cooling Fan
3. Test operation: Fan should start spinning smoothly
4. Check for: No unusual noise or vibration

5. Verify: Error message no longer appears

Error Codes: 0300-0600-0001-0001, 0300-0700-0001-0001, 0300-0800-0001-0001

Motor-A has an open-circuit. There may be a loose connection, or the motor may have failed.

What it is

There are several current sensors on the MC board of the printer to sample the current of the motor, when the printer is turned on each time, the resistance of the motor will be detected through the current sensor to determine whether the motor is broken. If the resistance of motor is abnormally high, it is considered that the motor has an open circuit error.

The error that Motor-A has an open-circuit may be caused by the following reasons:

1. The cable of the motor is loose - This is the most common issue, and it is solved by simply reconnecting the cable
2. The motor is broken - This issue is rare, and it is solved by replacing the motor
3. The MC board is broken - This issue is rare, and it is solved by replacing the MC board

IMPORTANT! It's crucial to power off the printer before conducting any maintenance work, including work on the printer's electronics and tool head wires. Performing tasks with the printer on can result in a short circuit, leading to electronic damage and safety hazards. During maintenance or troubleshooting, you may need to disassemble parts, including the hotend. This exposes wires and electrical components that could short circuit if they contact each other, other metal, or electronic components while the printer is still on. This can result in damage to the printer's electronics and additional issues.
Therefore, it's crucial to turn off the printer and disconnect it from the power source before conducting any maintenance. This prevents short circuits or damage to the printer's electronics, ensuring safe and effective maintenance. For any concerns or questions about following this guide, open a new ticket in our Support Page and we will do our best to respond promptly and provide the assistance you need.

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Troubleshooting Steps

1. Check whether the cable of the motor is loose

Start by following this wiki article to remove the rear cover of the printer, to access the MC board.

2. Locate the motor wires

As shown in the image below, the MC board has 3 motor cables connected.

Motor Z - Bottom left side
Motor B - Bottom right side
Motor A - Middle right side

3. Try to re-connect each motor to determine the fault

When your printer reports the message that “Motor ___ has an open circuit” it's very probable that one of the motor cables is loose and can be solved by simply re-connecting the appropriate cable.

- If the error message reports that Motor A has an open circuit, you would need to try and re-connect the A motor cable (Motor A - Middle right side)
- If the error message reports that Motor B has an open circuit, you would need to try and re-connect the B motor cable (Motor B - Bottom right side)
- If the error message reports that Motor Z has an open circuit, you would need to try and re-connect the Z motor cable (Motor Z - Bottom left side)

It's recommended to follow these steps for accurate troubleshooting:

1. Turn off the printer
2. Remove the power cable and the back cover of the printer
3. Disconnect and re-connect the motor cable shown in the error message
4. Re-connect the Power Cable and confirm if the error message occurs again or not.

If the issue is solved, disconnect the power cable, then re-attach the back cover to the printer before using it.
If the issue persists, follow Step 2 and Step 3 again to try the other cables.

4. Further troubleshooting for less common faults

Please note that Motor A is the same as Motor B, but motor Z is different. If the HMS reports “Motor-A has an open-circuit”, follow these steps.

1. Turn off the printer
2. Remove the power cable and the back cover of the printer
3. Swap the cable of Motor A and Motor B
4. Re-connect the Power Cable, and turn on the printer
5. If the HMS still reports that Motor A has an open-circuit, the MC board is broken and will need to be replaced.

If the HMS changes to Motor B has an open-circuit, motor A is broken.

If the HMS reports that “Motor-Z has an open-circuit”, please follow these steps:

1. Turn off the printer
2. Remove the power cable and the back cover of the printer
3. Swap the cable of Motor Z and Motor A
4. Re-connect the Power Cable, and turn on the printer
5. Export the log file and share it with the Customer Support team so we can provide a solution

Error Codes: 0300-0600-0001-0002, 0300-0700-0001-0002, 0300-0800-0001-0002

Motor-A has a short-circuit. It may have failed.

Observation

The printer launches a motor self-check program each time it is switched on. The control board applies voltage to the motor and checks the current passing through the motor. If the current is unusually large, it indicates that the motor is short-circuited.

The error of motor short circuit may be caused by the following reasons:

1. Motor failure
2. MC board failure(X1/P1 Series: MC board, A1/A1 Mini: Main board)

Troubleshooting

Safety warning

Please turn off the printer and disconnect the AC cable during operation

Actions

Step 1: Expose the MC board and swap the plugs of Motor A and Motor B on the MC board.

Please refer to this Wikis to remove the printer back cover and swap Motor A and Motor B plugs.
X1/P1 Series Wiki

Connector: 2- Motor B, 3- Motor A

A1 Wiki

A1 Mini Wiki

• Suppose the original HMS error was "Motor-A has a short-circuit. It may have failed". After swapping Motor A and B, the new error is "Motor B short-circuited. It may have failed", that is, the fault changes with the switch of the motor. Root cause: Motor failure.
• If the fault remains unchanged, that is, the fault does not change with the switching of the motor. Root cause: MC board failure.

• Motor failure

please refer to the following Wiki to replace the motor.
X1 Series Motor
P1 Series Motor
A1 A Motor
A1 B Motor
A1 mini A Motor
A1 mini B Motor

• MC board faulty

please refer to the following Wiki to replace the MC board.
X1 Series MC Board

P1 Series MC Board
A1 Mainboard
A1 mini Mainboard

If HMS reports “motor Z has a short-circuit”, this is usually due to the motor being broken but there is also a small probability that the MC board is broken.

Equivalent code

HMS_0300-0600-0001-0002: Motor-A has a short-circuit. It may have failed.

0300-0600-0001-0002 A motor

0300-0700-0001-0002 B motor

0300-0800-0001-0002 C motor

Error Code: 0300-0600-0001-0003

The resistance of Motor-A is abnormal, the motor may have failed.

What it is

There are several current sensors on the control board of the printer to sample the current of the motor, when the printer is turned on each time, the resistance of the motor will be detected through the current sensor to determine whether the motor is broken.

This error is usually caused by excessive resistance due to the following reasons:

1. The cable of the motor is loose.
2. The motor is broken.
3. The control board (MC board for X1/P1/H2 series printer, Mainboard for A1/A1mini printer) is broken.

Safety warning and Machine state before starting operation

Please turn off the printer and disconnect the AC cable during operation.

Troubleshooting guide

- X1/P1 series printer

Please refer to this wiki to remove the rear cover of the printer and check the MC board:

- A1 printer

Please refer to this wiki to remove the bottom cover of the printer and check the Mainboard.

- A1 mini printer

Please refer to this wiki to remove the bottom cover of the printer and check the Mainboard.

- H2 series printer

Please refer to this Wiki to remove the back cover of the printer and check the motor connector on the MC board:

X1/P1 series printer

Please refer to this wiki to remove the rear cover of the printer and check the MC board:

A1 printer

Please refer to this wiki to remove the bottom cover of the printer and check the Mainboard.

A1 mini printer

Please refer to this wiki to remove the bottom cover of the printer and check the Mainboard.

H2 series printer

Please refer to this Wiki to remove the back cover of the printer and check the motor connector on the MC board:

Step 2 Check whether it is a motor failure or MC board failure

If HMS reports “Motor-A/B is abnormal”(Assume that motor A is abnormal):

1. Turn off the printer and disconnect the AC cable.
2. Exchange the cable of motors A and B and turn on the printer.
3. If the HMS still reports that motor A is abnormal, the control board is broken; if the HMS changes to motor B is abnormal, motor A is broken.

Note: In early versions of the A1 series printers, the cable for motor B is shorter and cannot be transferred to the circuit board interface for motor A. If you encounter this situation, please get in touch with the customer service team.

If HMS reports “Motor-Z is abnormal”, please:

1. Turn off the printer and disconnect the AC cable.
2. Exchange the cable of motors Z and A and turn on the printer.
3. Export the log and contact the customer service team (It will still report errors because motor Z and motor A parameters are different).

Error message

HMS_0300-0600-0001-0003: The resistance of Motor-A is abnormal, the motor may have failed.

The error code below shows a similar issue:

0300-0600-0001-0003

0300-0700-0001-0003

0300-0800-0001-0003

Error Code: 0300-0100-0001-0007

The heatbed temperature is abnormal; the sensor may have an open circuit.

What it is

The bed temperature is displayed as 0 C. This means that the temperature sensor of the heat bed is abnormal.

Error message

HMS_0300-0100-0001-0007: The heatbed temperature is abnormal; the sensor may have an open circuit.
" The heatbed temperature is abnormal; the sensor may have an open circuit."

0300-0100-0001-0007

IMPORTANT! It's crucial to power off the printer before performing any maintenance work on the printer and its electronics, including tool head wires, because leaving the printer on while conducting such tasks can cause a short circuit, which can lead to additional electronic damage and safety hazards. When you perform maintenane or troubleshooting on the printer, you may be required to disassemble some parts, including the hotend. This process can expose wires and electrical components that could potentially short circuit if they come into contact with each other or with other metal or electronic components while the printer is still on. This can damage the electronics of the printer and cause further damage. Therefore, it's essential to switch off the printer and disconnect it from the power source before doing any maintenance work. This will prevent any short circuits or damage to the printer's electronics. By doing so, you can avoid potential damage to the printer's electronic components and ensure that the maintenance work is performed safely and effectively. If you have any concerns or questions about following this guide, open a new ticket in our Support Page and we will do our best to respond promptly and provide you with the assistance you need.

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Troubleshooting guide

Please refer to these wikis for troubleshooting.

X1 Series: Troubleshooting link(Scenario one)

P1 Series: Troubleshooting link(Scenario one)

A1: Troubleshooting guide for A1 heatbed not heating up

A1 mini: Troubleshooting of Heatbed not Heating up for A1 mini

If the message still shows up, contact the after-sales team.

Error Code: 0300-0100-0001-0006

The heatbed temperature is abnormal; the sensor may have an short circuit.

What it is

" The heatbed temperature is abnormal; the sensor may have a short circuit. "

Safety warning and Machine state before starting operation

1. The heat bed is directly heated by mains electricity, and there is a danger of high voltage. It is necessary to ensure that troubleshooting is carried out with the power off.

2. Set the heat bed to a middle height from the LCD screen when the printer is still powered on.

    - Troubleshooting of heat bed not heating up

Error message

HMS_0300-0100-0001-0006: The heatbed temperature is abnormal; the sensor may have a short circuit.
" The heatbed temperature is abnormal; the sensor may have a short circuit. "

0300-0100-0001-0006

Error Codes: 0300-1300-0001-0001, 0300-1400-0001-0001, 0300-1500-0001-0001

The current sensor of Motor-A is abnormal. This may be caused by a failure of the hardware sampling circuit.

What it is

There are several current sensors on the MC board for sampling motor current. Every time the printer is turned on, it will check whether the current sensor is normal.

Safety warning and Machine state before starting operation

Troubleshooting guide

if HMS reports “The current sensor of Motor-A/B/Z is abnormal”, it may be caused by a failure of the hardware sampling circuit, the MC board is broken.

Error message

HMS_0300-1300-0001-0001: The current sensor of Motor-A is abnormal. This may be caused by a failure of the hardware sampling circuit.

0300-1300-0001-0001

0300-1400-0001-0001

Error Code: 0300-4000-0002-0001

Data transmission over the serial port is abnormal; the software system may be faulty.

Issue Description

Serial port data transmission lines:

- H2 series: TH - MC, AP - MC
- X1/P series: TH - (via Type C) - AP - MC
- A1/A1 mini: TH - (via Type C) - Mainboard

H2 series: TH - MC, AP - MC

X1/P series: TH - (via Type C) - AP - MC

A1/A1 mini: TH - (via Type C) - Mainboard

> **Note: H2D is used as an example.**

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Step 1: Restart the printer

Restart the printer and check if the error message has disappeared.

> **Note: Please disconnect the power before plugging and unplugging.**

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- Check if the TH-MC intermediate connector is loose. Please refer to: Replace H2D Toolhead to MC Board Cable;
- Check if the connectors on both sides of the 12-pin cable between the AP board and the MC board are loose. Please refer to: Replace H2D MC-AP Cable;
- Check if the MC side connector of the TH-MC is loose. Please refer to: Replace H2D Toolhead to MC Board Cable;

Step 3: If the problem persists after re-plugging and unplugging the connector, please replace the MC board and related cables.

Error Code: 0300-4100-0001-0001

The system voltage is unstable; triggering the power failure protection function.

What it is

When the voltage of the printer is unstable during operation, this error may occur.

Troubleshooting guide

Restart the printer and choose to continue printing or cancel the printing task.

Note: If you pause or resume without restarting the printer, the printer will not be able to continue printing normally.

Error Code: 0300-0100-0001-0008

An abnormality occurs during the heating process of the heatbed. The heating modules may be broken.

Issue Description

During the heatbed heating process, the printer strictly monitors the machine's internal state. When the printer detects that the heating status of the heatbed does not meet expectations (such as AC control cable disconnection, AC board failure, heatbed cable cracking, heatbed failure, etc.), the printer will stop heating to ensure system safety and display a message indicating a potential heatbed malfunction.

Possible reasons

- The AC control cable is loose or disconnected
- The AC board is damaged
- The heatbed signal cable is cracked
- The heatbed fails

The AC control cable is loose or disconnected

The AC board is damaged

The heatbed signal cable is cracked

The heatbed fails

Troubleshooting

If this error occurs, please immediately stop using the printer and turn off the printer's power to ensure safety.

Safety warning The heatbed is directly heated by mains electricity, and there is a danger of high voltage. It is necessary to ensure that troubleshooting is carried out with the power off. Set the heatbed to a middle height from the LCD screen when the printer is still powered on.

Safety warning

1. The heatbed is directly heated by mains electricity, and there is a danger of high voltage. It is necessary to ensure that troubleshooting is carried out with the power off.
2. Set the heatbed to a middle height from the LCD screen when the printer is still powered on.

The heatbed is directly heated by mains electricity, and there is a danger of high voltage. It is necessary to ensure that troubleshooting is carried out with the power off.

Set the heatbed to a middle height from the LCD screen when the printer is still powered on.

- H2.0 Hex Wrench
- H1.5 Hex Wrench
- Phillips screwdriver
- Tweezers
- Multimeter

H2.0 Hex Wrench

H1.5 Hex Wrench

Phillips screwdriver

Tweezers

Multimeter

- Check the heatbed power cable and circuit board connections.
- With the power off, use a multimeter to measure the heatbed resistance.
- With the power on, check the AC board output (high voltage is dangerous; electrical proficiency is required).

Check the heatbed power cable and circuit board connections.

With the power off, use a multimeter to measure the heatbed resistance.

With the power on, check the AC board output (high voltage is dangerous; electrical proficiency is required).

- Check heatbed resistance

  A1

  A1 mini
- Replace the AC board

  A1

  For A1 mini, it is similar to replace the mainboard.
- Replace the MC board

  For A1 mini, it is similar to replace the mainboard.

  For A1 mini, it is similar to replace the mainboard.
- Replace the heatbed

  A1

  A1 mini

Check heatbed resistance
A1
A1 mini

Replace the AC board
A1
For A1 mini, it is similar to replace the mainboard.

Replace the MC board

For A1 mini, it is similar to replace the mainboard.
For A1 mini, it is similar to replace the mainboard.

Replace the heatbed
A1
A1 mini

X1 & P1 Series

Disassembly

> **NOTE:**
>
> **There are a lot of screws involved in this procedure. Please label them and group them in separate sections to avoid issues.**

NOTE:

There are a lot of screws involved in this procedure. Please label them and group them in separate sections to avoid issues.

Remove the 10 screws and 4 screws from the rear cover shown in the picture. There are 2 types of screws, so keep them separate and remember which ones go where.

Step 2. Remove the rear cover

Remove the rear cover by unlocking the left side belt tension port first, and then the right side one to avoid getting stuck.

Step 3. Remove the excess chute

Remove 2 screws, and remove the excess chute.

Step 4. Remove the power module protective cover

Remove the 6 screws shown, move the power module protective cover to the side, and disconnect the power cable of the heat bed from the AC power board.

Inspection

Step 5. Check the 2PIN cable and its connection

After confirming that the 2 pins are normal, reconnect the power cable to the AC power board. Temporarily connect the power cord to start the printer, and set the bed temperature to confirm whether the problem is solved. If not resolved, after disconnecting the power cord, proceed to the next step.

> **Reminder:**
>
> **If you have a multimeter at home, you can also test the resistance value of the two pins. Under normal circumstances, the resistance value is about 50~60 ohms. If the resistance value is not displayed, the power connection at the heat bed end is likely open or the internal components of the heat bed are burned out.**

Reminder:

If you have a multimeter at home, you can also test the resistance value of the two pins. Under normal circumstances, the resistance value is about 50~60 ohms. If the resistance value is not displayed, the power connection at the heat bed end is likely open or the internal components of the heat bed are burned out.

& þ IMPORTANT: WARNING:

Always confirm that the power supply is disconnected before operation.

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Because the connector is usually glued with white silicone gel, it may not be directly clear whether the connection is normal. Just use the tip of the tweezers to push the connector several times. After the pressing, connect the power to start the printer and set the bed temperature to confirm whether the problem is solved. If the problem is not resolved, continue to the next step to check the AC board.

Step 7 - Check the appearance of the AC power board

Visually check the appearance of the AC board, including shaking the input cables to check whether they are loose (if loose, reconnect and fix them) and whether the components on the board have obvious burn marks (if there are, please refer to this wiki to replace the AC power board). Confirm that there is no abnormality, then proceed to the next step to confirm.

)

& þ IMPORTANT: WARNING:

&¡Shock Hazard&¡ - can cause serious injury or death. This operation detects the mains voltage, and it is not recommended for personnel without relevant knowledge to operate. Skip this step if you don't know what you're doing.

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When the printer is turned on, and the heatbed is heating, use the AC voltage test file of the multimeter to check the output voltage, which should be the same as the local voltage. If the output voltage is abnormal, the AC board is faulty. Please refer to this wiki to replace the AC power board.

If the above operation fails to eliminate the fault, you need to contact the service team for confirmation.

Assembly

After troubleshooting, the dismantled parts need to be reassembled. The steps are as follows:

Step 1 - Install the power module protective cover

Install the power module protective cover, and secure it with the 6 screws.

Step 2 - Install the excess chute

Install the excess chute, pay attention to the buckles on both sides and use the 2 screws to secure it.

Step 3 - Install the rear cover

Pass the PTFE tube through the tube bracket, first install the right-side belt tension port, then the left-side to install the rear cover.

Step 4 - Install the 10+4 screws

Install the 10 + 4 screws (see picture).

H2 Series

You can adjust the order of steps as needed.

Disassembly

Step 1. Remove the printer's rear cover and AC board cover.

For detailed steps, please refer to Replace H2 Series AC Board/AC board Cover.

Inspection

Step 2. Check the AC board power connection.

Unplug the power cord, confirm the pins are working properly, and reconnect. Turn on the printer and set the heatbed temperature to see if the issue is resolved. If not, disconnect the power cord and proceed to the next step.

Step 3. Check the heatbed resistance.

Disconnect the cables shown in the image below, and use a multimeter to measure the resistance between the cables.

Normal resistance values are shown in the table below.

If the resistance is zero or excessively high, the heatbed power connection may be open or the heatbed components may be burned out. Please refer to the wiki Replace H2S Heatbed Unit or Replace H2D Heatbed Unit to replace the heatbed.

Step 4. Check the AC board.

Check the AC board, shake it to check if the power input cable is loose (if so, reconnect and secure it). Then, check for any visible signs of burnout on the components on the board (if so, refer to Replace H2 Series AC Board/AC board Cover to replace the AC board). If there are no abnormalities, please proceed to the next step.

Step 5. Check the heatbed temperature resistor.

Check that the heatbed temperature resistor connector (No. 24) on the MC board is fully inserted. If necessary, replug it.

If the above guides do not solve the problem, please contact our customer service team.

Assembly

Step 6. Install the AC board cover and printer rear panel.

Please refer to the Wiki Replacing the H2 Series AC Board/AC Board Cable Management Cover to reinstall the AC board cover and printer rear panel in sequence.

Error Message

HMS_0300-0100-0001-0008: An abnormality occurs during the heating process of the heatbed. The heating modules may be broken.

"An abnormality occurs during the heating process of the heatbed. The heating modules may be broken."

0300-0100-0001-0008

Error Code: 0300-0100-0003-0008

The temperature of the heated bed exceeds the limit and automatically adjusts to the limit temperature.

Observation

To ensure the safe use of the printer, when the heatbed temperature set in G-Code exceeds the limit, the printer will automatically adjust to the maximum allowable temperature of the heatbed and give an HMS warning.

The maximum allowable heatbed temperature for each model of printers are:

- X1 Series: 110C
- P1 Series: 100C
- A1: 100C
- A1mini: 80C

Troubleshooting

Please adjust the heatbed temperature in G-Code within the required range.

Equivalent Codes

HMS_0300-0100-0003-0008: The temperature of the heated bed exceeds the limit and automatically adjusts to the limit temperature.

Error Code: 0300-9000-0001-0001

Chamber heating failed. The chamber heater may be failing to blow hot air.

Error message

HMS_0300-9000-0001-0001: Chamber heating failed. The chamber heater may be failing to blow hot air.

“Chamber heating failed. The chamber heater may be failing to blow hot air.”

0300-9000-0001-0001

The cause of the error

The air outlet of the chamber temperature control module has a temperature sensor that can measure the air temperature at the air outlet. However, the sensor measured and found that hot air was not being produced normally, which would cause the chamber to fail to heat.

Solution

Check whether the housing of the chamber temperature control module is damaged, causing hot air to flow out from other places. After that, please contact the after-sales team to solve the issue.

Error Code: 0300-9000-0001-0002

Chamber heating failed. The chamber may not be enclosed, or the ambient temperature may be too low, or the heat dissipation vent of the power supply may be blocked.

Error message

HMS_0300-9000-0001-0002: Chamber heating failed. The chamber may not be enclosed, or the ambient temperature may be too low, or the heat dissipation vent of the power supply may be blocked.

“Chamber heating failed. The chamber may not be enclosed, or the ambient temperature may be too low, or the heat dissipation vent of the power supply may be blocked.”

0300-9000-0001-0002

The cause of the error

When the outside temperature is too low or the chamber is not closed, the chamber may not be heated to the predetermined temperature due to excessive heat dissipation. In addition, when the power supply is derated due to its inability to dissipate heat, the cavity temperature controller may also lack sufficient power to heat the cavity to a predetermined temperature.

Solution

1. Check the printer chamber to make sure the chamber is closed (front door and top glass cover);

2. Check the ambient temperature to avoid using the printer in an environment that is too cold;

3. Check the heat dissipation vent of the power supply to ensure that it is not blocked by trash cans or other components.

If the problem cannot be solved after taking the above measures, please submit the ticket to contact the support team.

Error Code: 0300-9000-0001-0003

Chamber heating failed. The temperature of power supply may be too high.

What it is

HMS_0300-9000-0001-0003: Chamber heating failed. The temperature of power supply may be too high.

Error message

HMS_0300-9000-0001-0003: Chamber heating failed. The temperature of power supply may be too high.

The cause of the error

To ensure proper functionality, the printer power supply must be used within a safe temperature range. This error indicates that the power supply temperature has exceeded the temperature range that can be safely used and therefore can not operate properly. This usually occurs when the outside temperature is too high.

Solution

1. Check the heat dissipation vent of the power supply to ensure that it is not blocked by trash cans or other components.

2. Check the ambient temperature to avoid using the printer in an environment that is too hot.

If the problem can not be solved after taking the above measures, please contact the after-sales team.

Error Code: 0300-9000-0001-0004

Chamber heating failed. The speed of the heating fan is too low.

What it is

HMS_0300-9000-0001-0004: Chamber heating failed. The speed of the heating fan is too low.

Error message

HMS_0300-9000-0001-0004: Chamber heating failed. The speed of the heating fan is too low.

The cause of the error

The chamber temperature control module needs a fan to bring hot air into the chamber. This error means that the fan speed is lower than normal and therefore can not heat the cavity to the predetermined temperature.

Solution

After powering off, check the cable connector of the chamber temperature control module's heating fan, reconnect it properly then turn on the power and set the cavity temperature to 60 degC. If the error is no longer reported, the problem is fixed.

If the problem can not be solved after taking the above measures, please take a video of the issue and then contact the support team.

Error Code: 0300-9000-0001-0005

Chamber heating failed. The thermal resistance is too high.

Error message

HMS_0300-9000-0001-0005: Chamber heating failed. The thermal resistance is too high.

The cause of the error

The chamber will perform thermal resistance calculations during the heating process. When the thermal resistance value is significantly higher than the normal value, the chamber can not be heated normally. This condition is usually caused by blockage of heat transfer between the upper and lower parts of the chamber.

Solution

Avoid insulation of some areas inside the printer chamber to allow complete heat circulation.

If the problem can not be solved after taking the above measures, please contact the support team.

Error Code: 0300-9100-0001-000C

HMS_0300-9100-0001-000C: The chamber heater 1 works at full load for a long time. Temperature control system may be abnormal.

Error message

HMS_0300-9100-0001-000C: The chamber heater 1 works at full load for a long time. Temperature control system may be abnormal.

Error Codes: 0300-9100-0001-0008, 0300-9200-0001-0008

Chamber heater 1 failed to rise to target temperature.

Error message

HMS_0300-9100-0001-0008: Chamber heater 1 failed to rise to target temperature.

HMS_0300-9200-0001-0008: Chamber heater 2 failed to rise to target temperature.

Error Codes: 0300-9100-0001-000A, 0300-9200-0001-000A

HMS_0300-9100-0001-000A: The temperature of chamber heater 1 is abnormal. The AC board may be broken.

Error message

HMS_0300-9100-0001-000A: The temperature of chamber heater 1 is abnormal. The AC board may be broken.

HMS_0300-9200-0001-000A: The temperature of chamber heater 2 is abnormal. The AC board may be broken.

Error Codes: 0300-9100-0001-0003, 0300-9200-0001-0003

The temperature of chamber heater 1 is abnormal. The heater is over temperature.

Error message

The temperature of chamber heater 1/2 is too high:

HMS_0300-9100-0001-0003: The temperature of chamber heater 1 is abnormal. The heater is over temperature.

HMS_0300-9200-0001-0003: The temperature of chamber heater 2 is abnormal. The heater is over temperature.

The cause of the error

There are two heaters in the chamber temperature control module. This error will be reported if the temperature of the heaters is too high.

Solution

Turn off the power immediately, and then disconnect the following two connectors(as show below). Then power on again and export the Log to the SD card. Submit the ticket to contact the support team to solve the problem.

Error Codes: 0300-9100-0001-0001, 0300-9200-0001-0001

The temperature of chamber heater 1 is abnormal. The heater may have a short circuit.

Error message

The temperature control of chamber heater 1/2 is abnormal.

HMS_0300-9100-0001-0001: The temperature of chamber heater 1 is abnormal. The heater may have a short circuit.

HMS_0300-9200-0001-0001: The temperature of chamber heater 2 is abnormal. The heater may have a short circuit.

The cause of the error

There are two heaters in the chamber temperature control module. A short circuit will cause the chamber temperature control module to fail to work properly. This error code indicates a short circuit problem in the heaters.

Solution

Please turn off the power and check whether there is a short circuit between the heater connectors. After the problem is eliminated, you can power it on again and see if the problem still exists. Note that powering on again in this situation may cause a short circuit. It is recommended to operate with caution while ensuring safety.

If the problem can not be solved after taking the above measures, please contact the support team.

Error Codes: 0300-9100-0001-0002, 0300-9200-0001-0002

The temperature of chamber heater 1 is abnormal. The heater may have an open circuit or the thermal fuse may have taken effect.

Error message

The temperature control of chamber heater 1/2 is abnormal.

HMS_0300-9100-0001-0002: The temperature of chamber heater 1 is abnormal. The heater may have an open circuit or the thermal fuse may have taken effect.

HMS_0300-9200-0001-0002: The temperature of chamber heater 2 is abnormal. The heater may have an open circuit or the thermal fuse may be in effect.

The cause of the error

There are two heaters in the chamber temperature control module. A open circuit of the heater will cause the chamber temperature control module to fail to work properly. This error code indicates an open circuit problem in the heaters.

Solution

Please turn off the power and check for poor contact at the heater connectors. After eliminating the problem, you can power it on again and observe whether the error code still exists.

If the problem can not be solved after taking the above measures, please contact the support team.

Error Codes: 0300-9100-0001-0006, 0300-9200-0001-0006

The temperature of chamber heater 1 is abnormal. The sensor may have a short circuit.

Error message

The temperature sensor of chamber heater 1/2 is abnormal:

HMS_0300-9100-0001-0006: The temperature of chamber heater 1 is abnormal. The sensor may have a short circuit.

HMS_0300-9200-0001-0006: The temperature of chamber heater 2 is abnormal. The sensor may have a short circuit.

The cause of the error

There are two heaters in the chamber temperature control module. They rely on the temperature sensor for closed-loop control. When the temperature sensor is damaged, the heater can not control the temperature. This error indicates that the temperature sensor has a short circuit.

Solution

Please check whether there is a short circuit in the Chamber Heater Unit NTC shown in the following figure. After troubleshooting, you can power it on again and check whether the error code still exists.

If the error still persists, the temperature sensor may be damaged. Please contact the support team for assistance.

Error Codes: 0300-9100-0001-0007, 0300-9200-0001-0007

The temperature of chamber heater 1 is abnormal. The sensor may have an open circuit.

Error message

The temperature sensor of chamber heater 1/2 is abnormal:

HMS_0300-9100-0001-0007: The temperature of chamber heater 1 is abnormal. The sensor may have an open circuit.

HMS_0300-9200-0001-0007: The temperature of chamber heater 2 is abnormal. The sensor may have an open circuit.

The cause of the error

There are two heaters in the chamber temperature control module. They rely on the temperature sensor for closed-loop control. When the temperature sensor is damaged, the heater can not control the temperature. This error indicates that the temperature sensor has an open circuit.

Solution

Please check whether there is an open circuit in the Chamber Heater Unit NTC shown in the following figure. After troubleshooting, you can power it on again and check whether the error code still exists.

If the error persists, the temperature sensor may be damaged. Please contact the support team for assistance.

Error Code: 0300-9300-0001-0003

Chamber temperature is abnormal. The chamber heater's temperature sensor at the air outlet may have a short circuit.

Error message

HMS_0300-9300-0001-0003: Chamber temperature is abnormal. The chamber heater's temperature sensor at the air outlet may have a short circuit.

Error Code: 0300-9300-0001-0004

Chamber temperature is abnormal. The chamber heater's temperature sensor at the air outlet may have an open circuit.

Error message

HMS_0300-9300-0001-0004: Chamber temperature is abnormal. The chamber heater's temperature sensor at the air outlet may have an open circuit.

Error Code: 0300-9300-0001-0007

Chamber temperature is abnormal. The temperature sensor at the power supply may have a short circuit.

Error message

HMS_0300-9300-0001-0007: Chamber temperature is abnormal. The temperature sensor at the power supply may have a short circuit.

Error Code: 0300-9300-0001-0008

Chamber temperature is abnormal. The temperature sensor at power supply may have an open circuit.

Error message

HMS_0300-9300-0001-0008: Chamber temperature is abnormal. The temperature sensor at power supply may have an open circuit.

Error Code: 0300-9300-0001-0001

HMS_0300_9300_0001_0001: The chamber temperature sensor at heater is short circuited

Error message

The temperature sensor of chamber temperature control is abnormal:
HMS_0300-9300-0001-0001: Chamber temperature is abnormal. The chamber heater's temperature sensor may have a short circuit.

The cause of the error

The chamber temperature control module relies on four temperature sensors to provide temperature feedback. They are located at the heater, air outlet, air inlet, and power supply. This error code indicates whether there is a short circuit or open circuit problem with these temperature sensors.

Solution

1. When the temperature sensors at heater, air outlet, and air inlet are abnormal, please cut off the power and check whether there is a short circuit or poor contact in the Chamber Heater Unit Temperature sensor shown in the figure. Please check it and reconnect it

2. When the power temperature sensor is abnormal, please cut off the power and check whether there is a short circuit or poor contact in the Power Supply Temperature sensor shown in the figure. Please check it and reconnect it.

If the error persists, the corresponding temperature sensor may be damaged, please contact support team for assistance.

Error Code: 0300-9300-0001-0002

Chamber temperature is abnormal. The chamber heater's temperature sensor may have an open circuit.

Error message

HMS_0300-9300-0001-0002: Chamber temperature is abnormal. The chamber heater's temperature sensor may have an open circuit.

Troubleshooting Guide

The chamber temperature sensor is built into the button board, so any sensor issue is usually related to this component. To fix the problem, start by checking whether the ribbon cable connecting the board is firmly in place. If the cable is properly connected but the sensor still does not work, the button board itself may need to be replaced.

You can find detailed instructions on how to access the button board, inspect the ribbon cable connection, and replace the board if necessary on the replacement button board wiki page.

Error Code: 0300-9300-0001-0005

Chamber temperature is abnormal. The chamber heater's temperature sensor at the air inlet may have a short circuit.

Error message

HMS_0300-9300-0001-0005: Chamber temperature is abnormal. The chamber heater's temperature sensor at the air inlet may have a short circuit.

Error Code: 0300-9300-0001-0006

Chamber temperature is abnormal. The chamber heater's temperature sensor at the air inlet may have an open circuit.

Error message

HMS_0300-9300-0001-0006: Chamber temperature is abnormal. The chamber heater's temperature sensor at the air inlet may have an open circuit.

Error Code: 0300-9400-0003-0001

Chamber cooling may be too slow. You can open the chamber to help cooling if the gas in chamber is non-toxic.

Error message

HMS_0300-9400-0003-0001: Chamber cooling may be too slow. You can open the chamber to help cooling if the gas in chamber is non-toxic.

The cause of the error

When the chamber temperature is set and waited, and the set chamber temperature is lower than the current temperature of the chamber, the chamber will enter a natural cooling state until it reaches the predetermined temperature. At this time, if the chamber temperature controller finds that the cooling is too slow, this reminder will pop up. Following the suggestion can help the chamber quickly cool down to the predetermined temperature.

Solution

You can open the front door and the top glass cover to help cooling if the gas in chamber is non-toxic.

Error Code: 0300-0200-0001-0001

The nozzle temperature is abnormal, the heater may be short circuit.

What it is

Nozzle temperature control mainly includes a heater and a NTC sensor. When the heater is short-circuited, the nozzle cannot heat up, which will cause the hotend to be unable to extrude the filament, and the filament may be jammed in the heatbreak.

Below are the possible reasons that this HMS error occurs:

1. A short circuit has occurred at the heater's connector or wires
2. The heater is broken
3. The TH board is broken

IMPORTANT! The nozzle temperature is very high; please avoid touching the nozzle. It's crucial to switch off the printer and disconnect it from the power source before doing any maintenance work. Otherwise, it could damage the electronics of the printer and cause further damage. It is crucial to ensure that the plug is aligned with the slot.

IMPORTANT!

1. The nozzle temperature is very high; please avoid touching the nozzle.
2. It's crucial to switch off the printer and disconnect it from the power source before doing any maintenance work. Otherwise, it could damage the electronics of the printer and cause further damage.
3. It is crucial to ensure that the plug is aligned with the slot.

Troubleshooting guide

X1 Series Troubleshooting

Re-plug the cable of NTC

Re-plug the cable of NTC and then check whether the error prompt disappears.

Disassemble the hotend

Please refer to this wiki to disassemble the hotend and check if the NTC sensor or wire is broken.

Measure with a multimeter

If you have a multimeter available, you can try measuring the resistance of the hotend component and follow the steps below to determine if there is a hotend malfunction.

The resistance of the ceramic heater at room temperature is typically around 12:’Â v†–ÆR F†R &W6—7F æ6R öb F†R åD2 is around 50-100 K:“

Replug the 10Pin connector

If you have a V8 TH board, replug the 10Pin connector (sometimes it needs to be replugged several times); also check for any signs of burnout on the 10-pin plug. You need to remove the silicone with tweezers before replugging it. Then try again.

If you have a V9 TH board, reseat both ends of the FPC.

Please refer to the following wiki for more information: Replacing the toolhead TH board assembly (V8)| Bambu Lab Wiki Replace the toolhead boards (V9) | Bambu Lab Wiki

Please refer to the following wiki for more information:

- Replacing the toolhead TH board assembly (V8)| Bambu Lab Wiki
- Replace the toolhead boards (V9) | Bambu Lab Wiki

P1 Series Troubleshooting

Re-plug the cable of NTC

Re-plug the cable of NTC and then check whether the error prompt disappears.

Disassemble the hotend

Please refer to this wiki to disassemble the hotend and check if the NTC sensor or wire is broken.

Measure with a multimeter

If you have a multimeter available, you can try measuring the resistance of the hotend component and follow the steps below to determine if there is a hotend malfunction.

The resistance of the ceramic heater at room temperature is typically around 10:' ‡Gpo pins indicated by the red arrows in the image below), while the resistance of the NTC is around 100 K:' ‡Gpo pins indicated by the orange arrows):

Reseat the FPC

If the problem persists, reseat both ends of the FPC (sometimes you need to reseat it several times).

!9þ NOTE: Please refer to the following wiki for more information:
PCBs on the toolhead of P1P | Bambu Lab Wiki

Please refer to the following wiki for more information:
PCBs on the toolhead of P1P | Bambu Lab Wiki

A1 Series Troubleshooting

Re-plug the cable of NTC

Re-plug the cable of NTC and then check whether the error prompt disappears.

Disassemble the hotend

Please refer to this wiki to disassemble the hotend and check if the NTC sensor or wire is broken.

Measure with a multimeter

If you have a multimeter available, you can try measuring the resistance of the hotend component and follow the steps below to determine if there is a hotend malfunction.

Typically, the error message for A1 series printers is 'Nozzle Temperature Malfunction'. To measure the resistance of the heating component, please remove the rear cover of the toolhead. The plug to be measured is the number 1 plug on the TH board.

The resistance of the ceramic heater at room temperature is typically around 7:' ‡Gpo pins indicated by the red arrows in the image below), while the resistance of the NTC is around 100 K:' ‡Gpo pins indicated by the orange arrows, the resistance of NTC varies widely with temperature fluctuations):

Contact the after-sales

If the message still shows up, contact the after-sales team.

Error message

HMS_0300-0200-0001-0007: The nozzle temperature is abnormal, the sensor may be open circuit.

"The nozzle temperature is abnormal, the sensor may be open circuit."

0300-0200-0001-0007

Error Code: 0300-0200-0001-0002

The nozzle temperature is abnormal,the heater may be open circuit.

What it is

The nozzle temperature control system primarily consists of a heater and an NTC sensor. If the heater experiences an open circuit, the nozzle will fail to heat up, preventing the hotend from extruding filament and potentially causing filament jams. The TH board monitors the heater's current to detect whether an open circuit has occurred.

Here are the possible reasons for this HMS error:

- The heater's connector has an open circuit.
- The heater is damaged.
- The TH board is faulty.

IMPORTANT! The nozzle temperature is very high; please avoid touching the nozzle. It's crucial to switch off the printer and disconnect it from the power source before doing any maintenance work. Otherwise, it could damage the electronics of the printer and cause further damage. It is crucial to ensure that the plug is aligned with the slot.

IMPORTANT!

1. The nozzle temperature is very high; please avoid touching the nozzle.
2. It's crucial to switch off the printer and disconnect it from the power source before doing any maintenance work. Otherwise, it could damage the electronics of the printer and cause further damage.
3. It is crucial to ensure that the plug is aligned with the slot.

Troubleshooting guide

X1 Series Troubleshooting

Re-plug the cable of NTC

Re-plug the cable of NTC and then check whether the error prompt disappears.

Disassemble the hotend

Please refer to this wiki to disassemble the hotend and check if the NTC sensor or wire is broken.

Measure with a multimeter

If you have a multimeter available, you can try measuring the resistance of the hotend component and follow the steps below to determine if there is a hotend malfunction.

The resistance of the ceramic heater at room temperature is typically around 12:'Â v†–ÆR F†R &W6—7F æ6R öb F†R åD2 is around 50-100 K:“

Replug the 10Pin connector

If you have a V8 TH board, replug the 10Pin connector (sometimes it needs to be replugged several times); also check for any signs of burnout on the 10-pin plug. You need to remove the silicone with tweezers before replugging it. Then try again.

If you have a V9 TH board, reseat both ends of the FPC.

Please refer to the following wiki for more information: Replacing the toolhead TH board assembly (V8)| Bambu Lab Wiki Replace the toolhead boards (V9) | Bambu Lab Wiki

Please refer to the following wiki for more information:

- Replacing the toolhead TH board assembly (V8)| Bambu Lab Wiki
- Replace the toolhead boards (V9) | Bambu Lab Wiki

P1 Series Troubleshooting

Re-plug the cable of NTC

Re-plug the cable of NTC and then check whether the error prompt disappears.

Disassemble the hotend

Please refer to this wiki to disassemble the hotend and check if the NTC sensor or wire is broken.

Measure with a multimeter

If you have a multimeter available, you can try measuring the resistance of the hotend component and follow the steps below to determine if there is a hotend malfunction.

The resistance of the ceramic heater at room temperature is typically around 10:’ ‡Gpo pins indicated by the red arrows in the image below), while the resistance of the NTC is around 100 K:’ ‡Gpo pins indicated by the orange arrows):

Reseat the FPC

If the problem persists, reseat both ends of the FPC (sometimes you need to reseat it several times).

!9þ NOTE: Please refer to the following wiki for more information:
PCBs on the toolhead of P1P | Bambu Lab Wiki

Please refer to the following wiki for more information:
PCBs on the toolhead of P1P | Bambu Lab Wiki

A1 Series Troubleshooting

Re-plug the cable of NTC

Re-plug the cable of NTC and then check whether the error prompt disappears.

Disassemble the hotend

Please refer to this wiki to disassemble the hotend and check if the NTC sensor or wire is broken.

Measure with a multimeter

If you have a multimeter available, you can try measuring the resistance of the hotend component and follow the steps below to determine if there is a hotend malfunction.

Typically, the error message for A1 series printers is 'Nozzle Temperature Malfunction'. To measure the resistance of the heating component, please remove the rear cover of the toolhead. The plug to be measured is the number 1 plug on the TH board.

The resistance of the ceramic heater at room temperature is typically around 7:’ ‡Gpo pins indicated by the red arrows in the image below), while the resistance of the NTC is around 100 K:’ ‡Gpo pins indicated by the orange arrows, the resistance of NTC varies widely with temperature fluctuations):

Contact the after-sales

If the message still shows up, contact the after-sales team.

The characteristic curve of an NTC thermistor.

This thermistor is a Negative Temperature Coefficient (NTC) thermistor with a resistance value of 100K at room temperature (25 degC). In winter, the resistance value increases as the ambient temperature decreases, while in summer, the resistance value decreases as the temperature rises.

In practical measurements, slight deviations from the theoretical curve may occur due to factors such as the accuracy limitations of the measuring instrument, instantaneous fluctuations in the ambient temperature, the thermal response time of the thermistor, and the parasitic parameters of the measurement circuit. These deviations are considered normal and will not affect the normal operation of the thermistor or its temperature sensing function.

Error message

HMS_0300-0200-0001-0007: The nozzle temperature is abnormal, the sensor may be open circuit.

"The nozzle temperature is abnormal, the sensor may be open circuit."

0300-0200-0001-0007

Error Code: 0300-0200-0001-0003

The nozzle temperature is abnormal, the heater is over temperature.

Issue Description

Nozzle temperature control mainly includes a heater and an NTC sensor. When the heater is operating at full power but fails to bring the nozzle up to the target temperature within time, the TH board will trigger a "heater over-temperature" error. This is a safety mechanism designed to prevent potential damage to the device or other risks in case of a temperature control failure.

The NTC is a temperature sensor that continuously monitors the actual nozzle temperature in real time. If there is a fault with the heater, the NTC itself, or a poor connection between components, the system may misread the temperature and trigger this error.

Common Scenarios Causing the Error

- The silicone sleeve is broken or improperly installed.
- The heater or NTC is not installed properly or is faulty.
- The connectors of the heater or NTC are in poor contact.
- The TH board is faulty.

IMPORTANT! It's crucial to power off the printer before performing any maintenance work on the printer and its electronics, including tool head wires, because leaving the printer on while conducting such tasks can cause a short circuit, which can lead to additional electrical damage and safety hazards. When you perform maintenance or troubleshooting on the printer, you may be required to disassemble some parts, including the hotend. This process can expose wires and electrical components that could potentially short circuit if they come into contact with each other or with other metal or electrical components while the printer is still on. This can damage the electronics of the printer and cause further damage. Therefore, it's essential to power off the printer and disconnect it from the power source before doing any maintenance work. This will prevent any short circuits or damage to the printer's electronics. By doing so, you can avoid potential damage to the printer's electronic components and ensure that the maintenance work is performed safely and effectively. If you have any concerns or questions about following this guide, open a new ticket in our Support Page and we will do our best to respond promptly and provide you with the assistance you need.

IMPORTANT!

It's crucial to power off the printer before performing any maintenance work on the printer and its electronics, including tool head wires, because leaving the printer on while conducting such tasks can cause a short circuit, which can lead to additional electrical damage and safety hazards.

When you perform maintenance or troubleshooting on the printer, you may be required to disassemble some parts, including the hotend. This process can expose wires and electrical components that could potentially short circuit if they come into contact with each other or with other metal or electrical components while the printer is still on. This can damage the electronics of the printer and cause further damage.

Therefore, it's essential to power off the printer and disconnect it from the power source before doing any

maintenance work. This will prevent any short circuits or damage to the printer's electronics. By doing so, you can avoid potential damage to the printer's electronic components and ensure that the maintenance work is performed safely and effectively.

If you have any concerns or questions about following this guide, open a new ticket in our Support Page and we will do our best to respond promptly and provide you with the assistance you need.

Troubleshooting guide

X1 Series Troubleshooting

Quick Check: The easiest way to troubleshoot this issue is to try replacing the complete hotend with a new one to see if the error is resolved.

Remove the Toolhead Front Cover

The X1 toolhead uses a magnetic front cover design, which can be removed directly by hand. Turn off the printer before operation. Gently open the front cover and rest it on the carbon rod as shown in the image. Handle with care to avoid damaging the connected cables.

Reseat the cable of NTC

Reseat the cable of NTC and then check whether the error prompt disappears. The NTC for X1 series printers is located on the extrusion interface board.

Disassemble the hotend

Please refer to this wiki to disassemble the hotend and check if the NTC sensor or wire is broken.

Measure with a multimeter

If you have a multimeter available, you can try measuring the resistance of the hotend component and follow the steps below to determine if there is a hotend malfunction.

The resistance of the ceramic heater at room temperature is typically around 12:'Â v†–ÆR F†R &W6—7F æ6R öb F†R åD2 is around 50-100 K:“

Replug the 10Pin connector

If you have a V8 TH board, replug the 10Pin connector (sometimes it needs to be replugged several times); also check for any signs of burnout on the 10-pin plug. You need to remove the silicone with tweezers before replugging it. Then try again.

If you have a V9 TH board, reseat both ends of the FPC.

Please refer to the following wiki for more information: Replacing the toolhead TH board assembly (V8)| Bambu Lab Wiki Replace the toolhead boards (V9) | Bambu Lab Wiki

Please refer to the following wiki for more information:

- Replacing the toolhead TH board assembly (V8)| Bambu Lab Wiki
- Replace the toolhead boards (V9) | Bambu Lab Wiki

P1 Series Troubleshooting

Quick Check: The easiest way to troubleshoot this issue is to try replacing the complete hotend with a new one to see if the error is resolved.

Remove the Toolhead Front Cover

Turn off the printer and then open the toolhead front cover.

Reseat the cable of NTC

Reseat the cable of NTC as shown below and then check whether the error prompt disappears.

Disassemble the hotend

Please refer to this wiki to disassemble the hotend and check if the NTC sensor or wire is broken. For the P1 series, the ceramic heater and the NTC are integrated into a single module through a combined terminal connector.

Measure with a multimeter

If you have a multimeter available, you can try measuring the resistance of the hotend component and follow the steps below to determine if there is a hotend malfunction.

The resistance of the ceramic heater at room temperature is typically around 10:’ ‡Gpo pins indicated by the red arrows in the image below), while the resistance of the NTC is around 100 K:’ ‡Gpo pins indicated by the orange arrows):

Reseat the FPC

If the problem persists, reseat both ends of the FPC (sometimes you need to reseat it several times).

> **!9þ NOTE: Please refer to the following wiki for more information:**
> **PCBs on the toolhead of P1P | Bambu Lab Wiki**

Please refer to the following wiki for more information:
PCBs on the toolhead of P1P | Bambu Lab Wiki

A1 Series Troubleshooting

Quick Check: The easiest way to troubleshoot this issue is to try replacing the complete hotend with a new one to see if the error is resolved.

Remove the Toolhead Rear Cover

Pull outward on the bottom of the rear cover as shown below, and open it.

Reseat the cable of NTC

Reseat the cable of NTC as shown below and then check whether the error prompt disappears.

Disassemble the hotend

Please refer to the wikis below to disassemble the hotend and check if the NTC sensor or wire is broken:

- hotend-heating-assembly
- hotend-heating-assembly-replacement

Measure with a multimeter

If you have a multimeter available, you can try measuring the resistance of the hotend component and follow the steps below to determine if there is a hotend malfunction.

Typically, the error message for A1 series printers is 'Nozzle Temperature Malfunction'. To measure the resistance of the heating component, please remove the rear cover of the toolhead. The plug to be measured is the number 1 plug on the TH board.

The resistance of the ceramic heater at room temperature is typically around 7:’ ‡Gpo pins indicated by the red arrows in the image below), while the resistance of the NTC is around 100 K:’ ‡Gpo pins indicated by the orange arrows, the resistance of NTC varies widely with temperature fluctuations):

Contact the after-sales

If the message still shows up, please contact the after-sales team for further assistance.

Error message

HMS_0300-0200-0001-0007: The nozzle temperature is abnormal, the sensor may be open circuit.

Error Code: 0300-0200-0001-0006

The nozzle temperature is abnormal, the sensor may be short circuit.

What it is

Nozzle temperature control mainly includes a heater and a NTC sensor. When the NTC sensor is short-circuited, TH board cannot get the right temperature of the nozzle, which will cause the heater to stop heating and the filament may be jammed in the heatbreak.

Below are the possible reasons that this HMS error occurs:

1. A short circuit has occurred at the NTC's connector or wires
2. The NTC is broken
3. The TH board is broken

IMPORTANT! The nozzle temperature is very high; please avoid touching the nozzle. It's crucial to switch off the printer and disconnect it from the power source before doing any maintenance work. Otherwise, it could damage the electronics of the printer and cause further damage. It is crucial to ensure that the plug is aligned with the slot.

IMPORTANT!

1. The nozzle temperature is very high; please avoid touching the nozzle.
2. It's crucial to switch off the printer and disconnect it from the power source before doing any maintenance work. Otherwise, it could damage the electronics of the printer and cause further damage.
3. It is crucial to ensure that the plug is aligned with the slot.

Troubleshooting guide

X1 Series Troubleshooting

Re-plug the cable of NTC

Re-plug the cable of NTC and then check whether the error prompt disappears.

Disassemble the hotend

Please refer to this wiki to disassemble the hotend and check if the NTC sensor or wire is broken.

Measure with a multimeter

If you have a multimeter available, you can try measuring the resistance of the hotend component and follow the steps below to determine if there is a hotend malfunction.

The resistance of the ceramic heater at room temperature is typically around 12:'Â v†–ÆR F†R &W6—7F æ6R öb F†R åD2 is around 50-100 K:“

Replug the 10Pin connector

If you have a V8 TH board, replug the 10Pin connector (sometimes it needs to be replugged several times); also check for any signs of burnout on the 10-pin plug. You need to remove the silicone with tweezers before replugging it. Then try again.

If you have a V9 TH board, reseat both ends of the FPC.

Please refer to the following wiki for more information: Replacing the toolhead TH board assembly (V8)| Bambu Lab Wiki Replace the toolhead boards (V9) | Bambu Lab Wiki

Please refer to the following wiki for more information:

- Replacing the toolhead TH board assembly (V8)| Bambu Lab Wiki
- Replace the toolhead boards (V9) | Bambu Lab Wiki

P1 Series Troubleshooting

Re-plug the cable of NTC

Re-plug the cable of NTC and then check whether the error prompt disappears.

Disassemble the hotend

Please refer to this wiki to disassemble the hotend and check if the NTC sensor or wire is broken.

Measure with a multimeter

If you have a multimeter available, you can try measuring the resistance of the hotend component and follow the steps below to determine if there is a hotend malfunction.

The resistance of the ceramic heater at room temperature is typically around 10:’ ‡Gpo pins indicated by the red arrows in the image below), while the resistance of the NTC is around 100 K:’ ‡Gpo pins indicated by the orange arrows):

Reseat the FPC

If the problem persists, reseat both ends of the FPC (sometimes you need to reseat it several times).

!9þ NOTE: Please refer to the following wiki for more information:
PCBs on the toolhead of P1P | Bambu Lab Wiki

Please refer to the following wiki for more information:
PCBs on the toolhead of P1P | Bambu Lab Wiki

A1 Series Troubleshooting

Re-plug the cable of NTC

Re-plug the cable of NTC and then check whether the error prompt disappears.

Disassemble the hotend

Please refer to this wiki to disassemble the hotend and check if the NTC sensor or wire is broken.

Measure with a multimeter

If you have a multimeter available, you can try measuring the resistance of the hotend component and follow the steps below to determine if there is a hotend malfunction.

Typically, the error message for A1 series printers is 'Nozzle Temperature Malfunction'. To measure the resistance of the heating component, please remove the rear cover of the toolhead. The plug to be measured is the number 1 plug on the TH board.

The resistance of the ceramic heater at room temperature is typically around 7:’ ‡Gpo pins indicated by the red arrows in the image below), while the resistance of the NTC is around 100 K:’ ‡Gpo pins indicated by the orange arrows, the resistance of NTC varies widely with temperature fluctuations):

Contact the after-sales

If the message still shows up, contact the after-sales team.

Error message

HMS_0300-0200-0001-0007: The nozzle temperature is abnormal, the sensor may be open circuit.

"The nozzle temperature is abnormal, the sensor may be open circuit."

0300-0200-0001-0007

Error Code: 0300-0200-0001-0007

The nozzle temperature is abnormal, the sensor may be open circuit.

What it is

Nozzle temperature control mainly includes a heater and an NTC sensor. When the NTC sensor is open-circuited, the TH board cannot get the right nozzle temperature, which will cause the heater to heat continuously and cause printing failure.

Below are the possible reasons that this HMS error occurs:

1. Poor contact between cable and connector
2. The NTC sensor is broken
3. The wire of the sensor is broken

IMPORTANT! The nozzle temperature is very high; please avoid touching the nozzle. It's crucial to switch off the printer and disconnect it from the power source before doing any maintenance work. Otherwise, it could damage the electronics of the printer and cause further damage. It is crucial to ensure that the plug is aligned with the slot.

IMPORTANT!

1. The nozzle temperature is very high; please avoid touching the nozzle.
2. It's crucial to switch off the printer and disconnect it from the power source before doing any maintenance work. Otherwise, it could damage the electronics of the printer and cause further damage.
3. It is crucial to ensure that the plug is aligned with the slot.

Troubleshooting guide

X1 Series Troubleshooting

Re-plug the cable of NTC

Re-plug the cable of NTC and then check whether the error prompt disappears.

Disassemble the hotend

Please refer to this wiki to disassemble the hotend and check if the NTC sensor or wire is broken.

Measure with a multimeter

If you have a multimeter available, you can try measuring the resistance of the hotend component and follow the steps below to determine if there is a hotend malfunction.

The resistance of the ceramic heater at room temperature is typically around 12:'Â v†–ÆR F†R &W6—7F æ6R öb F†R åD2 is around 50-100 K:"

Replug the 10Pin connector

If you have a V8 TH board, replug the 10Pin connector (sometimes it needs to be replugged several times); also check for any signs of burnout on the 10-pin plug. You need to remove the silicone with tweezers before replugging it. Then try again.

If you have a V9 TH board, reseat both ends of the FPC.

Please refer to the following wiki for more information: Replacing the toolhead TH board assembly (V8)| Bambu Lab Wiki Replace the toolhead boards (V9) | Bambu Lab Wiki

Please refer to the following wiki for more information:

- Replacing the toolhead TH board assembly (V8)| Bambu Lab Wiki
- Replace the toolhead boards (V9) | Bambu Lab Wiki

P1 Series Troubleshooting

Re-plug the cable of NTC

Re-plug the cable of NTC and then check whether the error prompt disappears.

Disassemble the hotend

Please refer to this wiki to disassemble the hotend and check if the NTC sensor or wire is broken.

Measure with a multimeter

If you have a multimeter available, you can try measuring the resistance of the hotend component and follow the steps below to determine if there is a hotend malfunction.

The resistance of the ceramic heater at room temperature is typically around 10:' ‡Gpo pins indicated by the red arrows in the image below), while the resistance of the NTC is around 100 K:' ‡Gpo pins indicated by the orange arrows):

Reseat the FPC

If the problem persists, reseat both ends of the FPC (sometimes you need to reseat it several times).

!9þ NOTE: Please refer to the following wiki for more information:
PCBs on the toolhead of P1P | Bambu Lab Wiki

Please refer to the following wiki for more information:
PCBs on the toolhead of P1P | Bambu Lab Wiki

A1 Series Troubleshooting

Re-plug the cable of NTC

Re-plug the cable of NTC and then check whether the error prompt disappears.

Disassemble the hotend

Please refer to this wiki to disassemble the hotend and check if the NTC sensor or wire is broken.

Measure with a multimeter

If you have a multimeter available, you can try measuring the resistance of the hotend component and follow the steps below to determine if there is a hotend malfunction.

Typically, the error message for A1 series printers is 'Nozzle Temperature Malfunction'. To measure the resistance of the heating component, please remove the rear cover of the toolhead. The plug to be measured is the number 1 plug on the TH board.

The resistance of the ceramic heater at room temperature is typically around 7:' ‡Gpo pins indicated by the red arrows in the image below), while the resistance of the NTC is around 100 K:' ‡Gpo pins indicated by the orange arrows, the resistance of NTC varies widely with temperature fluctuations):

The characteristic curve of an NTC thermistor.

This thermistor is a Negative Temperature Coefficient (NTC) thermistor with a resistance value of 100K at room temperature (25 degC). In winter, the resistance value increases as the ambient temperature decreases, while in summer, the resistance value decreases as the temperature rises.

In practical measurements, slight deviations from the theoretical curve may occur due to factors such as the accuracy limitations of the measuring instrument, instantaneous fluctuations in the ambient temperature, the thermal response time of the thermistor, and the parasitic parameters of the measurement circuit. These deviations are considered normal and will not affect the normal operation of the thermistor or its temperature sensing function.

Contact the after-sales

If the message still shows up, contact the after-sales team.

Error message

HMS_0300-0200-0001-0007: The nozzle temperature is abnormal, the sensor may be open circuit.

"The nozzle temperature is abnormal, the sensor may be open circuit."

0300-0200-0001-0007

Error Code: 0300-1200-0002-0001

The front cover of the toolhead fell off.

What it is

The front cover and the toolhead are held together by magnets, with a total of 8 magnets on both sides.

V9 TH board:

V8 TH board:

There is a Hall sensor at the top right of the TH board, which determines whether the front cover is off by detecting the hall value of the magnet in the top right corner.

V9 TH board:

V8 TH board:

Troubleshooting guide

Step 1

Please mount it back and click the “resume” icon to resume the print job

If the printing model is normal and the front cover does not fall off, please click the "Resume" button to continue.

Step2

If the front cover doesn't fall off, but there are repeated false alarms of the front cover felling off, sometimes it is caused by a falling magnet or loose screws holding the TH board. You need to ensure the toolhead magnets are still there. Then check if the screw holding the PCB (in the picture below) is loose; if so, please tighten it.

V9 TH board:

V8 TH board:

Step 3

The false alarm could also be caused by poor contact or damage to the connector. Please try to re-plug the 10-pin connector(sometimes it needs to be re-plugged several times) in the picture below. You need to remove the silicone with tweezers before re-plugging it. If the error persists, please contact the customer support team.

If your X1 series printer has a V9 TH board, please reinstall both ends of this FPC.

X1 series V9 TH board

Replacing the toolhead TH board assembly (V8)| Bambu Lab Wiki

Replace the toolhead boards (V9) | Bambu Lab Wiki

If your printer model is P1 Series printer, please replug both ends of this FPC.

TH board of P1 Series Printer

Step 4

If the front cover does fall off (as shown below), please manually put the front cover back to the toolhead first. If the front cover fell off during the printing, it is possible that the toolhead hit something on the heat bed and the front cover was touched off. We suggest you turn on the video function before printing, and if the front cover comes off again, you can send the corresponding video and log files to our customer support team for analysis.

Error message

HMS_0300-1200-0002-0001: The front cover of the toolhead fell off.

"The front cover of the toolhead fell off."

0300 8005

0300-1200-0002-0001

Error Code: 0300-1D00-0001-0001

HMS_0300-1D00-0001-0001: The position sensor of extrusion motor is abnormal. The connection to sensor may be loose.

Error message

HMS_0300-1D00-0001-0001: The position sensor of extrusion motor is abnormal. The connection to sensor may be loose.

Error Code: 0C00-0100-0001-0001

HMS_0C00-0100-0001-0001: Micro Lidar camera is offline.

What it is

The Micro Lidar camera can't be found due to loss of I2C signal. The possible causes include, with probability from high to low:

1. the camera-to-TH connector is unstable (image data flows from this connector)
2. USB-C cable is broken or the connection is unstable (image data flows in this cable)
3. the TH board is broken
4. the camera is broken

Safety warning and Machine state before starting operation

When you perform maintenance or troubleshooting on the printer, you may be required to disassemble some parts. This process can expose wires and electrical components that could potentially short circuits if they come into contact with each other or with other metal or electronic components while the printer is still on. This can damage the electronics of the printer and cause further damage. Always turn off the printer and remove the power cable when performing any operation.

Operation guide

1. Reboot and re-print

Reboot the printer and start printing several times and see if the message is consistently reported. If after rebooting the message disappears, it can be ignored. Otherwise, go to the following steps.

2. Unplug and then plug in the camera-to-TH connector

Power off the printer. Unplug and plug in the connector as shown below. Then power on the printer and start a new print job, to see if the message disappears.

3. Unplug and then plug in USB-C cable

Power off the printer. Unplug and then plug in the type-c cable. Restart a print job and see if the message disappears. If yes, the problem is solved.

USB-C Cable Connection Issue | Bambu Lab Wiki

4. Unplug and then plug in the MC-to-AP cable

If the above step cannot solve the problem, you may need to replace the spare parts. Please contact our support team for solutions.

5. Replace the USB-C cable

If the problem still exists, replace the type-c cable following the video guide:

https://wiki.bambulab.com/x1/maintenance/replace-typec-cable

After replacement, turn on the printer and start a print job to see if the message still pops up.

6. Replace Micro Lidar

If the above five steps cannot solve the problem, please follow the instruction to replace Micro Lidar:

https://wiki.bambulab.com/x1/maintenance/replace-micro-lidar

7. Replace the TH board assembly

If the above six steps cannot solve the problem, please follow the instruction to replace the TH board assembly:

https://wiki.bambulab.com/en/x1/maintenance/toolhead_boards_v9

8. Replace the AP board

If the above seven steps cannot solve the problem, please follow the instruction to replace the AP board:

https://wiki.bambulab.com/en/x1/maintenance/replace-ap-board

How to verify completion/success

Reboot the printer, and start a new print. If it succeeds, the error message should be gone.

Error

HMS_0C00-0100-0001-0001: Micro Lidar camera is offline.

- If you have completed the above 8 steps but the issue still persists, please record a video demonstrating the motion of the heatbed and the tool head during the printer's homing process. After that, allow the printer to sit idle for 10 minutes, upload the recorded video, and submit the printer's log file. Finally, please submit a technical service ticket for further troubleshooting and problem resolution.
- If your printer encounters an issue related to the micro LiDAR unit or AI, we strongly recommend selecting both of these options. This will greatly assist the technical support team in accurately diagnosing and addressing the problem.

Error Code: 0C00-0100-0002-0002

HMS_0C00-0100-0002-0002: Micro Lidar camera is malfunctioning.

What it is

The Micro Lidar camera is malfunctioning and can't get any valid images. The possible causes include, with probability from high to low:

1. USB-C cable is broken, or the connection is unstable (image data flows in this cable)
2. the camera-to-TH connector is unstable (image data flows from this connector)
3. the TH board is broken
4. the camera is broken

Note that the cables are torn gradually, meaning that at the beginning, the error only occurs occasionally, and the Micro Lidar function is not hampered too much.

Safety warning and Machine state before starting operation

When you perform maintenance or troubleshooting on the printer, you may be required to disassemble some parts. This process can expose wires and electrical components that could potentially short circuits if they come into contact with each other or with other metal or electronic components while the printer is still on. This can damage the electronics of the printer and cause further damage. Always turn off the printer and remove the power cable when performing any operation.

Operation guide

1. Reboot and re-print

Reboot the printer and start printing several times and see if the message is consistently reported. If after rebooting the message disappears, it can be ignored. Otherwise, go to the following steps.

2. Unplug and then plug in USB-C cable

Power off the printer. Unplug and then plug in the type-c cable. Restart a print job and see if the message disappears. If yes, the problem is solved.

USB-C Cable Connection Issue | Bambu Lab Wiki

3. Unplug and then plug in the camera-to-TH connector

4. Unplug and then plug in the MC-to-AP cable

If the above step cannot solve the problem, you may need to replace the spare parts. Please contact our support team for solutions.

Please refer to this wiki: Replace the MC-AP cable | Bambu Lab Wiki

5. Replace the USB-C cable

If the problem still exists, replace the type-c cable following the video guide:

https://wiki.bambulab.com/x1/maintenance/replace-typec-cable

After replacement, turn on the printer and start a print job to see if the message still pops up.

6. Replace the TH board assembly

If the above five steps cannot solve the problem, please follow the instructions to replace the TH board assembly:

https://wiki.bambulab.com/en/x1/maintenance/toolhead_boards_v9

7. Replace Micro Lidar

If the above six steps cannot solve the problem, please follow the instructions to replace Micro Lidar:

https://wiki.bambulab.com/x1/maintenance/replace-micro-lidar

8. Replace the AP board

If the above seven steps cannot solve the problem, please follow the instructions to replace the AP board:

https://wiki.bambulab.com/en/x1/maintenance/replace-ap-board

How to verify completion/success

Reboot the printer, and start a new print. If it succeeds, the error message should be gone.

Error message

HMS_0C00-0100-0002-0002: Micro Lidar camera is malfunctioning.
0C00-0100-0002-0002

Error Code: 0C00-0100-0001-0003

HMS_0C00-0100-0001-0003: Micro Lidar Synchronization abnormal.

What it is

This error message occurs when the synchronization between different Micro Lidar subsystems (mainly the mc and camera sensor) is lost.

When to use

It should not happen on normal printers. When this message shows up, there is an abnormal synchronization signal between the Micro Lidar subsystems, which will cause the related functions of the lidar system to work abnormally.

Safety warning and Machine state before starting operation

None

Operation guide

step1

Please try to start the printing task several times to observe whether the message prompt occurs frequently. If the message disappears and only occurs occasionally after restarting printing, it means that the software time synchronization fails. In this case, it is a software error and can be ignored.

step2

If restarting multiple printing tasks, the message still prompts, indicating that it is a hardware problem. Usually, the hardware synchronization signal is abnormal due to the failure of the circuit board. You need to contact the after-sales service for troubleshooting and hardware replacement.

How to verify completion/success

The message should disappear if the problem is solved.

Error message

HMS_0C00-0100-0001-0003: Micro Lidar Synchronization abnormal.

0C00-0100-0001-0003

Error Code: 0C00-0100-0001-0004

HMS_0C00-0100-0001-0004: Micro Lidar Lens dirty.

What it is

The Micro Lidar lens is dirty, possibly due to dirt, oil, or organic vapor (such as ABS vapor) on the lens. A dirty lens reduces the resolution and contrast of the observed images and should be cleaned.

When to use

When printing volatile materials such as ABS for a long time, volatile substances will adhere to the lens surface of the camera module, resulting in a decrease in clarity and making lidar-related functions unreliable.

Safety warning and Machine state before starting operation

It is suggested to power down the printer before you wipe the lens.

Operation guide

step1

a. Make sure the calibration board is clean and in good condition.

b. Power down the printer, and use alcohol and other cleaning tools to wipe the module lens. The lens positions are as follows:

step2

Turn on the printer, trigger a printing task, and see if the message disappears or not.

step3

If several attempts to clean the lens do not solve the problem, please contact the after-sales service to replace the Micro Lidar accessories.

https://wiki.bambulab.com/x1/maintenance/replace-micro-lidar

How to verify completion/success

The message should disappear if the problem is solved.

Error message

HMS_0C00_0100_0001_0004: Micro Lidar Lens dirty.

0C00-0100-0001-0004

Error Code: 0C00-0100-0001-0005

HMS_0C00-0100-0001-0005: Micro Lidar OTP parameter abnormal.

What it is

The calibration parameters of the lidar are abnormal. This message indicates that there is a problem with the calibration parameters of the lidar at the factory.

When to use

This problem occurs when the calibration parameters of the camera module are lost or there is a problem with the storage of internal parameters. The machine

uses the default parameters to work, which will affect the performance of lidar-related functions.

Safety warning and Machine state before starting operation

None

Operation guide

step1

Contact after-sales to replace Micro Lidar accessories.

https://wiki.bambulab.com/x1/maintenance/replace-micro-lidar

How to verify completion/success

The message should disappear if the problem is solved.

Error message

HMS_0C00-0100-0001-0005: Micro Lidar OTP parameter abnormal.

0C00-0100-0001-0005

Error Code: 0C00-0100-0002-0006

HMS_0C00-0100-0002-0006: Micro Lidar extrinsic parameter abnormal.

HMS_0C00-0100-0002-0006: Micro Lidar extrinsic parameter abnormal.
Not implemented yet.

0C00-0100-0002-0006

Error Code: 0C00-0100-0002-0007

HMS_0C00-0100-0002-0007: Micro Lidar laser parameter drifted.

What it is

Since we have two lasers on the Micro Lidar, measuring the depth of the same point using the two lasers should give the same result. If the results are different than a given threshold, this message will appear, indicating that the parameters of the two lasers are no longer correct.

Why does this happen

The laser parameter encodes the spatial distance and angle (i.e. transformation) between the laser and camera. They are supposed to rigidly connected. However, the connection is not strong enough to resist the vibration of toolhead. During long-time usage, the transformation may change, and the laser parameter is no longer accurate. We call this phenomenon “drifting”.

Operation guide

Solving the issue is easy, just run the calibration.

step1

Enter the screen calibration page, click to start calibration, and wait for the calibration to end.

step2

Restart printing and check if the error message disappears

step3

If the message is still there, check if the calibration marker is clean and uncovered.

How to verify completion/success

The message should disappear if the problem is solved.

Error message

HMS_0C00-0100-0002-0007: Micro Lidar laser parameter drifted.

0C00-0100-0002-0007

Error Code: 0C00-0100-0002-0008

HMS_0C00-0100-0002-0008: Failed to get image from chamber camera. Spaghetti and waste chute pileup detection is not available for now..

What it is

The chamber camera failed to get image indicating the chamber camera hardware failure.

When to use

In the X1C printer, this error message will be reported when the chamber camera fails to read the image correctly.

Safety warning and Machine state before starting operation

Be sure to power down the printer before you check and replace the camera.

Operation guide

step1

Turn off the printer, and follow the instructions below to plug and unplug the camera cable.

https://wiki.bambulab.com/x1/maintenance/replace-chamber-camera

step2

Power on the printer, and use the app or studio software to connect the printer to check whether the real-time video of the chamber camera is normal.

step3

If the above operations cannot solve the problem, please contact the after-sales service for the replacement of the chamber camera accessories.

https://wiki.bambulab.com/x1/maintenance/replace-chamber-camera

How to verify completion/success

The message should disappear if the problem is solved.

Error message

HMS_0C00-0100-0002-0008: Failed to get image from chamber camera. Spaghetti and waste chute pileup detection is not available for now.

0C00-0100-0002-0008

Error Code: 0C00-0100-0001-0009

HMS_0C00-0100-0001-0009: Chamber camera dirty.

HMS_0C00-0100-0001-0009: Chamber camera dirty.

0C00-0100-0001-0009

Error Code: 0C00-0100-0001-000A

HMS_0C00-0100-0001-000A: The Micro Lidar LED may be broken.

What it is

There is a LED light on the Micro Lidar to help capture RGB image. If this light is broken, the captured RGB image will be all blank, and functions like calibration and build plate tag localization, will be unavailable.

What to do

We use image analysis to check if the LED is good. There might be false alarams. For example, if the Micro Lidar camera is ill-functional, or the command of opening LED is delayed, we'll also get a black image. It's suggested to double-check if the LED is really broken.

1. Check if the HMS message keeps showing up in new print jobs. If it's gone, the LED is good and the previous message is a false alarm.
2. Start a print job, check if the LED is bright during camera check and build plate tag localization.

1. If it's confirmed that LED is broken, please contact our support team for solutions.

Error message

HMS_0C00-0100-0001-000A: The Micro Lidar LED may be broken.

0C00-0100-0001-000A

If your printer encounters an issue related to the micro LiDAR unit or AI, we strongly recommend selecting both of these options. This will greatly assist the technical support team in accurately diagnosing and addressing the problem.

Error Code: 0C00-0100-0001-000B

HMS_0C00-0100-0001-000B: Failed to calibrate Micro Lidar.

What is it

The Micro Lidar parameters are calibrated during machine calibration. Basically, we move the toolhead over the calibration board, capture several images with laser on and off, and then calibrate the parameters.

This calibration may fail for several reasons:

1. The calibration board is broken, dirty, or covered.

2. The lens is dirty.

3. The USB-C cable is not working well.

4. The camera itself is broken (very rare case).

5. Firmware bug.

What to do

According to the different reasons, these are the steps to follow:

1. Check if the calibration board is in a good state. Clean it if necessary.

2. Clean the lens if step 1 doesn't work.

3. Unplug and then plug in the USB-C cable. Please see steps 2-5 in this article to unplug and plug in again.

4. Initiate a print job and perform micro lidar calibration again.

This print job requires using the latest version of Bambu Studio to slice and generate the corresponding model. After the printer completes the print preparation process, you can choose to cancel the print job or let the printer complete the model printing. Once the task is completed, perform micro lidar calibration again.

5. Contact our support team if all the above steps don't work.

Error message

HMS_0C00-0100-0001-000B: Failed to calibrate Micro Lidar.

0C00-0100-0002-000B

If your printer encounters an issue related to the micro LiDAR unit or AI, we strongly recommend selecting both of these options. This will greatly assist the technical support team in accurately diagnosing and addressing the problem.

Error Code: 0C00-0200-0001-0001

HMS_0C00-0200-0001-0001: The horizontal laseris not lit.Pleasecheckifit's covered or hardware connection has aproblem.

What it is

This error occurs when the lidar does not detect the reflected laser. This could be a hardware failure of the lidar, or it could be caused by external factors.

Before each print starts, the machine will let the tool head return to the home position, checking the lidar in the process. You can also tap the Home icon on the screen to check the lidar.

Troubleshooting

Step 1

Turn off the printer power, use a lens wiper, and use alcohol or other cleaning tools to wipe the module lens to eliminate the lens dirty problem.

The lens position is as follows:

Step 2

1. There is no reflected light after the laser is emitted into the heated bed. Residual black, clear, exceptionally bright silk filaments on the heated bed may cause this problem. Please clean the residual filament on the heated bed.

2. This problem can be caused by using an unofficial heated bed or by applying a film to the heated bed. Please make sure that the build plate used is official and that no special film is applied.

3. The laser of the lidar is not lit. Lidar emits both red and blue lasers. The red laser is invisible to the naked eye. If your printer has chamber camera, you can check whether the red laser is on or off normally during the process of homing. You can also send a log to our after-sales team and we will help you analyze whether the laser is normal through the log.

step 3

If it is confirmed that this is a hardware failure of the lidar, please contact our after-sales team for a replacement. You can refer to this article to replace it:

https://wiki.bambulab.com/en/x1/maintenance/replace-micro-lidar

How to verify completion/success

The message should disappear if the problem solved

Error message

HMS_0C00-0200-0001-0001: Laser not lit.
"The laser is not lit. Please check the hardware connection [0C00 0200 0001 0001]"

0C00-0200-0001-0001

Error Code: 0C00-0200-0002-0002

HMS_0C00-0200-0002-0002: Laser too thick

What it is

We use multiple exposure time to capture the laser line and choose the best exposure. Under this exposure time the laser line should be bright and thin. This is crucial for accurate depth measuring. However, if the build plate is dirty (e.g. covered with white filament residual), the laser line may be too thick.

Troubleshooting

This is just a warning message, which means it won't affect the functions of Micro Lidar in general. We recommend you check if the build plate is clean and undamaged.

Error message

HMS_0C00-0200-0002-0002: Laser too thick
The horizontal laser line is too thick in the camera view.

0C00-0200-0002-0002

Error Code: 0C00-0200-0002-0003

HMS_0C00-0200-0002-0003: Laser not bright enough.

What it is

Every time the printing system will check the status of the lidar sensor, when it is detected that the brightness of the lidar sensor is insufficient, it will

prompt this error message. However, the sensor measurement data is affected by the clarity of the lens and the cleanliness of the hot bed, so it is

necessary to comprehensively analyze the current fault type.

Troubleshooting

Step 1

Start the print task multiple times to check that there is no error prompt indicating that the lens is dirty.

Step 2

a. Confirm whether to use an official hotbed. If you use an unofficial hotbed with low surface material reflectivity, this problem may occur. Please replace it with an official hotbed to check whether the problem still occurs. Click the home button to check:

b. If using an official hot bed, please check if there is a lot of residual material sticking to the hot bed, please try to clean the hot bed and test again.

Step 3

If the lens is not dirty and the error message does not disappear after repeated printing after cleaning the hot bed for many times, please contact the after-sales service to replace the Micro Lidar parts.

https://wiki2.bambulab.com/x1/maintenance/replace-micro-lidar

How to verify completion/success

The message should disappear if the problem is solved.

Error message

HMS_0C00-0200-0002-0003: Laser not bright enough.
The laser not bright enough indicates that the lidar has a software or hardware failure.

0C00-0200-0002-0003

Error Code: 0C00-0200-0002-0004

HMS_0C00-0200-0002-0004: Nozzle height seems too low.

What it is

After using the printer for a long time, the nozzle may be worn, and its height is lower than the saved value when the printer is produced.

Another case is, when the toolhead crashes with printed objects or something, and the nozzle is tilted, then the detected nozzle height is also different from the saved value.

What to do

Step 1

Check if the nozzle is severely worn as below:

Step 2

If the nozzle is OK, run "Calibration" to dismiss the warning message.

How to verify completion/success

Restart the printer and start a print job. The message should disappear if the problem is solved.

Error message

HMS_0C00-0200-0002-0004: Nozzle height seems too low.

0C00-0200-0002-0004

Error Code: 0C00-0200-0001-0005

HMS_0C00-0200-0001-0005: A new micro lidar is detected.

What it is

The new Micro Lidar detected occurs when replacing a new lidar accessory. The system will detect the configuration and calibration parameters of the lidar and perform online calibration.

Safety warning and Machine state before starting operation

None

Operation guide

Step 1

Power on the printer, click Machine Calibration and wait for the calibration to complete.

Please note that device self-test and factory reset calibration do not include micro lidar calibration. If you need to calibrate the micro lidar, you need to select the micro lidar calibration option in the calibration menu and start calibration.

Step 2

Trigger the machine to print to see if the lidar-related functions work normally.

How to verify completion/success

The message should disappear if the problem is solved.

Error message

HMS_0C00-0200-0001-0005: A new micro lidar is detected.

0C00-0200-0001-0005

Error Code: 0C00-0200-0002-0006

HMS_0C00-0200-0002-0006: Nozzle height seems too high.

What it is

HMS_0C00-0200-0002-0006: Nozzle height seems too high.
The detected nozzle height is larger than the stored value. This is usually caused by accumulated filament residual around the nozzle tip.

Troubleshooting

Step 1

Check if there is a filament residual around the nozzle tip.

Step 2

Remove the residual filament if there is any. Heating the nozzle might be needed.

Step 3

Restart a print job and see if the error message disappears. If not, run a calibration.

Error message

HMS_0C00-0200-0002-0006: Nozzle height seems too high.
0C00-0200-0002-0006

Error Code: 0C00-0300-0002-0001

HMS_0C00-0300-0002-0001: Filament exposure metering failed.

What it is

Before printing, the printer must meter the laser reflection of the first used filament and choose the best camera exposure value. If the filament is pure transparent or black, this metering may fail, and the exposure value may not be accurate. Thus first layer inspection may not be accurate.

Safety warning and Machine state before starting operation

None

Troubleshooting

Pay extra attention to the first layer, as the inspection module may miss some first-layer failure.

Error message

HMS_0C00-0300-0002-0001: Filament exposure metering failed.

0C00-0300-0002-0001

Error Code: 0C00-0300-0002-0002

HMS_0C00-0300-0002-0002: First layer inspection terminated due to abnormal lidar data.

What it is

First layer inspection terminated due to abnormal lidar data indicates that the data is abnormal during the scanning process. This error message will appear when there are software problems such as abnormal scan data

Safety warning and Machine state before starting operation

None

Troubleshooting

Step 1

Trigger printing multiple times or restart the device to see if the error persists.

Step 2

If the above operations cannot solve the problem, please contact the after-sales service for the replacement of Micro Lidar.

https://wiki.bambulab.com/x1/maintenance/replace-micro-lidar

How to verify completion/success

The message should disappear if the problem is solved.

Error message

HMS_0C00-0300-0002-0002: First layer inspection terminated due to abnormal lidar data.

0C00-0300-0002-0002

Error Code: 0C00-0300-0002-0004

HMS_0C00-0300-0002-0004: First layer inspection not supported for current print job.

What it is

Two main issues can cause this error:

GCode parsing issue: The first-layer scanning path cannot be correctly obtained during GCode parsing.

The micro lidar camera fails to start properly.

Troubleshooting

1. GCode parsing issue:

Typically, GCode parsing issues are intermittent, and initiating a new print job usually resolves the problem.

If the issue persists, it might be due to slicing GCode problems or firmware bugs.

Consider updating or changing your slicing software and ensure that you are using the latest firmware version.

2. Failure to start the micro lidar camera:

Observe if there are any subsequent micro lidar camera-related HMS alarms, such as the following HMS error message:

HMS_0C00_0100_0001_0001: Micro Lidar camera is offline. X1 series

HMS_0C00_0100_0001_0001: Micro Lidar camera is offline.

- X1 series

HMS_0C00_0100_0002_0002: Micro Lidar camera is malfunctioning. X1 series

HMS_0C00_0100_0002_0002: Micro Lidar camera is malfunctioning.

- X1 series

Error message

HMS_0C00-0300-0002-0004: First layer inspection not supported for current print job.
If there are no camera-related HMS alarms, it might be an intermittent issue. In such cases, initiating a new print job can help avoid the problem.

0C00-0300-0002-0004

Error Code: 0C00-0300-0002-0005

HMS_0C00-0300-0002-0005: First layer inspection timeout.

When to use

It should not happen on normal printers. There are two possible causes for this issue:

1. A software problem is causing the calculation to time out.
2. The camera system hardware system has failed.

Safety warning and Machine state before starting operation

Be sure to power down the printer before you unplug the type-c cable.

Troubleshooting guide

Step 1

Try to restart the printer, trigger the print job multiple times, observe whether the error message has a high probability of occurrence, if only occasionally indicates that it is a software problem, it can be ignored and will not affect the current printing.

Step 2

If this error message appears frequently, contact the after-sales team to replace the type-c cable.

https://wiki.bambulab.com/x1/maintenance/replace-typec-cable

After the replacement is complete, turn on the printer and start a print job to see if the message still pops up.

Step 3

If the above two steps cannot solve the problem, please refer to the following after-sales video to replace Micro Lidar accessories.

https://wiki.bambulab.com/x1/maintenance/replace-micro-lidar

How to verify completion/success

The message should disappear if problem solved.

Error message

HMS_0C00-0300-0002-0005: First layer inspection timeout.
This error message may be caused by software or hardware.

0C00-0300-0002-0005

Error Code: 0C00-0300-0003-0006

HMS_0C00-0300-0003-0006: Purged filaments piled up.

What it is

This message indicates the purged filament is piled up at the excess chute and may affect printing. This function is enabled on printers with chamber cameras. We use AI to detect excess pileup, and pause printing when pileup is detected.

Troubleshooting

Clean the excess chute and resume printing.

Error message

HMS_0C00-0300-0003-0006: Purged filaments piled up.

0C00-0300-0003-0006

Error Code: 0C00-0300-0003-0007

HMS_0C00-0300-0003-0007: Possible first layer defects.

What it is

Possible first layer defects. The micro-lidar detects small defects in first layer, but not severe enough to trigger print pause action. As a system information, the user may want to check the quality to decide whether to stop the printing.

Troubleshooting

Check the first layer quality either through bare eyes or chamber camera live view. If the quality is not acceptable, stop the printing.

Error message

HMS_0C00-0300-0003-0007: Possible first layer defects.

0C00-0300-0003-0007

Error Code: 0C00-0300-0003-0008

HMS_0C00-0300-0003-0008: Possible spaghetti defects.

What it is

There may be spaghetti failure in current printing process, which is considered not severe enough to pause the printing. But the user may want to check the quality, since AI can make mistake.

When to use

When the Spaghetti Detection function is enabled (only can be used with Chamber Camera), the intelligent analysis system will analyze the video stream in real time and notify the user when the risk of spaghetti failure is detected.

Note: it is strongly advised not to shut down the camber light when Spaghetti Detection is on. This function needs good light condition to work.

Troubleshooting

Check print quality and decide whether to stop printing.

Error message

HMS_0C00-0300-0003-0008: Possible spaghetti defects.
0C00-0300-0003-0008

Error Code: 0C00-0300-0001-0009

HMS_0C00-0300-0001-0009: First layer inspection module rebooted.

What it is

HMS_0C00_0300_0001_0009: First layer inspection module rebooted
This is a warning of the lidar system software error.

When to use

This error message pops up when the radar system restarts due to a problem with the system software.

Safety warning and Machine state before starting operation

None

Troubleshooting

Does not affect the current print.

How to verify completion/success

The message should disappear if problem solved.

Error message

HMS_0C00-0300-0001-0009: First layer inspection module rebooted.
0C00-0300-0001-0009

Error Code: 0C00-0300-0003-000B

HMS_0C00-0300-0003-000B: Inspecting first layer.

What it is

This is a progress status message for the first layer inspection.

When to use

When the lidar system calculates the scan results of the first layer printing, a prompt for information will appear to help users understand the current operating state and execution progress.

Safety warning and Machine state before starting operation

None

Operation guide

None

How to verify completion/success

The message should disappear if calculating finished.

Error message

HMS_0C00-0300-0003-000B: Inspecting first layer.
0C00-0300-0003-000B

Error Code: 0C00-0300-0002-000C

HMS_0C00-0300-0002-000C: Build plate marker not detected.

What it is

The build plate localization marker is not detected.

Why does this happen

You may notice there is a 2D code on Bambu Lab build plates. These are AruCo tags, and are used to detect where the build plate is properly place and aligned. The HMS warning message appears if:

1. the build plate is not well aligned (the common issue is that the build plate is slightly off the heatbed towards the door side)
2. the marker is damaged or broken
3. 3rd party build plate is used

Troubleshooting

If you are using the official Bambu Lab build plate:

1. check if the build plate is placed in the correct direction and well-aligned with the heatbed.

2. check if the marker is good and clean.

If you are using 3rd party build plate, just ignore the message.

Error message

HMS_0C00-0300-0002-000C: Build plate marker not detected.

0C00-0300-0002-000C

Error Code: 0500-0100-0002-0001

The media pipeline is malfunctioning.

What it is

Media pipeline is for video file recording, timelapse recording, live video streaming. This HMS error code indicates there is problem in this pipeline.

When to use

This HMS error code is reported when:

1. There is no enough resource for media pipeline.
   a. There are too many clients connected to live video streaming.
   b. There is no enough memory for media pipeline in this scenaria.
2. There is problem in hardware
   a. Chamber camera is not installed properly. Please check according to this guide: https://wiki.bambulab.com/en/x1/maintenance/replace-chamber-camera
   b. Other cable or power supply is not connected properly.

1. Keep only one application which could be one of video file recording, timelapse recording, live video streaming with only one client.
2. Reconnect chamber camera and related cables.
3. Reboot the printer again.
4. Factory resetting.

Error message

HMS_0500-0100-0002-0001: The media pipeline is malfunctioning.
0500-0100-0002-0001

Error Code: 0500-0100-0002-0002

USB camera is not connected.

What it is

The HMS indicates USB camera is not connected.

When to use

The HMS means USB camera is not connected.

Troubleshooting

Please check or install chamber camera according to https://wiki.bambulab.com/en/x1/maintenance/replace-chamber-camera

Error message

HMS_0500-0100-0002-0002: USB camera is not connected
The HMS means USB camera is not connected.

Error Code: 0500-0100-0002-0003

USB camera is malfunctioning.

What it is

The HMS indicates chamber camera is malfunctioning.

When to use

Chamber camera is detected but it does not work.

Troubleshooting

Please check or install chamber camera according to https://wiki.bambulab.com/en/x1/maintenance/replace-chamber-camera

Error message

HMS_0500-0100-0002-0003: USB camera is malfunctioning.
0500-0100-0002-0003

Error Code: 0500-0100-0003-0004

Not enough space in SD Card.

What it is

This HMS indicates insufficient storage space on the SD card.

Why does this error occur

There is not enough space in the SD card and it may not be possible to record video files or time-lapse files properly. Trigger conditions for this HMS of each memory SD card:

1. 128G memory card, remaining memory space is less than 10G;

2. 64G memory card, remaining memory space is less than 8G;

3. 32G memory card, remaining memory space is less than 6G;

4. 8/16G memory card, remaining memory space is less than 3G.

    1. Delete some files to get more free space in the SD card.
    2. Format SD card on the printer or PC. The file system must be FAT32.
    3. Replace a new SD card.

For more information on using SD cards in the printer, please refer to the wiki: FAQ for Printing from micro SD Card | Bambu Lab Wiki.

What it is

HMS_0500-0100-0003-0004: Not enough space in SD Card.

0500-0100-0003-0004

Error Code: 0500-0100-0003-0005

Error in SD Card.

What it is

The HMS indiates error about writing data to SD card for video file recording or timelapse file recording.

1. Format the SD card on the printer or PC. The file system must be FAT32.
2. Replace a new SD card.

Error message

HMS_0500-0100-0003-0005: Error in SD Card.
0500-0100-0003-0005

Error Code: 0500-0100-0003-0006

Unformatted SD Card.

What it is

The HMS indicates info for SD card has not being formatted yet.

When to use

The file system in the SD card is not recognized or not supported. The printer supports FAT32 only.

Troubleshooting

Format the SD card on the printer or PC. The file system must be FAT32.

Format the SD card on the printer:
You can see the "Format" button in the SD card information. The following picture uses an X1 series machine as an example.

Error message

HMS_0500-0100-0003-0006: Unformatted SD Card.
0500-0100-0003-0006

Error Codes: 0500-0200-0002-0001, 0500-0200-0002-0003, 0500-0200-0002-0005

Failed to connect internet, please check the network connection.

What it is

The HMS shows it failed to connect to the internet, please check the network connection.

Troubleshooting

For X1E: Network Connection Guide.

For X1/X1C: Network Connection Guide.

For P1 Series: Network Connection Guide.

For H2 Series: Network Connection Guide

Error Code: 0500-0200-0002-0002

Failed to login device.

What it is

The HMS message is to notify you there is a problem when binding the printer. Normally, it is because:

1. Network is not good enough
2. The connection to the cloud service is not stable

1. Make the printer closer to the router and optimize network for better network performace.
2. Reboot the printer to make the printer login the cloud.
3. Unbind the user and bind again.

Error Code: 0500-0200-0002-0004

Unauthorized user.

What it is

This HMS message indicates a problem when getting user information that is bound to the printer or there is no correct user information when binding the printer.

1. Login on Studio or Handy
2. Unbind the user and bind

Error message

HMS_0500-0200-0002-0004: Unauthorized user.
0500-0200-0002-0004

Error Code: 0500-0200-0002-0006

Liveview service is malfunctioning.

Issue Description

Liveview helps users remotely video monitor printers with Studio and Handy. The video stream is transmitted through WiFi, connected to the local router, and then relayed over the public or local network to your Studio or Handy device.

The liveview malfunction may occur with black screen and errors at the screen bottom right corner when:

1. There is problem when initialize liveview service.
2. There is a problem when creating a liveview connection between the printer and client (Handy or Studio).
3. The account information is not correct.

Troubleshooting

The following troubleshooting process and actions aim to solve the most frequent issues, which are:

- Initialization failed error
- Loading failed error
- Connecting error
- Black screen

If the issue you met is not listed in this wiki, please jump to the LiveView troubleshooting details or submit a customer support ticket.

- Tweezers (Flat)
- Silicone glue

Fault Isolation Procedure

This chart illustrates the general troubleshooting procedure, including critical judgments, checks and actions.

Actions

Operators may use discretion in determining the actual procedures used, and the order in which these steps are applied.

- Upgrade or reinstall Handy

Handy download

Handy quick start guide

- Studio troubleshooting

LiveView troubleshooting

- Memory overflow check Test by printing a small model, such as the Bed scraper model that comes with the machine. If the Liveview is good in the test, reset printer and format SD card. If the Liveview is malfunctioning in the test, upgrade to the latest firmware.
- Test by printing a small model, such as the Bed scraper model that comes with the machine.
- If the Liveview is good in the test, reset printer and format SD card.
- If the Liveview is malfunctioning in the test, upgrade to the latest firmware.
- Reset printer Restore to factory settings Update LAN-only access ID when you are using LAN Mode Liveview
- Restore to factory settings
- Update LAN-only access ID when you are using LAN Mode Liveview

1. Test by printing a small model, such as the Bed scraper model that comes with the machine.
2. If the Liveview is good in the test, reset printer and format SD card.
3. If the Liveview is malfunctioning in the test, upgrade to the latest firmware.

1. Restore to factory settings
2. Update LAN-only access ID when you are using LAN Mode Liveview

- Power cycle the printer
- Open the front door to cool the compartment
- Wireless router connection check Link printer with mobile 2.4G hotpot and check liveview in the Studio. If liveview with hotspot is normal, reboot the wifi router, move the printer closer to the Wifi router, or upgrade your home network bandwidth as nessary.
- Link printer with mobile 2.4G hotpot and check liveview in the Studio.
- If liveview with hotspot is normal, reboot the wifi router, move the printer closer to the Wifi router, or upgrade your home network bandwidth as nessary.

1. Link printer with mobile 2.4G hotpot and check liveview in the Studio.
2. If liveview with hotspot is normal, reboot the wifi router, move the printer closer to the Wifi router, or upgrade your home network bandwidth as nessary.

- Wi-Fi communication check Check wifi signal strength, If there are only 1 to 2 bars, it indicates a relatively poor Wi-Fi connection. Check cable connection between Wifi antenna and AP board(main board for A series), re-insert the connector if it is loose. X Series, P Series, A1 mini, A1, H2 Series Replace the wifi antenna if it was damaged.
- Check wifi signal strength, If there are only 1 to 2 bars, it indicates a relatively poor Wi-Fi connection.
- Check cable connection between Wifi antenna and AP board(main board for A series), re-insert the connector if it is loose. X Series, P Series, A1 mini, A1, H2 Series
- Replace the wifi antenna if it was damaged.
- Camera connection check Printer power off. Check cable connection and re-insert as necessary. A1 Series, X1 Series, P1 Series, H2 Series Replace liveview camera or board if any broken or damage.
- Printer power off.
- Check cable connection and re-insert as necessary. A1 Series, X1 Series, P1 Series, H2 Series
- Replace liveview camera or board if any broken or damage.

1. Check wifi signal strength, If there are only 1 to 2 bars, it indicates a relatively poor Wi-Fi connection.
2. Check cable connection between Wifi antenna and AP board(main board for A series), re-insert the connector if it is loose. X Series, P Series, A1 mini, A1, H2 Series
3. Replace the wifi antenna if it was damaged.

1. Printer power off.
2. Check cable connection and re-insert as necessary. A1 Series, X1 Series, P1 Series, H2 Series
3. Replace liveview camera or board if any broken or damage.

Error Code: 0500-0200-0002-0008

HMS_0500-0200-0002-0008 Time synchronization failed.

Observation

The device cannot refresh the QR code or the screen shows a "Time synchronization failed" error once the device fails to synchronize the time. The issue is because the device failed to access the time synchronization service through the network, generally due to poor network conditions.

Troubleshooting

The following troubleshooting procedures and actions aim to isolate the issue's root cause and guide you to solve it step by step.

- PC with Windows operation system
- Network router for X1 series printer except X1E

Fault Isolation Procedure

This chart illustrates the general troubleshooting procedure, including critical judgments, and checks and actions.

Actions

Operators may use discretion in determining the actual procedures used, and the order in which these steps are applied.

- Change the hotspot or network router, and then reboot the printer Link printer with mobile 2.4G hotpot.

  Or change with another network router. Reboot the printer.
- Link printer with mobile 2.4G hotpot.
- Or change with another network router.
- Reboot the printer.
- NTP server conection test Search “CMD” in the Windows start menu for Command Prompt. Execute the command, "w32tm /stripchart /computer:Domain name /dataonly /samples:5", the domain name could be anyone in the below list. Domain name pool.ntp.org 0.pool.ntp.org 1.pool.ntp.org 2.pool.ntp.org 3.pool.ntp.org If there is a time synchronization delay output in your screen, the connect test is passed. Otherwise, you will see the errors. Take a screenshot of the connection test result.
- Search “CMD” in the Windows start menu for Command Prompt.
- Execute the command, "w32tm /stripchart /computer:Domain name /dataonly /samples:5", the domain name could be anyone in the below list.
- If there is a time synchronization delay output in your screen, the connect test is passed. Otherwise, you will see the errors.
- Take a screenshot of the connection test result.
- Modify the DNS server As for X1E, X1E Network Connection Guide. As for X1C, please prepare a network router and set the DNS server of the router to anyone in the table below. Then, connect your printer to this router. District Provider Primary DNS Secondary DNS Global Google 8.8.8.8 8.8.4.4 Global Control D 76.76.2.0 76.76.10.0 Global Quad9 9.9.9.9 149.112.112.112 Global OpenDNS Home 208.67.222.222 208.67.220.220 Global Cloudflare 1.1.1.1 1.0.0.1 Global AdGuard DNS 94.140.14.14 94.140.15.15 Global CleanBrowsing 185.228.168.9 185.228.169.9 China Mainland 114 114.114.114.114 114.114.114.115 China Mainland Ali DNS 223.5.5.5 223.5.5.6
- As for X1E, X1E Network Connection Guide.
- As for X1C, please prepare a network router and set the DNS server of the router to anyone in the table below. Then, connect your printer to this router.
- Manually Set DNS

Change the hotspot or network router, and then reboot the printer

1. Link printer with mobile 2.4G hotpot.
2. Or change with another network router.
3. Reboot the printer.

NTP server conection test

1. Search “CMD” in the Windows start menu for Command Prompt.
2. Execute the command, "w32tm /stripchart /computer:Domain name /dataonly /samples:5", the domain name could be anyone in the below list.

1. If there is a time synchronization delay output in your screen, the connect test is passed. Otherwise, you will see the errors.
2. Take a screenshot of the connection test result.

Modify the DNS server

1. As for X1E, X1E Network Connection Guide.
2. As for X1C, please prepare a network router and set the DNS server of the router to anyone in the table below. Then, connect your printer to this router.

Manually Set DNS

Error Code: 0500-0300-0001-0001

The MC module is malfunctioning. Please restart the device.

What it is

The MC module is malfunctioning, and the root cause could be:

1. Communication between AP and MC does not work;
2. The data sent from MC to AP is incorrect.

When this issue happens, all operations related to MC will malfunction, such as:

1. Move x/y/z;
2. Set the temperature of hotend and heatbed.

Troubleshooting

Check the communication between AP and MC and check if MC board works well.

Operation Guide for X1/P1 Series Printers

Depending on your printer model, re-plug the connection cable between the AP board and MC board. If the issue persists, the MC-AP connection cable, MC board, or AP board may be faulty. Please follow the guides below for replacement. If the problem still exists, contact the Bambu Lab customer support team.

- Replace MC-AP Cable for X1 Series Printers
- Replace MC Board for X1 Series Printers
- Replace AP Board for X1 Series Printers

- Replace MC-AP Cable for P1 Series Printers
- Replace MC Board for P1 Series Printers
- Replace AP Mainboard for P1 Series Printers

Operation Guide for A1 Series Printers

For A1 series printers, the AP and MC boards are integrated into the Mainboard. Check if the Mainboard is working properly. If the Mainboard is malfunctioning, follow the guide below for replacement. If the problem persists, contact the Bambu Lab customer support team.

A1

Replace Mainboard for A1 Printer

A1 mini

Replace MC Board for A1 mini Printer

Error Code: 0500-0300-0001-0002

The toolhead is malfunctioning. Please restart the device.

What it is

The Tool head module is malfunctioning and suggest rebooting the device.

1. Communication between the AP and the MC board is abnormal
2. Communication between the MC and the TH board is abnormal

When this issue happens, all operations related to MC will malfunction, such as:

1. Set the temperature of the hot end
2. The extruder does not work

Troubleshooting

Check the communication between the AP and TH board, including AP->MC, MC->TH.

X1 Series Printer

Replacing the toolhead TH board assembly | Bambu Lab Wiki
Replacing the AP board / Application Board | Bambu Lab Wiki
USB-C Cable Connection Issue | Bambu Lab Wiki

For the X1 printer, when installing the USB-C cable on the AP side, ensure that the letter 'A' faces outward. This can also be interpreted as the plastic protrusion being positioned at the front; otherwise, it may lead to communication issues and errors.

P1 Series Printer

Toolhead cable | Bambu Lab Wiki
PCBs on the toolhead of P1P | Bambu Lab Wiki

MC-AP cable installation tutorial

AP board installation tutorial

For P1 series printer, please check if the FPC is correctly inserted as shown in the picture below:

Besides, the MC-TH connector is directional, so please ensure the socket is inserted correctly. If inserted in reverse, it will cause a power short circuit, resulting in the printer reporting an HMS error or inability to power on properly.

Please also check whether the seat of the tool head cable is soldered. Please contact our after-sales service if the seat is floating and falsely welded.

Note: When installing the heat sink of the MC board, make sure the direction of the heat sink is correct. Installing the heat sink in reverse may result in direct contact between the heat sink and the 24V power supply, leading to a short circuit.

A1 mini

Please check if the USB-C cable connections on both ends are secure. You can re-plug the cable a few times to ensure a stable connection.

Please refer to this link to remove the base plate and check the USB-C cable connection on the mainboard. Note: There is no need to cut any zip ties on the cable.

A1 mini installation tutorial

A1

Check whether the USB-C plug under the base is fully inserted. Slide the cable management box upwards until the Type-C cable is securely inserted into its slot. You can refer to Step 10 in this wiki for details: Unboxing A1 combo | Bambu Lab Wiki

If the error persists, check whether the USB-C cables at both ends are properly connected and reinsert them several times to ensure that the connection is stable.

H2D

Refer to Wiki - Replacing the AP Board. Remove the AP board cover and inspect the latch of the MC-TH cable connector to see if it is loose. If so, please reconnect it with the power off.

P2S

If you are having communication problems between the Toolboard and the MC board, follow the next steps:

Let's start by verifying if the USB cable between the toolboard and the AP board is fully connected and doesn't present damage signals.

- H1.5 Allen key

& þ IMPORTANT: IMPORTANT! Always power off and disconnect power on the printer before performing maintenance work. Not doing so means there is a risk of electric shock, short circuit, and damage to the printer or surrounding area. When a maintenance task necessitates the printer being powered on, use insulated gloves for safety and pay special care not to pinch, damage, or put pressure on any exposed wires, connectors, or circuit boards. Additionally, the nozzle can be extremely hot so never touch it with exposed skin. If you have any questions or concerns related to the above or the steps in this guide, please open a new ticket in our Support Page for assistance.

IMPORTANT!

Always power off and disconnect power on the printer before performing maintenance work. Not doing so means there is a risk of electric shock, short circuit, and damage to the printer or surrounding area.

When a maintenance task necessitates the printer being powered on, use insulated gloves for safety and pay special care not to pinch, damage, or put pressure on any exposed wires, connectors, or circuit boards. Additionally, the nozzle can be extremely hot so never touch it with exposed skin.

If you have any questions or concerns related to the above or the steps in this guide, please open a new ticket in our Support Page for assistance.

Step 1: Remove the Toolhead Front Cover

Open the Toolhead Front Cover, press the connector latch to unplug the cable, and then remove the Toolhead Front Cover.

Step 2: Remove the Toolhead Rear Cover

Remove the four screws with an H1.5 Allen key.

Grip the Rear Cover near the upper rod, then pull it backward firmly to detach it.

Step 3: Check the TH board connections

Ensure that the USB cable (6) and the FPC cable and securely connected. These are the main common failure points for the errors your machine is displaying. Since we are here, take the opportunity to check if all other connectors are fully inserted and secure.

If any cable looks damaged, it will need to be replaced.

Step 4: Install the Toolhead rear cover back on

Install the Toolhead Rear Cover, then secure it with four screws.

Step 5: Install the Toolhead Front Cover

Connect the Toolhead Front Cover cable to the Extruder Connection Board connector.

The Toolhead Front Cover Assembly is magnetically secured. To reinstall, simply close it into position.

Step 6: Check the MC board connection

Refer to the Rear panel replacement guide and follow the Removing the Rear Panel section to get access to the MC board.

Step 7: Disconnect and reconnect the Toolhead USB cable

Disconnect the toolhead cable from the MC board, and plug it back in. Gently try to pull it out to verify if the connection is secure.

Step 8: Check Operation

Power on the machine and check if the error still appears.

- If the issue is resolved. Disconnect the machine's power and follow the Intalling the Rear Panel section.
- If the issue still remains, the problem might lie elsewhere, such as the toolboard it self or the MC Board, we advise contacting the Bambu Lab Support team.

Error Code: 0500-0300-0001-0003

The AMS module is malfunctioning. Please restart the device.

What it is

The AMS module is malfunctioning. Please restart the device.

Troubleshooting

To be added.

Error message

HMS_0500-0300-0001-0003: The AMS module is malfunctioning. Please restart the device.

Error Code: 0500-0300-0001-0004

The Filament Buffer module is malfunctioning. Please restart the device.

Issue Description

When the MC board fails to retrieve filament buffer information, an error indicating a buffer module malfunction will be displayed.

Possible Causes:

- Software issue
- Buffer mainboard failure
- MC board failure
- Loose connection between the MC board and the buffer

Troubleshooting

1. H2.0 Allen key
2. H1.5 Allen key

H2.0 Allen key

H1.5 Allen key

Troubleshooting Steps

Step 1. Restart the Printer

If the error persists, proceed to Step 2.

Step 2. Disconnect the AMS

If the error persists, proceed to Step 3.

Step 3. Check and Reconnect the Cable Between the Buffer and MC Board

Step 4. Replace the Buffer and 4-Pin Cable

If the issue remains unresolved, replace the buffer and the 4-pin cable.

Step 5. Replace the MC Board

If the issue still persists, please refer to the Wiki to replace the MC board.

Error Code: 0500-0300-0001-000A

HMS_0500-0300-0001-000A: System state is abnormal. Please restore factory settings.

What it is

This HMS message is notified when the system state is abnormal, it may be recovered with a factory resetting.

1. Reboot the printer to check if the issue can be recovered.
2. Restore factory settings
3. if the issue is still there, after step 2, please contact the support team.

Note: With firmware versions prior to 01.05.01.00, if Bambu Studio is set to some non-English language, the sliced file may cause the LiDAR and software to be incompatible and not work properly and prompt this error. Upgrading the printer firmware to 01.05.01.00 or switching the language of Bambu Studio to English can solve this.

Error Code: 0500-0300-0001-000B

HMS_0500-0300-0001-000B: The screen is malfunctioning.

What it is

The HMS message is notified when the screen is malfunctioning. The screen may remain black, only APP and Studio can report this HMS.

Troubleshooting

Please take this guide as a reference to check this issue https://wiki.bambulab.com/en/x1/troubleshooting/black-flicker-screen

Error message

The HMS message is notified when the screen is malfunctioning.
0500-0300-0001-000B

Error Code: 0500-0300-0001-0021

Hardware incompatible, please check the laser.

What it is

Currently, the X1 series is supplied with two types of micro LiDAR and two types of TH board. They are not fully compatible. The printer would perfrom hardware configration compatibility check before each print, and raise errors if the check failed.

Troubleshooting

Step 1. Check hardware configration

Referees this wiki to check and record hardware serial number(SN).
https://wiki.bambulab.com/e/en/x1/manual/micro-lidar-model

Step 2. Check hardware compatibility

The compatibility matrix of TH board and Micro Lidar is as follows:

x represents any number
Y represents a letter after the letter D, such as E,F
-- not Compatible
" compatible

Step 3. Replace either the Micro-Laser or TH board assembly to meet the compatibility

https://wiki.bambulab.com/en/x1/maintenance/replace-micro-lidar
https://wiki.bambulab.com/en/x1/maintenance/replace-the-th-board-assembly

Error message

HMS_0500-0300-0001-0021: Hardware incompatible, please check the laser.
0500-0300-0001-0021

Error Code: 0500-0300-0002-000C

HMS_0500-0300-0002-000C: Wireless hardware error: please turn off/on WiFi or restart the device.

What it is

The HMS message is reported when AP detects hardware issues in wifi module.

1. Turn off/on wifi module on the screen to check if the issue can be recovered.
2. Restart the device to check if the issue can be recovered.

Error message

HMS_0500-0300-0002-000C: Wireless hardware error: please turn off/on WiFi or restart the device.
0500-0300-0002-000C

Error Codes: 0500-0300-0002-000E, 0500-0500-0001-000E, 0500-0500-0001-000F, 0500-0500-0001-0011, 0500-0500-0001-0012, 0500-0400-0001-0046, 0500-0400-0001-0047, 0500-0400-0001-0048, 0500-0400-0001-0044, 0501-0400-0001-0044, 0502-0400-0001-0044, 0503-0400-0001-0044, 0500-0500-0001-0010, 0501-0500-0001-0010, 0502-0500-0001-0010, 0503-0500-0001-0010, 0580-0500-0001-0010, 0581-0500-0001-0010, 0582-0500-0001-0010, 0583-0500-0001-0010, 0584-0500-0001-0010, 0585-0500-0001-0010, 0586-0500-0001-0010, 0587-0500-0001-0010, 0580-0400-0001-0045, 0581-0400-0001-0045, 0582-0400-0001-0045, 0583-0400-0001-0045, 0584-0400-0001-0045, 0585-0400-0001-0045, 0586-0400-0001-0045, 0587-0400-0001-0045

HMS_0500-0300-0002-000E: Some modules are incompatible with the printer's firmware version, which may affect use. Please go to the "Firmware" page to update after connected to the internet, or you may update offline according to wiki.

Issue Description

The printer will automatically detect its accessories. If there is a version mismatch between any accessory and the printer firmware, a warning will be triggered. Common accessories include: AMS, laser module, cutting module, etc.

Troubleshooting

Solutions

Keep the accessories connected to the printer and try the following upgrade methods:

Option 1. Upgrade via the Printer (Online)

Take H2 Series as example.

Option 2. Upgrade via Handy App

Option 3. Offline Upgrade

H2D and H2C
X1 and X1C
X1E
A1 mini
A1
P1 Series
P2S

Equivalent Codes

The following HMS codes all represent the same type of fault.

Error Code: 0500-0400-0001-0001

Failed to download print job. Please check your network connection.

What it is

There is a problem when downloading the printing 3MF file. Normally, the signal quality or bandwidth is not good enough.

1. Make the printer closer to the router
2. Run the network test tool to check network performance
3. Reprint the job

Error Code: 0500-0400-0001-0002

Failed to report print state. Please check your network connection.

What it is

There is a problem when reporting printing state to cloud service. Normally, it is because the signal quality or bandwidth is not good enough.
This HMS will not be detected when in LAN only mode.

1. Make the printer closer to router
2. Run network test tool to check network performance, and optimize it if there is problem

Error message

HMS_0500-0400-0001-0002: Failed to report print state. Please check your network connection.
0500-0400-0001-0002

Error Code: 0500-0400-0001-0003

The content of print file is unreadable. Please resend the print job.

What it is

The HMS message is triggered when the print file is unreadable or if there is a problem with it. Usually, this is due to file corruption during download.

1. Make the printer closer to the router.
2. Run the network test tool to check network performance, and optimize it if there is a problem.
3. Restart the print job.

Error Code: 0500-0400-0001-0004

The print file is unauthorized.

What it is

The HMS message is not used now.

Troubleshooting

None

Error message

HMS_0500-0400-0001-0004: The print file is unauthorized.
0500-0400-0001-0004

Error Code: 0500-0400-0001-0006

Failed to resume previous print.

What it is

There is a feature that the last printing job can be recovered after a power off and on if it finished first layer.
The HMS message is notified when there is a problem with the last printing job recovery. The possible reason is:

1. The context of the last printing job is not stored successfully when powered off.
2. There is power off again in the last recovery.

Troubleshooting

The only way is to keep the power supply stable and restart the printing job again.

Error Code: 0500-0400-0002-0007

The bed temperature exceeds the filament's vitrification temperature, which may cause a nozzle clog.

What Is This?

When the printer is fully enclosed with no air exchange between the chamber and the outside, the heatbed temperature will continuously heat the chamber environment. If the heatbed temperature exceeds the softening temperature of the filament, the chamber temperature after radiant heating can easily cause the filament to soften prematurely.

In the extruder, the filament is subjected to the squeezing force of the extruder gear. Once the filament has softened, the squeezing process will cause the filament to deform. Eventually, the irregular filament will cause the extruder to clog, commonly known as "heat creep".

Troubleshooting Methods

It is recommended to keep the printer's front door open or appropriately lower the heatbed temperature to prevent the chamber temperature from rising.

Error Code: 0500-0500-0001-0007

MQTT Command verification failed, please update Studio or Handy.

Issue Description

MQTT command verification failed. Please update Studio or Handy.
Starting from firmware versions 01.08.03.00beta/01.08.05.00, the X1 series printers have introduced an authorization and authentication protection mechanism, aiming to comprehensively enhance the security of Bambu products. To ensure compatibility between the printer firmware and Bambu Studio/Bambu Handy, it is necessary to upgrade Studio and Handy to the specified versions.

- Upgrade Bambu Studio

  Please refer to the Bambu Studio Quick Start Guide to download and install version 01.10.02.64 or above.
- Upgrade Bambu Handy

  Please refer to the Bambu Handy Quick Start Guide to download and install version 2.17.0 or above.

Additionally, we have developed an optional developer mode feature for Bambu Lab printers. When this mode is enabled, the authorization and authentication functions are disabled, and the printer will accept all control commands without verifying their source. How to use developer mode on Bambu Lab printers?

Error Code: 0700-0100-0001-0001

The AMS assist motor has slipped.The extrusion wheel may be worn down,or the filament may be too thin.

Error message

HMS_0701-0100-0001-0001: The AMS2 assist motor has slipped. The extrusion wheel may be worn down, or the filament may be too thin.

HMS_0702-0100-0001-0001: The AMS3 assist motor has slipped. The extrusion wheel may be worn down, or the filament may be too thin.

HMS_0703-0100-0001-0001: The AMS4 assist motor has slipped. The extrusion wheel may be worn down, or the filament may be too thin.

HMS_0700-0100-0001-0001: The AMS1 assist motor has slipped. The extrusion wheel may be worn down, or the filament may be too thin.

Error Code: 0700-0100-0001-0003

The AMS assist motor torque control is malfunctioning.The current sensor may be faulty.

Error message

HMS_0701-0100-0001-0003: The AMS2 assist motor torque control is malfunctioning. The current sensor may be faulty.

HMS_0702-0100-0001-0003: The AMS3 assist motor torque control is malfunctioning. The current sensor may be faulty.

HMS_0703-0100-0001-0003: The AMS4 assist motor torque control is malfunctioning. The current sensor may be faulty.

HMS_0700-0100-0001-0003: The AMS1 assist motor torque control is malfunctioning. The current sensor may be faulty.

Error Code: 0700-0100-0001-0004

The AMS assist motor speed control is malfunctioning.The speed sensor may be faulty.

What it is

HMS_0701-0100-0001-0004: The AMS2 assist motor speed control is malfunctioning. The speed sensor may be faulty.

HMS_0702-0100-0001-0004: The AMS3 assist motor speed control is malfunctioning. The speed sensor may be faulty.

HMS_0703-0100-0001-0004: The AMS4 assist motor speed control is malfunctioning. The speed sensor may be faulty.

HMS_0700-0100-0001-0004: The AMS1 assist motor speed control is malfunctioning. The speed sensor may be faulty.

Error Codes: 0700-0100-0002-0002, 0701-0100-0002-0002, 0702-0100-0002-0002, 0703-0100-0002-0002

The AMS assist motor is overloaded,The filament may be tangled or stuck.

What it is

HMS_0700-0100-0002-0002: The AMS assist motor is overloaded,The filament may be tangled or stuck.
During the printing process, the motor of the filament hub of AMS assists in feeding the filament to the printer, reducing the resistance of the tool head extruder to extrude the filament, and the motor will detect the resistance of the filament, when the resistance continues to be too large and the speed of filament is too small, it will prompt overload error. Then the filament will return to AMS and printing will be paused.

This error is usually caused by excessive resistance due to the following reasons:

1. filament tangled or sagging and stuck
2. the spool size exceeds AMS specifications or the cardboard spool is used
3. the feed funnel of the first-stage feeder is seriously worn
4. the roller support of the first-stage feeder was broken
5. the spring of the AMS buffer is stuck
6. the PTFE tube inside AMS is worn
7. the PTFE tube between AMS and printer is overbending

Safety warning and Machine state before starting operation

None

Operation guide

Step 1

Please refer to this wiki to check if the spool used are within the AMS specifications, check if the width of the spool is between 50~68mm and the diameter is between 197~202mm; some spools may be deformed, please rotate the spool back and forth to check whether there is too much resistance at any position.

Step 2

Check if the filament is tangled or sagged and stuck, and avoid using filaments that are not supported by AMS, such as TPU/damp PVA.

Step 3

Check if the feed funnel of the first-stage feeder is seriously worn, If there is wear, please refer to the wiki for replacement.

Step 4

Check if the spring of the AMS buffer or Hub is stuck:

Step 5

Please refer to this wiki to disassemble AMS and check if the PTFE tube inside AMS is worn.

Step 6

Check if the roller support is broken, and refer to this wiki to replace the main frame.

Step 7

Check if the PTFE tube between AMS and printer is overbending.

Step 8

Check whether the bearing of the AMS Active Support Shaft Assembly is too close to the black rubber, which leads to high resistance to rotation of the AMS Active Support Shaft Assembly. You can hold one end of the bearing and then rotate the rubber to see whether it is obvious to feel the resistance. You can also directly exchange the position with other AMS Active Support Shaft Assembly to see if the problem is transferred with the AMS Active Support Shaft Assembly.

Error message:

HMS_0700-0100-0002-0002: The AMS1 assist motor is overloaded. The filament may be tangled or stuck.

HMS_0701-0100-0002-0002: The AMS2 assist motor is overloaded. The filament may be tangled or stuck.

HMS_0702-0100-0002-0002: The AMS3 assist motor is overloaded. The filament may be tangled or stuck.

HMS_0703-0100-0002-0002: The AMS4 assist motor is overloaded. The filament may be tangled or stuck.

Error Code: 0700-0200-0001-0001

AMS filament speed and length error.The filament odometry may be faulty.

What it is

HMS_0701-0200-0001-0001: AMS2 Filament speed and length error: The filament odometry may be faulty.

HMS_0702-0200-0001-0001: AMS3 Filament speed and length error: The filament odometry may be faulty.

HMS_0703-0200-0001-0001: AMS4 Filament speed and length error: The filament odometry may be faulty.

HMS_0700-0200-0001-0001: AMS1 Filament speed and length error: The filament odometry may be faulty.

Error Code: 0700-1000-0001-0001

The AMS slot1 motor has slipped. The extrusion wheel may be malfunctioning,or the filament may be too thin.

What it is

HMS_0700-1000-0001-0001:The AMS slot1 motor has slipped. The extrusion wheel may be malfunctioning,or the filament may be too thin.
0700-1000-0001-0001

Error Code: 0700-1000-0001-0003

HMS_0700_1000_0001_0003: The AMS slot1 motor torque control is malfunctioning.The current sensor may be faulty.

What it is

HMS_0700_1000_0001_0003: The AMS slot1 motor torque control is malfunctioning.The current sensor may be faulty.
0700-1000-0001-0003

Error Code: 0700-1000-0002-0002

The AMS slot1 motor is overloaded.The filament may be tangled or stuck.

What it is

HMS_0700-1000-0002-0002: The AMS slot1 motor is overloaded.The filament may be tangled or stuck.
0700-1000-0002-0002

Error Codes: 0700-2000-0002-0001, 0700-2100-0002-0001, 0700-2200-0002-0001, 0700-2300-0002-0001, 0701-2000-0002-0001, 0701-2100-0002-0001, 0701-2200-0002-0001, 0701-2300-0002-0001, 0702-2000-0002-0001, 0702-2100-0002-0001, 0702-2200-0002-0001, 0702-2300-0002-0001, 0703-2000-0002-0001, 0703-2100-0002-0001, 0703-2200-0002-0001, 0703-2300-0002-0001, 0704-2000-0002-0001, 0704-2100-0002-0001, 0704-2200-0002-0001, 0704-2300-0002-0001, 0705-2000-0002-0001, 0705-2100-0002-0001, 0705-2200-0002-0001, 0705-2300-0002-0001, 0706-2000-0002-0001, 0706-2100-0002-0001, 0706-2200-0002-0001, 0706-2300-0002-0001, 0707-2000-0002-0001, 0707-2100-0002-0001, 0707-2200-0002-0001, 0707-2300-0002-0001

AMS1 Slot1 filament has run out.

What it is

HMS_0700-2000-0002-0001: AMS1 Slot1 filament has run out.

This warning appears when the filament is used up in one of the slots during printing, i.e. the filament detection switch of the extruder detects that there is no filament left. You can refer to this wiki about the working principle of the filament detection switch of the extruder: HMS_0700_2000_0002_0004: The AMS slot1's filament may be broken in the tool head | Bambu Lab Wiki

Safety warning and Machine state before starting operation

Operation guide

Step1

If you see that the spool's filament in the slot runs out, please insert the required filament and press the “Retry” button on the screen/APP/Slicer to continue.

Step2

If the message still shows up, please contact the after-sales team.

Error message:

Error Codes: 0700-2000-0002-0002, 0700-2100-0002-0002, 0700-2200-0002-0002, 0700-2300-0002-0002, 0701-2000-0002-0002, 0701-2100-0002-0002, 0701-2200-0002-0002, 0701-2300-0002-0002, 0702-2000-0002-0002, 0702-2100-0002-0002, 0702-2200-0002-0002, 0702-2300-0002-0002, 0703-2000-0002-0002, 0703-2100-0002-0002, 0703-2200-0002-0002, 0703-2300-0002-0002, 0704-2000-0002-0002, 0704-2100-0002-0002, 0704-2200-0002-0002, 0704-2300-0002-0002, 0705-2000-0002-0002, 0705-2100-0002-0002, 0705-2200-0002-0002, 0705-2300-0002-0002, 0706-2000-0002-0002, 0706-2100-0002-0002, 0706-2200-0002-0002, 0706-2300-0002-0002, 0707-2000-0002-0002, 0707-2100-0002-0002, 0707-2200-0002-0002, 0707-2300-0002-0002

The AMS A slot 1 is empty.

What it is

HMS_0700-2000-0002-0002: The AMS slot1 is empty.
This error occurs when the AMS starts to feed the filament of one slot, but this slot is empty, and the light of the corresponding slot will turn red with slow breathing.

Each slot of AMS has two sensors to detect the position of the filament, one sensor is at the first-stage feeder and the other is at the filament hub, as shown below:

When AMS detects that both sensors are “OFF”, it considers that the slot is empty.

Below are the possible reasons why this HMS error occurs:

1. The slot is empty;
2. The sensor of the first-stage feeder is abnormal.

Safety warning and Machine state before starting operation

Operation guide

Step1

Find the slot with red light and if this slot is empty, please insert the required filament and press “Retry” button on the screen/APP/Slicer to continue.

[picture]

Step2

If this slot already has filament but still shows a red light,the sensor of first-stage feeder may be abnormal.

AMS:

Please refer to steps 1~3 of this Wiki to disassemble the AMS:Replacing the AMS first stage feeder | Bambu Lab Wiki

and check if the cable of first-stage feeder plugged in tightly:

AMS 2 Pro:

Please refer to steps 1~4 of this Wiki to disassemble the AMS 2 Pro:AMS 2 Pro Disassembly and Assembly Guide

and check if the cable of first-stage feeder plugged in tightly:

- Replace AMS 2 Pro Feeder Unit

If the sensor returns to normal, the red light on the slot will change from slow breathing to single flashing,then please press “Retry” button on the screen/APP/Slicer to continue.

Step3

if the problem still exist, the sensor may be abnormal due to filament debris,which usually come from filaments that are easy to grind.please use a brush to clean the debris near the sensor.

Step4

If the message still shows up, contact the after-sales team.

Error message:

"The AMS[n] slot[x] is empty.." n=1,2,3,4 corresponding to 4 AMS,x indicates the number of slot.

These error code below share the same issue as this one:

0700-2000-0002-0002

0700-2100-0002-0002

0700-2200-0002-0002

0700-2300-0002-0002

0701-2000-0002-0002

0701-2100-0002-0002

0701-2200-0002-0002

0701-2300-0002-0002

0702-2000-0002-0002

0702-2100-0002-0002

0702-2200-0002-0002

0702-2300-0002-0002

0703-2000-0002-0002

0703-2100-0002-0002

0703-2200-0002-0002

0703-2300-0002-0002

Error Codes: 0700-2000-0002-0003, 0700-2100-0002-0003, 0700-2200-0002-0003, 0700-2300-0002-0003, 0701-2000-0002-0003, 0701-2100-0002-0003, 0701-2200-0002-0003, 0701-2300-0002-0003, 0702-2000-0002-0003, 0702-2100-0002-0003, 0702-2200-0002-0003, 0702-2300-0002-0003, 0703-2000-0002-0003, 0703-2100-0002-0003, 0703-2200-0002-0003, 0703-2300-0002-0003, 0704-2000-0002-0003, 0704-2100-0002-0003, 0704-2200-0002-0003, 0704-2300-0002-0003, 0705-2000-0002-0003, 0705-2100-0002-0003, 0705-2200-0002-0003, 0705-2300-0002-0003, 0706-2000-0002-0003, 0706-2100-0002-0003, 0706-2200-0002-0003, 0706-2300-0002-0003

The AMS A Slot 1 filament may be broken in AMS.

What it is

HMS_0700-2000-0002-0003: The AMS slot1's filament may be broken in AMS.
Each slot of AMS has two sensors to detect the position of the filament, one sensor is at the first-stage feeder and the other is at the filament hub,as shown below:

Some brittle filaments may break inside AMS and it usually causes the hub sensor at “ON” state and the feeder sensor at “OFF” state, and AMS will pop up with the message “...filament may be broken in AMS”, the light in this slot will also turn red.

Safety warning and Machine state before starting operation

Operation guide

step1

Please refer to this video to remove the broken filament:

AMS:

- AMS 2 Pro Disassembly and Assembly Guide

Error message:

"The AMS slot[n] filament may be broken in the PTFE tube."

The error codes below share the same issue as this one:

HMS_0700-2000-0002-0003: AMS1 Slot1's filament may be broken in AMS.

HMS_0700-2100-0002-0003: AMS1 Slot2's filament may be broken in AMS.

HMS_0700-2200-0002-0003: AMS1 Slot3's filament may be broken in AMS.

HMS_0700-2300-0002-0003: AMS1 Slot4's filament may be broken in AMS.

HMS_0701-2000-0002-0003: AMS2 Slot1's filament may be broken in AMS.

HMS_0701-2100-0002-0003: AMS2 Slot2's filament may be broken in AMS.

HMS_0701-2200-0002-0003: AMS2 Slot3's filament may be broken in AMS.

HMS_0701-2300-0002-0003: AMS2 Slot4's filament may be broken in AMS.

HMS_0702-2000-0002-0003: AMS3 Slot1's filament may be broken in AMS.

HMS_0702-2100-0002-0003: AMS3 Slot2's filament may be broken in AMS.

HMS_0702-2200-0002-0003: AMS3 Slot3's filament may be broken in AMS.

HMS_0702-2300-0002-0003: AMS3 Slot4's filament may be broken in AMS.

HMS_0703-2000-0002-0003: AMS4 Slot1's filament may be broken in AMS.

HMS_0703-2100-0002-0003: AMS4 Slot2's filament may be broken in AMS.

HMS_0703-2200-0002-0003: AMS4 Slot3's filament may be broken in AMS.

HMS_0703-2300-0002-0003: AMS4 Slot4's filament may be broken in AMS.

Error Codes: 0700-2000-0002-0004, 0700-2100-0002-0004, 0700-2200-0002-0004, 0700-2300-0002-0004, 0701-2000-0002-0004, 0701-2100-0002-0004, 0701-2200-0002-0004, 0701-2300-0002-0004, 0702-2000-0002-0004, 0702-2100-0002-0004, 0702-2200-0002-0004, 0702-2300-0002-0004, 0703-2000-0002-0004, 0703-2100-0002-0004, 0703-2200-0002-0004, 0703-2300-0002-0004, 0704-2000-0002-0004, 0704-2100-0002-0004, 0704-2200-0002-0004, 0704-2300-0002-0004, 0705-2000-0002-0004, 0705-2100-0002-0004, 0705-2200-0002-0004, 0705-2300-0002-0004, 0706-2000-0002-0004, 0706-2100-0002-0004, 0706-2200-0002-0004, 0706-2300-0002-0004, 0707-2000-0002-0004, 0707-2100-0002-0004, 0707-2200-0002-0004, 0707-2300-0002-0004

The AMS slot1's filament may be broken in the toolhead.

What it is

HMS_0700-2000-0002-0004: The AMS slot 1's filament may be broken in the toolhead.

After unloading the filament, the filament has already been pulled back to AMS, but the detection switch of the extruder still detects the filament, and then the printer determines that the filament is broken in the extruder.

Possible causes:

1. A filament is broken near the detection switch and keeps it triggered.

2. The magnet’s torsion spring is stuck, preventing it from returning properly. As a result, the magnet remains close to the Hall sensor even without filament, causing a false detection.

X1/P1 detection switch position

H2D detection switch position

The H2D filament detection switches are located on the dual extruder idlers (on the side of the dual extruder unit).

Regardless of the model, the filament detection switch consists of a magnet and a Hall sensor.

Working principle (X1 as an example)

When the filament is inserted, the magnet moves closer to the Hall sensor and automatically springs back when there is no filament. The printer determines whether the filament is inside the extruder based on the detected Hall value.

The magnet moves closer to the Hall sensor after inserting the filament.

When the display screen shows a green dot as shown below, it indicates that the extruder has detected a filament inside. When there is no filament, it shows a white dot. So you can also judge whether there is a filament broken in the extruder by the dot color.

Safety warning and Machine state before starting operation

Always power off the printer before performing any maintenance.

Operation guide

1. Please refer to the links below to remove the filament detection switch and check if there is a broken filament. Also, please check whether the magnet of the detection switch can spring back smoothly.

Hall sensor replacement - P1 Series

Hall sensor replacement - X1 Series

Replace H2D Dual Extruder Idlers and Filament Sensor

If there is something wrong with the detection switch, please contact the customer support team for further help.

2. If you find the filament stuck inside the extruder, you need to remove the extruder for unclogging. You can refer to the videos in this wiki to check and unclog it:

Extruder Clog of X1/P1

How to Disassemble and Assemble H2D Extruder

3. If you find that the filaments are broken in the tube, please refer to Wiki: Filament break in the path.

Error message:

"The AMS slot[n] filament may be broken in the toolhead."

The error codes below share the same issue as this one:

0700-2000-0002-0004

0700-2100-0002-0004

0700-2200-0002-0004

0700-2300-0002-0004

0701-2000-0002-0004

0701-2100-0002-0004

0701-2200-0002-0004

0701-2300-0002-0004

0702-2000-0002-0004

0702-2100-0002-0004

0702-2200-0002-0004

0702-2300-0002-0004

0703-2000-0002-0004

0703-2100-0002-0004

0703-2200-0002-0004

0703-2300-0002-0004

Error Codes: 0700-2000-0002-0005, 0700-2100-0002-0005, 0700-2200-0002-0005, 0700-2300-0002-0005, 0701-2000-0002-0005, 0701-2100-0002-0005, 0701-2200-0002-0005, 0701-2300-0002-0005, 0702-2000-0002-0005, 0702-2100-0002-0005, 0702-2200-0002-0005, 0702-2300-0002-0005, 0703-2000-0002-0005, 0703-2100-0002-0005, 0703-2200-0002-0005, 0703-2300-0002-0005, 0704-2000-0002-0005, 0704-2100-0002-0005, 0704-2200-0002-0005, 0704-2300-0002-0005, 0705-2000-0002-0005, 0705-2100-0002-0005, 0705-2200-0002-0005, 0705-2300-0002-0005, 0706-2000-0002-0005, 0706-2100-0002-0005, 0706-2200-0002-0005, 0706-2300-0002-0005, 0707-2000-0002-0005, 0707-2100-0002-0005, 0707-2200-0002-0005, 0707-2300-0002-0005

AMS1 Slot1 filament has run out, and purging the old filament went abnormally; please check whether the filament is stuck in the tool head.

This page is under construction, coming soon!

Click here to open a new ticket in our Support Page.

We will do our best to respond promptly and provide you with the assistance you need.

What it is

Safety warning and Machine state before starting operation

Troubleshooting guide

Error message

HMS_0700-2000-0002-0005: AMS1 Slot1 filament has run out, and purging the old filament went abnormally; please check whether the filament is stuck in the tool head.

HMS_0700-2100-0002-0005: AMS1 Slot2 filament has run out, and purging the old filament went abnormally; please check whether the filament is stuck in the tool head.

HMS_0700-2200-0002-0005: AMS1 Slot3 filament has run out, and purging the old filament went abnormally; please check whether the filament is stuck in the tool head.

HMS_0700-2300-0002-0005: AMS1 Slot4 filament has run out, and purging the old filament went abnormally; please check whether the filament is stuck in the tool head.

HMS_0701-2000-0002-0005: AMS2 Slot1 filament has run out, and purging the old filament went abnormally; please check whether the filament is stuck in the tool head.

HMS_0701-2100-0002-0005: AMS2 Slot2 filament has run out, and purging the old filament went abnormally; please check whether the filament is stuck in the tool head.

HMS_0701-2200-0002-0005: AMS2 Slot3 filament has run out, and purging the old filament went abnormally; please check whether the filament is stuck in the tool head.

HMS_0701-2300-0002-0005: AMS2 Slot4 filament has run out, and purging the old filament went abnormally; please check whether the filament is stuck in the tool head.

HMS_0702-2000-0002-0005: AMS3 Slot1 filament has run out, and purging the old filament went abnormally; please check whether the filament is stuck in the tool head.

HMS_0702-2100-0002-0005: AMS3 Slot2 filament has run out, and purging the old filament went abnormally; please check whether the filament is stuck in the tool head.

HMS_0702-2200-0002-0005: AMS3 Slot3 filament has run out, and purging the old filament went abnormally; please check whether the filament is stuck in the tool head.

HMS_0702-2300-0002-0005: AMS3 Slot4 filament has run out, and purging the old filament went abnormally; please check whether the filament is stuck in the tool head.

HMS_0703-2000-0002-0005: AMS4 Slot1 filament has run out, and purging the old filament went abnormally; please check whether the filament is stuck in the tool head.

HMS_0703-2100-0002-0005: AMS4 Slot2 filament has run out, and purging the old filament went abnormally; please check whether the filament is stuck in the tool head.

HMS_0703-2200-0002-0005: AMS4 Slot3 filament has run out, and purging the old filament went abnormally; please check whether the filament is stuck in the tool head.

HMS_0703-2300-0002-0005: AMS4 Slot4 filament has run out, and purging the old filament went abnormally; please check whether the filament is stuck in the tool head.

Error Codes: 0700-2000-0002-0006, 0700-2100-0002-0006, 0700-2200-0002-0006, 0700-2300-0002-0006, 0701-2000-0002-0006, 0701-2100-0002-0006, 0701-2200-0002-0006, 0701-2300-0002-0006, 0702-2000-0002-0006, 0702-2100-0002-0006, 0702-2200-0002-0006, 0702-2300-0002-0006, 0703-2000-0002-0006, 0703-2100-0002-0006, 0703-2200-0002-0006, 0703-2300-0002-0006, 0704-2000-0002-0006, 0704-2100-0002-0006, 0704-2200-0002-0006, 0704-2300-0002-0006, 0705-2000-0002-0006, 0705-2100-0002-0006, 0705-2200-0002-0006, 0705-2300-0002-0006, 0706-2000-0002-0006, 0706-2100-0002-0006, 0706-2200-0002-0006, 0706-2300-0002-0006, 0707-2000-0002-0006, 0707-2100-0002-0006, 0707-2200-0002-0006, 0707-2300-0002-0006

PTFE tube detached during the feeding process. Please check if the PTFE tube is connected from AMS to extruder.

Observation

Under the drive of AMS, the filament is delivered to the toolhead through multiple sections of Teflon tubes. During the transportation process, various sensors detect whether the filament passes through. This alarm is triggered when the printer finds that the sensor cannot always detect the filament.

Troubleshooting

- None

Actions

You can try feeding or re-initiating printing to check whether the filament can be delivered to the toolhead. If the filament comes out from the interface, please reconnect the AMS to the buffer, buffer to the printer back, and printer back to the toolhead, ensuring that the Teflon tube is inserted into the pneumatic connector.

If the issue persists after troubleshooting, please submit a customer support ticket.

Equivalent Codes

The following HMS codes represent the same type of fault.

0700-2000-0002-0006

0701-2000-0002-0006

0702-2000-0002-0006

0703-2000-0002-0006

Error Codes: 0700-2000-0003-0001, 0700-2100-0003-0001, 0700-2200-0003-0001, 0700-2300-0003-0001, 0701-2000-0003-0001, 0701-2100-0003-0001, 0701-2200-0003-0001, 0701-2300-0003-0001, 0702-2000-0003-0001, 0702-2100-0003-0001, 0702-2200-0003-0001, 0702-2300-0003-0001, 0703-2000-0003-0001, 0703-2100-0003-0001, 0703-2200-0003-0001, 0703-2300-0003-0001

AMS1 Slot1 filament has run out. Please wait while old filament is purged.

What it is

HMS_0700_2000_0003_0001: AMS1 Slot1 filament has run out. Please wait while old filament is purged.
When switching filaments during multicolor printing, if one slot is about to run out of filament, there is still a section of filament left in the PTFE tube. At this point, the filament can no longer be pulled back so the remaining filament will be flushed. After flushing, the AMS switches to the next slot to load the filament.

In this process, the filament detection sensor of the first stage feeder is used to detect whether the filament in the slot has been used up.

Operation guide

You just need to wait for it to flush the old filament and insert a new one when the flushing is done.

If anything goes wrong, please contact the customer support team.

Error message:

HMS_0700-2000-0003-0001: AMS1 Slot1 filament has run out. Please wait while old filament is purged.

HMS_0700-2100-0003-0001: AMS1 Slot2 filament has run out. Please wait while old filament is purged.

HMS_0700-2200-0003-0001: AMS1 Slot3 filament has run out. Please wait while old filament is purged.

HMS_0700-2300-0003-0001: AMS1 Slot4 filament has run out. Please wait while old filament is purged.

HMS_0701-2000-0003-0001: AMS2 Slot1 filament has run out. Please wait while old filament is purged.

HMS_0701-2100-0003-0001: AMS2 Slot2 filament has run out. Please wait while old filament is purged.

HMS_0701-2200-0003-0001: AMS2 Slot3 filament has run out. Please wait while old filament is purged.

HMS_0701-2300-0003-0001: AMS2 Slot4 filament has run out. Please wait while old filament is purged.

HMS_0702-2000-0003-0001: AMS3 Slot1 filament has run out. Please wait while old filament is purged.

HMS_0702-2100-0003-0001: AMS3 Slot2 filament has run out. Please wait while old filament is purged.

HMS_0702-2200-0003-0001: AMS3 Slot3 filament has run out. Please wait while old filament is purged.

HMS_0702-2300-0003-0001: AMS3 Slot4 filament has run out. Please wait while old filament is purged.

HMS_0703-2000-0003-0001: AMS4 Slot1 filament has run out. Please wait while old filament is purged.

HMS_0703-2100-0003-0001: AMS4 Slot2 filament has run out. Please wait while old filament is purged.

HMS_0703-2200-0003-0001: AMS4 Slot3 filament has run out. Please wait while old filament is purged.

HMS_0703-2300-0003-0001: AMS4 Slot4 filament has run out. Please wait while old filament is purged.

Error Codes: 0700-2000-0003-0002, 0700-2100-0003-0002, 0700-2200-0003-0002, 0700-2300-0003-0002, 0701-2000-0003-0002, 0701-2100-0003-0002, 0701-2200-0003-0002, 0701-2300-0003-0002, 0702-2000-0003-0002, 0702-2100-0003-0002, 0702-2200-0003-0002, 0702-2300-0003-0002, 0703-2000-0003-0002, 0703-2100-0003-0002, 0703-2200-0003-0002, 0703-2300-0003-0002

AMS1 Slot1 filament has run out and automatically switched to the slot with the same filament.

What it is

AMS has an automatic filament switching function when a certain slot's filament is used up. When the filament properties of different slots are consistent and the AMS Filament Backup option is enabled. After one of the slot filaments is used up, it will automatically switch to another roll of the same filament for printing.

For details about AMS features, you can refer to the wiki: AMS main functions and workflow introduction | Bambu Lab Wiki

This HMS message is just a reminder, indicating that the automatic filament switching has been successful.

Error message

HMS_0700-2000-0003-0002: AMS1 Slot1 filament has run out and automatically switched to the slot with the same filament.

HMS_0700-2100-0003-0002: AMS1 Slot2 filament has run out and automatically switched to the slot with the same filament.

HMS_0700-2200-0003-0002: AMS1 Slot3 filament has run out and automatically switched to the slot with the same filament.

HMS_0700-2300-0003-0002: AMS1 Slot4 filament has run out and automatically switched to the slot with the same filament.

HMS_0701-2000-0003-0002: AMS2 Slot1 filament has run out and automatically switched to the slot with the same filament.

HMS_0701-2100-0003-0002: AMS2 Slot2 filament has run out and automatically switched to the slot with the same filament.

HMS_0701-2200-0003-0002: AMS2 Slot3 filament has run out and automatically switched to the slot with the same filament.

HMS_0701-2300-0003-0002: AMS2 Slot4 filament has run out and automatically switched to the slot with the same filament.

HMS_0702-2000-0003-0002: AMS3 Slot1 filament has run out and automatically switched to the slot with the same filament.

HMS_0702-2100-0003-0002: AMS3 Slot2 filament has run out and automatically switched to the slot with the same filament.

HMS_0702-2200-0003-0002: AMS3 Slot3 filament has run out and automatically switched to the slot with the same filament.

HMS_0702-2300-0003-0002: AMS3 Slot4 filament has run out and automatically switched to the slot with the same filament.

HMS_0703-2000-0003-0002: AMS4 Slot1 filament has run out and automatically switched to the slot with the same filament.

HMS_0703-2100-0003-0002: AMS4 Slot2 filament has run out and automatically switched to the slot with the same filament.

HMS_0703-2200-0003-0002: AMS4 Slot3 filament has run out and automatically switched to the slot with the same filament.

HMS_0703-2300-0003-0002: AMS4 Slot4 filament has run out and automatically switched to the slot with the same filament.

Error Code: 0700-4000-0002-0001

The filament buffer signal lost,the cable or position sensor may be malfunctioning.

What it is

HMS_0700_4000_0002_0001: The filament buffer signal lost, and the cable or position sensor may be malfunctioning.
When using AMS, the buffer (or AMS Hub) is connected between the AMS and the printer.

It has a slide, a spring, and a hall sensor. When the AMS pushes the filament into the tool head extruder, the slide will move forward due to the pressure of the filament. The hall sensor can then detect the displacement and location of the slide, and feedback that signal back to the AMS and the printer. If the buffer signal is lost, AMS will not be able to feed the filament smoothly.

Troubleshooting

1. Check&Reconnect the 6pin cable

Usually, this is a 6P cable issue; please check whether the Pins of the 6P cable are open:

And try reconnecting the 6-pin cable:

• If the problem persists, check if the bus cable inside the AMS is loose:

• It is recommended to replug the bus cable and try to load the filament again, to see if it can be recovered so that we can narrow the fault to the bus cable.

3. Measure the voltage & resistance

1. If you have a multimeter, please unload the filament back to AMS and check the slider of the AMS buffer is on the left side as shown below:

• Then measure the voltage between the pins of the 6Pin cable (unplugged from AMS).

• The normal voltage should be as follows. Make sure the buffer slider is back to the left side, otherwise the voltage value will be different.

and the sum of the two voltages is 3.0~3.3V.

2. If the voltage measured in the previous step is in the normal range, the problem may be on the AMS side. Please measure the resistance of the 6-pin cable(plugged into AMS). Normal resistance values are as follows:

pin-pin

Resistance

GND-Buffe_A

4~6 K:•

GND-Buffe_B

4~6 K:•

• If the measured resistance values are abnormal, please disassemble the AMS and measure the resistance of Buffer A and B to GND on the AMS mainboard. The normal range is 4 ~ 6 K:’à If the resistance values are abnormal, the AMS mainboard is faulty.

• Then check the connectivity between buffer A and buffer B on the AMS mainboard and the corresponding pins on the 6-pin cable (plugged into the AMS). If the connectivity is abnormal, there is a problem with the AMS power board and it needs to be replaced.

Please follow the above steps to test, record the test results, and contact the customer support team for further assistance.

4. Check the internal magnet orientation in the AMS buffer

If you mistakenly remove the four screws shown in the left image during AMS buffer disassembly, the internal magnet (shown in the right image) may fall out. Please ensure that when reinstalling the magnet, its north-south polarity matches the orientation shown in the right image.

The side highlighted in red in the figure indicates the North Pole (N). Please use this marking as a reference when installing the magnet.

Error Code: 0700-4000-0002-0002

The filament buffer position signal error,the position sensor may be malfunctioning.

What it is

HMS_0700-4000-0002-0002: The filament buffer position signal error,the position sensor may be malfunctioning.
0700-4000-0002-0002 (0700400000020002)

Troubleshooting

Please refer to the Wiki HMS_0700-4000-0002-0001: The filament buffer signal lost,the cable or position sensor may be malfunctioning.

Error Code: 0700-4000-0002-0003

The AMS Hub communication is abnormal,the cable may be not well connected.

What it is

HMS_0700-4000-0002-0003: The AMS Hub communication is abnormal,the cable may be not well connected.

0700-4000-0002-0003

Error Code: 0700-4000-0002-0004

The filament buffer signal is abnormal,the spring may be stuck or the filament may be tangle.

What it is

The buffer (or AMS Hub) is a filament buffer machine connecting the AMS and the printer.

It has a slide, a spring, and a hall sensor. When the AMS pushes the filament into the tool head extruder, the slide will move forward due to the pressure of the filament. The hall sensor can then detect the displacement and location of the slide, and feedback that signal back to the AMS and the printer. If the spring of the buffer gets stuck, it prevents the slider from moving to the far right, which affects the AMS feed.

Troubleshooting

When you see that the spring of the buffer (or AMS Hub) is stuck, as shown below, please use a screwdriver to pick up the spring from the buffer window and adjust it back to its normal status to ensure that the slider moves smoothly from side to side.

Another possible reason is that the filament is tangled or the spool is stuck and causes excessive resistance:

If the error persists after adjusting the spring, please upload the log file and contact the customer support team.

Error message

HMS_0700-4000-0002-0004: The filament buffer signal is abnormal,the spring may be stuck or the filament may be tangle.

0700-4000-0002-0004

Error Code: 0700-4500-0002-0001

The filament cutter sensor is malfunctioning.The sensor may be disconected or damaged.

What it is

HMS_0700-4500-0002-0001: The filament cutter sensor is malfunctioning.The sensor may be disconected or damaged.

0700-4500-0002-0001

You can refer to the troubleshooting guide on this page to check if the connector is properly inserted:

https://wiki.bambulab.com/en/x1/troubleshooting/hmscode/0700_4500_0002_0003

Error Code: 0700-4500-0002-0002

The filament cutter's cutting distance is too large.The XY motor may lose steps.

Observation

When the printer either unloads or switches filaments, the cutter trims the filament prior to retracting it. During this process, the toolhead shifts, causing the upper lever block to press against the cutter lever. This action propels the blade to sever the filament within the extruder. Once the filament has been cut, the cutter lever will automatically spring back into place. The cutter lever is depicted in the image below.

X1/P1 series:

A1/A1mini

This error typically arises due to:

1. Incorrect installation of the magnet/sensor;
2. Inadequate contact between the sensor's cable and connector;
3. Misplacement of the toolhead often results from the XY motor missing a step before the filament is cut.

Troubleshooting

Step1 Check the magnet/sensor

X1 TH V8

Please also check the 10-pin terminal and reinsert the connector as needed. (Sometimes, it may require several attempts at reinsertion, and it's essential to remove any silicone residue on top using tweezers beforehand. If the error continues, please reach out to our customer support team.)

X1 TH V9/P1

Please verify that the magnet/sensor is correctly installed and that the sensor's cable is properly connected. After confirming these, click "Retry".

A1/A1mini

The feed, winding, and cutter hall sensors are integrated on a single circuit board. Please refer to this wiki to check. Cutter Hall Sensor Disassembly and Assembly

Step2 Check the lost step of the XY motor

You can activate the video recording function to observe if the toolhead encounters any foreign objects during printing, which may cause it to lose its step: Enable the video recording function and export videos | Bambu Lab Wiki.

If the printed model does not exhibit layer misalignment due to lost steps, you can temporarily restart the printer to activate the "resuming prints after power outage" function. This will recalibrate the printer and allow the printing process to continue.

Error message

HMS_0700-4500-0002-0002: The filament cutter's cutting distance is too large.The XY motor may lose steps.

0700-4500-0002-0002

Error Code: 0700-4500-0002-0003

The filament cutter handle has not been released. The handle or blade may be jammed, or there could be an issue with the filament sensor connection.

Fault Description

When the printer unloads the filament or switches filaments during printing with AMS, the cutter is used to cut the filament before pulling it back. When cutting, the tool head will move to the front of the printer and let the top lever block press the cutter lever, which will push the blade to cut the filament in the extruder. After the filament is cut, the cutter lever will automatically spring back. The cutter lever is shown in the picture below:

The cutter lever has a magnet inside, and the extruder has a Hall sensor in the corresponding position. The printer uses the Hall sensor to sense the cutter's position to determine whether it has successfully sprung back. When it is detected that the cutter lever does not spring back to its original position for two minutes, the prompt will pop up on the screen.

When the cutter is stuck, the lever will be stuck in the tool head without springing back, as shown:

& þ IMPORTANT: IMPORTANT! Always power off and disconnect power on the printer before performing maintenance work. Not doing so means there is a risk of electric shock, short circuit, and damage to the printer or surrounding area. When a maintenance task necessitates the printer being powered on, use insulated gloves for safety and pay special care not to pinch, damage, or put pressure on any exposed wires, connectors, or circuit boards. Additionally, the nozzle can be extremely hot so never touch it with exposed skin. If you have any questions or concerns related to the above or the steps in this guide, please open a new ticket in our Support Page for assistance.

IMPORTANT!

Always power off and disconnect power on the printer before performing maintenance work. Not doing so means there is a risk of electric shock, short circuit, and damage to the printer or surrounding area.

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If you have any questions or concerns related to the above or the steps in this guide, please open a new ticket in our Support Page for assistance.

Troubleshooting Guide

When the screen shows that the cutter is stuck, first check whether the cutter lever is actually stuck inside the toolhead.

If the cutter lever is stuck at its limit position

1. Lever may rebound after being stuck for a while, depending on the filament

This can occasionally happen when printing with Support or Silk filaments. Simply click "Retry". If the message no longer appears, you can continue printing.

If clicking "Retry" multiple times does not resolve the issue, the Hall sensor on the extruder may be malfunctioning.

If you have a spare extruder, try replacing it to see if the problem persists.

2. Hotend blockage

If the hotend is clogged, excessive filament pressure between the extruder and the hotend can cause the cutter to get stuck and fail to rebound smoothly. In this case, you need to remove and clean the hotend.

> **!9þ NOTE: Ø=ÜÌ Please refer to this wiki for hotend cleaning: Nozzle Clog**

Ø=ÜÌ Please refer to this wiki for hotend cleaning: Nozzle Clog

3. If the hotend is not clogged, but the cutter still frequently gets stuck during cutting

Retract the filament from the extruder, and manually press the cutter lever to see if it moves freely without filament.

If the cutter still gets stuck, it indicates that the friction between the cutter sides and the extruder walls is too high, preventing smooth rebound.

This may be caused by manufacturing tolerances. Try removing the cutter, lightly sanding a small portion on both sides using a knife or other tool, then reinstall it. Press a few times to check if the friction has been reduced.

> **!9þ NOTE: Ø=ÜÌ Please refer to this guide for removing the cutter blade: Replace Toolhead Cutter**

Ø=ÜÌ Please refer to this guide for removing the cutter blade: Replace Toolhead Cutter

4. Front cover cables interfering with the cutter lever

Check and organize the cables to ensure they do not obstruct the lever rebound.

5. Cutter spring not installed correctly

If the cutter lever spring is not properly seated, the lever may fail to rebound smoothly. Make sure one end of the spring is inserted into the small hole to ensure proper positioning.

> **Ø=ÜÌ For detailed instructions, refer to Replace Cutter Lever.**

Ø=ÜÌ For detailed instructions, refer to Replace Cutter Lever.

If the cutter lever is in its original position and not physically stuck

1. Hall sensor false detection

Check for debris on the Hall sensor and wipe the magnet and sensor surface with a clean tissue, then power on the printer.

2. Hall sensor malfunction or cable disconnection causing false detection

For P1 series printers, ensure all pins are fully and correctly inserted.

The image below shows the correct installation on the left; do not misalign the connector.

3. Poor contact or connector damage

Try re-plugging the 10-pin connector (sometimes several times).

Before re-plugging, use tweezers to remove any silicone.

If the error persists, contact customer support.

For X1 series printers with V9 TH board, refer to Toolhead Housing, remove the toolhead cover, and re-plug both ends of this FPC.

!9þ NOTE: Ø=ÜÌ X1 Series V9 TH Board
Replace Toolhead Extruder Board Assembly | Bambu Lab Wiki
Replace Toolhead PCB (New Version) | Bambu Lab Wiki

Ø=ÜÌ X1 Series V9 TH Board
Replace Toolhead Extruder Board Assembly | Bambu Lab Wiki
Replace Toolhead PCB (New Version) | Bambu Lab Wiki

For P1 series printers, re-plug both ends of this FPC.

!9þ NOTE: Ø=ÜÌ P1 Series TH Board
Replace Toolhead PCB | Bambu Lab Wiki

Ø=ÜÌ P1 Series TH Board
Replace Toolhead PCB | Bambu Lab Wiki

Error Message

HMS_0700-4500-0002-0003: The filament cutter handle has not been released. The handle or blade may be jammed, or there could be an issue with the filament sensor connection.

Related AMS error:"Fail to cut the filament.Please check the cutter.After troubleshooting,click the retry button."

Error Codes: 0700-5000-0002-0001, 0701-5000-0002-0001, 0702-5000-0002-0001, 0703-5000-0002-0001

AMS1 communication is abnormal; please check the connection cable.

Observation

AMS or AMS Lite is connected to the printer via a cable. When the signal transmission quality between it and the printer is poor, this error message will pop up.

This error is usually caused by the following reasons:

1. The connector between AMS/AMS Lite and the printer is loose
2. The connection cable between AMS/AMS Lite and the printer is broken
3. AMS circuit board malfunction

Troubleshooting

- Check connection between AMS and X1/P1 series printers
  AMS connection with X1 series printers
  AMS connection with P1 series printers
- Connection between AMS Lite and A1/A1mini printers
  AMS Lite connection with printers
- Connection between AMS Pro with printers
  AMS Pro connection with printers
- Connection between AMS HT with printers
  AMS Pro connection with printers

Check connection between AMS and X1/P1 series printers
AMS connection with X1 series printers
AMS connection with P1 series printers

Connection between AMS Lite and A1/A1mini printers
AMS Lite connection with printers

Connection between AMS Pro with printers
AMS Pro connection with printers

Connection between AMS HT with printers
AMS Pro connection with printers

Step 2. Check for connection cable faults

Use a multimeter to measure the resistance value of the connection cable Pin, and compare it with the standard value to determine whether there is a connection cable fault.
AMS Pin resistance measurement
AMS Lite Pin resistance measurement

If the resistance measurement of each Pin is normal, it may be a circuit board fault. Please submit a ticket to contact the customer service team and attach the resistance measurement results.

Equivalent Codes

Error Code: 0700-5100-0003-0001

AMS is disabled; please load filament from spool holder.

What it is

Troubleshooting

To be added.

Error message

HMS_1200-5100-0003-0001: AMS is disabled; please load filament from spool holder.

HMS_0700-5100-0003-0001: The AMS is disabled; please load filament from spool holder.

Error Code: 0700-6000-0002-0001

The AMS1 slot1 is overloaded. The filament may be tangled or the spool may be stuck.

What it is

During the printing process, the filament hub motor of AMS assists in feeding the filament to the printer, reducing the resistance of the toolhead extruder to extrude the filament, and the motor will detect the resistance of the filament when the resistance continues to be too large and the speed of filament is too small, it will prompt overload error.

The picture below: the motor of the filament hub

Possible reasons

This error is usually caused by excessive resistance due to the following reasons:

1. filament tangled or sagging and stuck
2. the spool size exceeds AMS specifications or the cardboard spool is used
3. the feed funnel of the first-stage feeder is seriously worn
4. the roller support of the first-stage feeder was broken
5. the spring of the AMS buffer is stuck
6. the PTFE tube inside AMS is worn
7. the PTFE tube between AMS and printer is overbending

Operation guide

Step 1: Check if the spool conforms to the specifications or if any deformation is present

Please refer to this wiki to check if the spool used are within the AMS specifications; check if the width of the spool is between 50~68mm and the diameter is between 197~202mm; some spools may be deformed(so the cardboard spool is not recommended); please rotate the spool back and forth to check whether there is too much resistance at any position.

Step 2: Check if the filament is tangled or sagged and stuck.

Please avoid using filaments unsupported by AMS, such as TPU/damp PVA.

Step 3: Check if the feed funnel of the first-stage feeder is seriously worn.

If there is wear, please refer to the wiki for replacement.

Step 4: Check if the spring of the AMS buffer or Hub is stuck

The example on the left shows a stuck spring, while the example on the right demonstrates a properly functioning spring.

Step 5: Check for wear on the PTFE tube inside the AMS.

Please refer to this wiki to disassemble AMS and check if the PTFE tube inside AMS is worn.

Step 6: Check for any damage to the support of the rollers.

If any damage is found, please get in touch with the Bambu Lab after-sales support team for replacement.

Step 7: Check if the PTFE tube between the AMS and the printer is excessively bent.

Excessive bending of the PTFE tube between the AMS and the printer can cause additional resistance during filament feeding and retraction. Please refer to the diagram below to check for any bends.

Step 8: After following the above 7 steps the error message still persists.

If you have followed the above 7 steps and the issue continues to occur, please get in touch with our after-sales team for assistance.

Related error codes:

HMS_0700-6000-0002-0001: The AMS1 slot1 is overloaded. The filament may be tangled or the spool may be stuck.

HMS_0700-6100-0002-0001: The AMS1 slot2 is overloaded. The filament may be tangled or the spool may be stuck.

HMS_0700-6200-0002-0001: The AMS1 slot3 is overloaded. The filament may be tangled or the spool may be stuck.

HMS_0700-6300-0002-0001: The AMS1 slot4 is overloaded. The filament may be tangled or the spool may be stuck.

HMS_0701-6000-0002-0001: The AMS2 slot1 is overloaded. The filament may be tangled or the spool may be stuck.

HMS_0701-6100-0002-0001: The AMS2 slot2 is overloaded. The filament may be tangled or the spool may be stuck.

HMS_0701-6200-0002-0001: The AMS2 slot3 is overloaded. The filament may be tangled or the spool may be stuck.

HMS_0701-6300-0002-0001: The AMS2 slot4 is overloaded. The filament may be tangled or the spool may be stuck.

HMS_0702-6000-0002-0001: The AMS3 slot1 is overloaded. The filament may be tangled or the spool may be stuck.

HMS_0702-6100-0002-0001: The AMS3 slot2 is overloaded. The filament may be tangled or the spool may be stuck.

HMS_0702-6200-0002-0001: The AMS3 slot3 is overloaded. The filament may be tangled or the spool may be stuck.

HMS_0702-6300-0002-0001: The AMS3 slot4 is overloaded. The filament may be tangled or the spool may be stuck.

HMS_0703-6000-0002-0001: The AMS4 slot1 is overloaded. The filament may be tangled or the spool may be stuck.

HMS_0703-6100-0002-0001: The AMS4 slot2 is overloaded. The filament may be tangled or the spool may be stuck.

HMS_0703-6200-0002-0001: The AMS4 slot3 is overloaded. The filament may be tangled or the spool may be stuck.

HMS_0703-6300-0002-0001: The AMS4 slot4 is overloaded. The filament may be tangled or the spool may be stuck.

Error Codes: 0700-7000-0002-0001, 0701-7000-0002-0001, 0702-7000-0002-0001, 0703-7000-0002-0001

Failed to pull out the filament from the extruder. Please check whether the extruder is clogged or whether the filament is broken inside the extruder.

Observation

This error occurs when the extruder can still detect filament even after the AMS has completed the unloading process for a while.

Possible root causes:

- Extruder clog
- Extruder hall switch abnormal
- High unloading resistance

Troubleshooting

Tools, Equipment, and Materials

Please refer to the info inside each detailed troubleshooting wiki.

Fault Isolation Procedure

Actions

Operators may use discretion in determining the actual procedures used, and the order in which these steps are applied.

Step 1. Check extruder clog

When filament cannot be successfully unloaded from the extruder it leads to an unloading failure, it is likely due to a clog in the extruder or high unloading resistance.

In this case, press the PTFE tube connector on the back remove the PTFE tube first, and then manually pull out the filament. If you can't pull out the filament, it indicates that the filament is stuck inside the extruder; if you can easily pull out the filament from the toolhead, it indicates high unloading resistance, please see the high unloading resistance section below for troubleshooting. If the extruder is clogged, please refer to this Wiki article to clear the clog:

Extruder clog

Step 2. Check extruder hall switch abnormal

If the filament has been successfully unloaded from the extruder, it still reports a failure to exit the extruder. Please refer to the abnormal extruder sensor section to check whether the green dot of the extruder is on.

If the green dot is on, it may be:

- A broken filament inside the extruder.
- The extruder hall switch is abnormal.

Please refer to this wiki article for detailed troubleshooting procedures: HMS_0700-2000-0002-0004: The AMS slot1's filament may be broken in the tool head

Step 3. Check high unloading resistance

Usually, excessive bending of the PTFE tube of the toolhead, or debris accumulation in the AMS internal hub unit, can create high unloading resistance for filament unloading. First, please make sure to update the printer firmware to the latest version where the unloading failure issue is fixed.

(1) The PTFE tube is bent too much

If the PTFE tube of the toolhead is bent too much, it can create high resistance for the filament in the tube so that it can't be pulled back, and leads to unloading failure.

You can click here to download a PTFE tube bracket that can be mounted above the toolhead to prevent the PTFE tube from bending too much.

See the image below to install the bracket:

On Makerworld, there are more models to help straighten the PTFE tube to reduce unloading resistance.

Extruder AMS Filament Pull Back Fix by SPStudio - MakerWorld

AMS Filament Pull Back Fix V2 Upgraded by SPStudio - MakerWorld

AMS Fix Stack The AMS on Top and Fix For Failed To Retract Filament Error by SPStudio - MakerWorld

(2) Debris accumulation in AMS internal hub unit

After using the AMS for a long time, excessive accumulation of filament debris inside the internal hub unit can result in increased resistance and potential unloading failure.

You can refer to this wiki article for instructions on how to clean the internal hub unit:

Dismantling and cleaning the AMS internal hub unit

(3) Unloading failure of special filament

If you have an unloading failure for a special filament, such as PLA-CF or other hard filament, we recommend using the spool holder at the back of the printer when printing with the filament.

If the issue persists after troubleshooting, please submit a customer support ticket.

Equivalent Codes

The following HMS codes represent the same type of fault on different AMSs.

0700-7000-0002-0001 AMS1

0701-7000-0002-0001 AMS2

0702-7000-0002-0001 AMS3

0703-7000-0002-0001 AMS4

Error Codes: 0700-7000-0002-0006, 0701-7000-0002-0006, 0702-7000-0002-0006, 0703-7000-0002-0006

Timeout purging old filament. Please check if the filament is stuck or the extruder/nozzle is clogged.

Observation

The flushing of old material exceeded the time limit, triggering an alarm.

Possible reasons:

- There was an abnormality in the AMS retraction process, causing a jam (if using AMS).
- The extruder or nozzle is blocked.

Troubleshooting

- Referring the detailed Wiki in the following Actions.

Actions

Operators may use discretion in determining the actual procedures used, and the order in which these steps are applied.

Step 1. Check abnormality in the AMS retraction process.

Step 1.1 High unloading resistance that leads to unloading failure.

Please refer to the high unloading resistance section to reduce the resistance.

Step 1.2 The broken filament inside the AMS internal hub unit or abnormal sensors.

After the filament has fully exited the first-stage feeder, the AMS is still performing the unloading process with the LED status of the feeder in solid red (as shown below). This is possible that there is a broken filament inside the internal hub unit that leads to the unloading failure. Please refer to this wiki article to dismantle and clean the internal hub unit, and check if the sensor is normal.

Dismantling and cleaning the AMS internal hub unit

Step 1.3 Abnormal spool rotation

When unloading the filament, the AMS internal hub unit motor pulls back the filament, and the first-stage feeders rotate the spool back. If the situation in the image below happens, the spool doesn't rotate, or its rotating speed is slower than the unloading speed of the internal hub unit motor.

(1) The spool doesn't rotate.

It could be because the active support shaft is not fully inserted into the AMS main frame. As a result, only the yellow gear of the feeder rotates, while the black active support shaft remains stationary. This prevents the spool from rotating properly and can lead to unloading failures, which are usually accompanied by a humming sound from the gears of the first-stage feeder. Please see the image below to fully insert the active support shaft into the AMS main frame.

(2) The spool rotating speed is slower than the unloading speed of the internal hub unit motor.

During the unloading process, if the spool can rotate, but the speed is slower than the unloading speed of the internal hub unit motor, please check whether this issue only happens in a slot by swapping the first-stage feeders of different slots. If this only happens to a slot, it means that the first-stage feeder of this slot is malfunctioning. If this happens to multiple slots, please record a video and contact our customer support team.

Step 2. Investigate the extruder or nozzle blocked issues.

Check which part is clogging by referring to this Wiki.

How to troubleshoot Extruder clog
How to troubleshoot Nozzle clog

If the issue persists after troubleshooting, please submit a customer support ticket.

Equivalent Codes

The following HMS codes represent the same type of fault. 07[XX]-7000-0002-0006, XX means the No. of AMS which triggered this warning.

0700-7000-0002-0006,

0701-7000-0002-0006,

0702-7000-0002-0006,

0703-7000-0002-0006

Error Codes: 0700-7000-0002-0007, 0701-7000-0002-0007, 0702-7000-0002-0007, 0703-7000-0002-0007

AMS filament ran out. Please put a new filament into AMS and click the "Retry" button.

Observation

The printer showed a warning that filament ran out during the printing. The printing was paused due to the filament sensor in the feeder not detecting filament in the current slot.

Troubleshooting

The following troubleshooting process and actions aim to solve the most frequent issues, which are:

- The filament ran out during printing.
- None

Actions

Operators may use discretion in determining the actual procedures used, and the order in which these steps are applied.

- Follow this Wiki to replace the empty AMS slot with another spool of filaments.
- Please make sure the new filament you replaced is the same as the previously used one.
- After the AMS loads the new filament successfully, please manually continue the printing work.

If the issue persists after troubleshooting, please submit a customer support ticket.

Equivalent Codes

The following HMS codes represent the same type of fault. 07[XX]-7000-0002-0007, XX means the number of AMS which triggered this warning.

0700-7000-0002-0007,

0701-7000-0002-0007,

0702-7000-0002-0007,

0703-7000-0002-0007

Error Codes: 0700-7000-0002-0008, 0701-7000-0002-0008, 0702-7000-0002-0008, 0703-7000-0002-0008, 0704-7000-0002-0008, 0705-7000-0002-0008, 0706-7000-0002-0008, 0707-7000-0002-0008, 07FE-7000-0002-0008, 07FF-7000-0002-0008

Failed to get AMS mapping table; please click "Resume" to retry.

Observation

The printer was unable to read the mapping sheet despite numerous attempts, consequently setting off an alarm.

Possible reasons:

- There is an issue with the slicing software.
- The wrong color or incompatible materials were selected in the print file/slice file.
- The SD card is damaged.

Troubleshooting

Actions

Operators may use discretion in determining the actual procedures used, and the order in which these steps are applied.

Click “Retry” or reboot the printer. If ineffective, please try the following troubleshooting steps:

Step 1. Investigate any issues with the slicing software.

Please upgrade to the latest version of Bambu Studio. If you are already using the latest version of Bambu Studio, try uninstalling and reinstalling it.

Step 2. Check if the wrong color or incompatible materials were selected in the print file/slice file.

Ensure that the color in AMS matches the UI display of the AMS printer, and confirm the color/material settings are correct when sending the print. Please refer to this Wiki for setting up multicolor printing.

Step 3. Check for damage to the SD card.

If you are printing from a file on the SD card, it's possible that an issue with the SD card is causing the machine to read the material color data incorrectly. Try using other file transfer methods, or replace the SD card.

If the issue persists after troubleshooting, please submit a customer support ticket.

Equivalent Codes

The following HMS codes represent the same type of fault. 07[XX]-7000-0002-0008, XX means the number of AMS which triggered this warning.

0700-7000-0002-0008,

0701-7000-0002-0008,

0702-7000-0002-0008,

0703-7000-0002-0008

Error Code: 07FF-2000-0002-0001

HMS_07FF-2000-0002-0001: External filament has run out; please load a new filament.

What it is

When printing with spool holder filament, this warning will be triggered if the filament runs out.

Troubleshooting guide

Feed the new filament into the extruder and click the “retry” (or “resume”, depending on the firmware version) button on the screen.

Error message

HMS_07FF-2000-0002-0001: External filament has run out; please load a new filament.
07FF-2000-0002-0001

Error Code: 07FF-2000-0002-0002

HMS_07FF-2000-0002-0002: External filament is missing; please load a new filament.

What it is

This error occurs if the filament never reaches the extruder when you feed the filament from the external spool holder. For an introduction to the external spool holder(also known as "virtual slot"), you can refer to firmware release history (version 01.05.01.00): X1/X1C and AMS Firmware Release History | Bambu Lab Wiki

Troubleshooting

Manually feed the filament into the extruder and click the “retry” (or “resume” depending on the firmware version) button on the screen.

If you want to load filament from AMS instead of the external spool holder, and the printer shows the error, please check:

1. If you are using an X1 series printer, check whether you have incorrectly selected "Virtual slot" on the display before loading. Select the correct AMS slot and click "load" again.

2. If you are using a P1 series printer, when you click "load" on the display screen, it loads the filament from the external spool holder, requiring you to manually insert the filament into the extruder. Even if the printer is connected to AMS, the filament cannot be loaded from AMS through this operation. If you want to feed from AMS, you need to select the corresponding slot on Bambu Studio and click "Load".

Error Code: 07FF-2000-0002-0004

HMS_07FF-2000-0002-0004: Please pull out the filament on the spool holder from the extruder.

What it is

This prompt will pop up when unloading from the spool holder if the filament is not pulled out of the extruder in time. If you don't pull out the filament, the extruder hall sensors will always be triggered, and the system will think that the "unload" instruction has not been completed, thus blocking all subsequent instructions.

Troubleshooting

Printing with spool holders

This message appears when printing with the spool holder because printing with the spool holder does not complete the automatic unloading process and you need to manually pull the filament out of the tool head to complete the unloading process.

Printing with AMS

This message appears when printing with the AMS, this is due to the last time the spool holder was used for printing, as printing with the spool holder does not allow for the automatic unloading process and the filament may not have been manually pulled out of the tool head, so we recommend checking to make sure that the filament has been pulled out of the tool head first, and that the PTFE tubes that are connected from the AMS to the printer are all connected.

Expand knowledge: If the filament in the current tool head is coming from one of the AMS slots before the filament is loaded through the AMS, the AMS will automatically unload the filament from the tool head, return it to the AMS and re-load the tool head, and the prompt will not appear either.

If the error persists after you remove the filament, there may be a broken filament inside the extruder, or the extruder's filament detection switch is triggered abnormally, please refer to the Wiki for troubleshooting:
HMS_0700_2000_0002_0004: The AMS slot1's filament may be broken in the tool head | Bambu Lab Wiki

You can also use the dot color on the control page of Bambu Studio or the display screen to determine if there is a filament inside the extruder.

Error message

HMS_07FF-2000-0002-0004: Please pull out the filament on the spool holder from the extruder.
07FF-2000-0002-0004